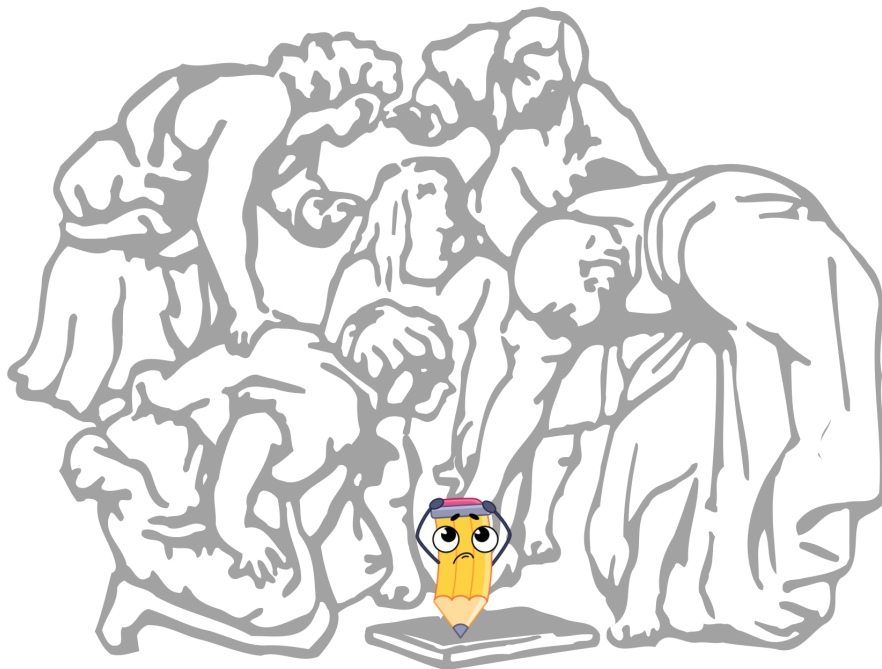


Risk of Dysgraphia Detection Using Raw Time Series Data from a Sensor Equipped Ink Pen



GROUP HST-25-ST-9-9403

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Abstract:

Background & Aim: Dysgraphia is characterized by difficulties with etc. legibility, writing, and fine motor skills. Dysgraphia has a negative impact on children's academic results, daily life, and self-esteem. Machine learning algorithms with tablets have been investigated to detect dysgraphia, but the surface seems insufficient for this task. However, this study aimed to develop a time series classifier to detect risk and no risk of dysgraphia using raw time series data collected with a sensor equipped ink pen.

Methods & Materials: Raw time series data recorded with a sensor equipped ink pen from typically developing children (Risk=729, No Risk=1933). Time series classifiers were trained and tested with different configurations. The performance of the time series classifiers was evaluated using accuracy, precision, recall, F1-score, AUC, ROC, confusion matrix, and school grade level confusion matrix.

Results: MINIROCKET-HYDRA with balanced class weights using accelerometer, gyroscope, pressure, and school grade level as features, with accuracy=76.54%, precision=55.73%, recall=69.86%, F1-score=62.00%, AUC=0.81.

Conclusion: The MINIROCKET-HYDRA classifier showed potential to detect the risk of dysgraphia in typically developing children by using raw time series data recorded by a sensor equipped ink pen.

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Preface

The project is written by group HST-25-st-9-9403 in the 3rd semester of the Master of Science in Biomedical Engineering and Informatics at Aalborg University, with the theme *Biomedical Research Collaboration*. The project was allocated 30 ECTS point and written from September 1st to January 2nd and has been developed under guidance from supervisor Erika G. Spaich, Associate Professor, Department of Health Science and Technology at Aalborg University. The project was written in collaboration with Politecnico di Milano University with guidance from Simona Ferrante, Associate Professor, Linda Greta Dui, Junior Researcher, and Simone Toffoli, PhD Student.

Reference method

The reference method used in the project is the Vancouver method. The reference will appear as a number in square brackets [X]. The number in the square brackets will correspond to the source reference in the bibliography, which is arranged chronologically. If the reference is positioned before the end of a sentence, it means that the reference belongs to that specific sentence. If the reference is referred to as study by XX et al. the year of the reference is defined in round brackets (XXXX).

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Part I

Problem Analysis

1. Introduction

Handwriting is a fundamental skill essential for academic success, influencing both legibility and writing speed [7, 20]. It develops gradually from early childhood, beginning with basic visual motor coordination and progressing to fluent, automatic writing by around age 15 [7, 20]. This process requires the integration of motor skills and higher-level cognitive functions, including planning, language, and working memory [30]. Difficulties in automatization or motor control can reduce writing efficiency and in some children, these challenges manifest as dysgraphia, a disorder affecting legibility, writing speed, and fine motor skills [23, 13, 7]. Affected children often show hesitant letter formation and poor pen control, reflecting increased movement variability [4]. Estimates suggest that 10–30% of children experience significant writing difficulties, with boys being more frequently affected [7, 23]. There is no standard for diagnosing dysgraphia but the evaluation is often based on the final products and not the motor process [7, 15]. Since written skills are developed in the early school years, dysgraphia is frequently diagnosed at a later stage [7]. Late diagnosing can lead to reduced academic performance, decreased self-esteem and difficulties in social relationships [23]. For diagnosing dysgraphia, Concise Assessment Method for Children’s Handwriting (BHK) is utilized as the most common assessment for writing skills focused on the final product [30].

The BHK evaluation requires approximately 15 minutes of manual scoring [4] and depends on subjective assessments, therefore more quantitative approaches have been investigated. Image analysis showed promising results[12], but did not eliminate the requirement of human evaluation. New measurement methods was developed, including the use of tablets, to also include motor process aspects in the quantitative diagnosing approach [4, 3]. In contrast, studies from Gerth et al.(2016), No and Choi(2021) and Alamargot et al.(2015) underline the importance of the writing medium to get reliable results. This leads to the interest of measuring the movements through a sensor equipped pen and not the writing medium. A sensor equipped pen provides access to raw time series data, making it possible to apply time series classification techniques directly [5, 25].

To prevent the consequences of writing difficulties and to investigate quantitative approaches for detecting dysgraphia using time series classification, the following initial problem formulation will be researched:

How do handwriting difficulties influence children’s writing development, and how can time series classification techniques enhance quantitative detection of dysgraphia?

2. Writing Development and Dysgraphia

This chapter describes how handwriting difficulties influence children’s writing development. In Section 2.1, the acquisition of handwriting skills for typically developing children is described, which is followed by the description of how writing difficulties affect the children in Section 2.2. The focus of this study is dysgraphia, defined in Section 2.3, where the symptoms and diagnosis of dysgraphia are further described in Section 2.4.

2.1 Acquisition of Handwriting Skills

Functional handwriting is defined by the legibility of text and evaluated using five features of letter formation, such as shape, slant, size, spacing, and alignment [30]. Acquisition of writing skills begins in early childhood, where the transcription training starts in the preschool with symbols and form drawing to develop visual motor coordination [7]. Handwriting is first developed qualitatively in terms of legibility and later quantitatively in terms of writing speed [4, 17]. In typically developing children (TDC), handwriting improves rapidly around ages 6–7, levels off around ages 7-8 year, and becomes more automatic around ages 8-9 years [30]. Letter awareness and understanding of letter sound relationships typically develop around ages 7-8 years, whereas the automaticity of letter writing, in which letters are produced as a rote motor response, is generally achieved around ages 8-9 years [7]. In the elementary school years, automaticity and handwriting are improved with additional skills for long-term outcome [7].

Handwriting is a complex skill that requires continuous interaction between lower-level and higher-level perceptual-motor processes [30, 4, 20]. Lower-level perceptual-motor processes are defined as motor planning and execution, and higher-level cognitive processes are defined as psycholinguistic and executive functions such as planning, language generation, reading, and editing [30, 4, 20]. Once the lower-level motor processes become relatively automatic, the higher-level functions can operate simultaneously [30].

Additionally, a study by Kiefer et al. (2016) [21] addresses that writing has shifted from handwriting to digital devices such as computers, tablets, and smartphones in recent years. Adults and children increasingly type at digital devices instead of writing by hand, and children often gain their first writing experiences through digital interaction. This early exposure to digital tools can affect the development of handwriting skills, as using keyboards requires less complex motor coordination.

2.2 Writing Difficulties

A study by Chung et al. (2020) emphasizes that the ability to write is essential in academic contexts, in fact writing tasks could occupy half of a school day. Students with written difficulties are frequently mislabeled as lazy or sloppy instead of being identified as having a learning disorder. In a study by Kushki et al. (2011), it is addressed that difficulties can lead to lower academic achievement, reduced self-esteem, and problems in social relationships. It is further supported by Asselborn et al. (2020), Asselborn et al. (2018), Dui et al. (2021) and Van Hoorn et al. (2013), highlighting that insufficient handwriting proficiency affects academic performance, self-confidence, and behavior.

During the handwriting development the motor aspects of handwriting become increasingly automated, but in cases of writing difficulties, this process can negatively affect the overall quality of handwriting [23]. Failure to develop writing automaticity by third grade is associated with an increased risk of difficulties in more complex writing tasks [7]. Consequently, extended writing tasks may negatively affect both legibility and writing speed, likely due to fluctuations in attention alongside muscular and psychological fatigue [23].

Writing difficulties have been defined in several ways [31]. In the Diagnostic and Statistical Manual of Mental Disorders, 5th edition (DSM-5), written difficulties are placed under the broader category of *specific learning disorder in written expression* characterized by problems with spelling, grammar, punctuation, and overall written expression [7, 13]. The International Statistical Classification of Disease and Related Health Problems (ICD-10) distinguishes between combined difficulties in reading and writing and isolated difficulties in writing, which are categorized as *developmental disorders of academic skills* [13]. In contrast, the Individuals with Disabilities Education Act (IDEA) in United States includes the written difficulties under the broader category *Specific Learning Disability* [7]. Despite these differences in terminology and classification, a common thread across the various frameworks is that written difficulties are often defined and referred to as *dysgraphia* [31, 7, 23]. Accordingly, this study is concerned with *dysgraphia* as its central focus.

2.3 Definition and Characteristics of Dysgraphia

Chung et al.(2020) characterized dysgraphia by difficulties with legibility, spelling, grammar, writing speed, fine motor skills, and composition. Dysgraphia is often acquired during the writing development phase, where children do not develop typical writing skills, which is the focus of this study. Asselborn et al. (2018) mentioned that children with dysgraphia (CD) show hesitant letter formation and reduced pen control, indicating poor motor control and increased movement variability. Rosenblum et al. (2004) describes that CD achieve higher neuro motor noise, which indicates a limited ability to

regulate muscle strength. In continuation of this, Van Hoorn et al. (2013) defines the legibility and speed as the most important elements of handwriting. The difference between handwriting of CD and TDC are illustrated in Figure 2.1.

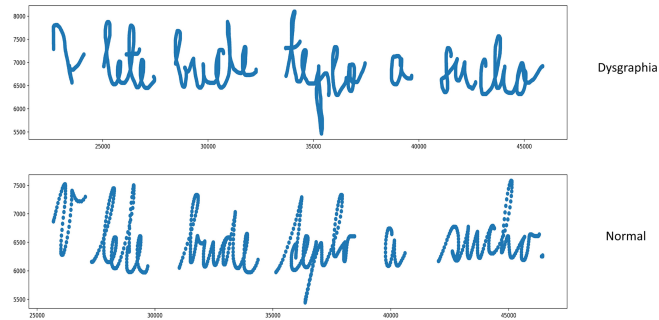


Figure 2.1: Visualization of handwriting acquired of (Dysgraphia) CD and (Normal) TDC. [22]

The prevalence of dysgraphia depends on the definition of dysgraphia as described in section 2.2, but 10-30% of children experience writing difficulties, most commonly among boys [7, 23]. The prevalence depends on the grade level, the criteria used to select students, and the assessment or testing methods applied [20], where grade level is referred to as the students school grade. Overvelde et al. (2011) emphasize that the timing of assessment may strongly influence the detection of handwriting difficulties and could partly explain the large variation in reported prevalence. It also underlines that the variation is clinically relevant, as illegible or slow handwriting is a frequent reason for referral to occupational or physiotherapy.

Dohla and Heim (2016) addressed that phonological processing difficulties are associated with reading and writing disabilities. It involves phonological awareness, including working with language sounds, phonological working memory, including storing and processing sounds, and phonological recoding, including rapidly retrieving sound codes. These abilities are crucial for reading fluency, word decoding, and written language skills. Dysgraphia is highly correlated with dyslexia, with 30-47% [7], which is defined as difficulties in reading, where the developmental phases indicate similarities, as the alphabet plays a key role in both writing and reading [13].

2.4 Symptoms and Diagnosis of Dysgraphia

A study by Kushki et. al (2011) observed that the increased grip force on the pen barrel and pressure on the writing surface in TDC and CD indicates compensatory strategies against fatigue, such as greater muscle stiffness and postural adjustments, reducing legibility by limiting fluency. These forces were measured using an instrumented pen capable of recording pressure and grip dynamics during a 10 minutes writing task. The study further found that the vertical speed decreased, indicating fast fatigue in the fingers, while the horizontal speed increased, due to the wrist-driven movements being

less affected by fatigue and allowing compensation through greater extensions. This is followed up by Chung et al. (2020), who defined the symptoms for preschool CD as awkward pencil grip, rapid fatigue, avoidance of writing tasks, inconsistent letter form, and problems with margins.

For diagnosing dysgraphia, there is no standard, but it is often diagnosed in an educational setting by interdisciplinary teams [7]. The diagnosis of dysgraphia often involves evaluating the handwriting process, focusing on both the speed and quality of the written product, but typically lacks a quantitative assessment of the underlying motor skills [15]. Written skills are often developed in the early school years [7]. Therefore, it is difficult to diagnose during the period, and dysgraphia is often diagnosed later [7]. According to ICD-10 and DSM-5, no specific recommendations are provided regarding the tests most suitable for diagnosing of reading or writing difficulties and there are no guidelines offered on appropriate therapeutic approaches [13]. The diagnose is based on multiple contexts, including observation, review of completed work, anecdotal report, and normative data [7]. The recommendation of diagnosis of dysgraphia is demonstrated by slow writing speed and illegible handwriting, with a notable discrepancy between spelling skills and verbal intelligence, as well as observed delays in graphomotor planning, orthographic awareness, and/or rapid automatized naming [7].

The Concise Assessment Method for Children’s Handwriting (BHK) is often used as an assessment of children’s writing skills addressed in the study by Overvelde et al.(2011). The BHK is suited for assessing the rapid development of handwriting in the early grades and for providing detailed profiles of risk or dysgraphia. It assesses broader aspects such as the organization of written work and includes items on letter formation, as well as checks of letter trace quality. Norms for handwriting quality are available only for grades 2 and 3, while speed norms exist for all grades. Visual motor integration (VMI) is an assessment that measures copying of geometric forms in a time span of 3 minutes. BHK is suitable to differentiate between normal writing, risk of handwriting difficulties, and dysgraphia handwriting, where VMI can not be used alone, but as a supplement. VMI is relevant in relation to the early writing process as the assessment analyzes the processing of visual information translated to motor and monitors the motor in the writing process [33]. By assessing motor difficulties, it is possible, with an earlier intervention of dysgraphia, as motor is one of the key elements of handwriting [33].

3. Quantitative Assessment of Dysgraphia

To investigate the possibilities of a more quantitative and less resource-intensive diagnosing method, Dimauro et al.(2020) developed a software system designed to support the diagnosis of dysgraphia by automating the BHK. The system processes scanned handwritten texts and automatically extracts 13 parameters, such as size, skewed writing, and space between words, with fully and semi-automatic document analysis algorithms. The software reached a Recall of 83%, a specificity of 98%, and an accuracy of 96%. The study demonstrates that automated assessment enables faster and more quantitative evaluation, with results comparable to those of expert manual scoring. However, the system is challenging with borderline cases and overlapping handwriting, where human evaluation remains essential.

3.1 Digitizer Based Assessment with Engineered Features

As highlighted by Asselborn et al. (2020) and Asselborn et al. (2018), digitizing tablets have opened new possibilities for acquiring continuous motor data during handwriting tasks, moving beyond the traditional focus on the final written product. Rosenblum et al. (2004) emphasize that digitizer based systems enable the handwriting evaluation in terms of temporal, spatial, and kinematic measures. Using a wireless pen and a personal computer, digitizer technology records x-y coordinates as well as kinematic and kinetic parameters throughout the writing process. Gargot et al. (2020) further note that digital tablets expand the possibilities for analysis by capturing dynamic features such as movement dynamics and pen pressure.

In a study by Asselborn et al.(2018), a tablet-based method for the clinical assessment of dysgraphia was developed. In contrast to the BHK assessment, which typically requires around 15 minutes of manual scoring, the tablet based system processes the handwriting data within milliseconds, as feature extraction and scoring are fully automated. It integrates static and dynamic features of handwriting for instance, spatial characteristics of the written text including letter size, spacing, and alignment, as well as temporal and kinematic parameters recorded during writing including pen trajectory, speed, pressure, and pen-lifts. By eliminating manual scoring, the system reduces subjectivity.

Since motor control is central to handwriting, Kushki et al.(2011) emphasized that temporal data becomes relevant, offering a dynamic view of the writing process and enabling early detection of dysgraphia. Building on this Drotar and Dobes (2020) [14] proposed a Machine Learning (ML)

approach using digitizing tablets to detect handwriting impairments related to dysgraphia. Their study was based on data from 120 schoolchildren aged 8-15, and extracted over 1.0000 temporal and kinematic features, including velocity, acceleration, pressure, and pen lifts. The best performance was found with the adaptive boost classifier, reporting an accuracy of 79.5%, a specificity of 76.5%, and a Recall of 79.7%. The study underlined the importance of task difficulty and feature selection, showing that complex handwriting tasks facilitate more dysgraphia related information in the temporal and kinematic features. The study hypothesized that a smaller age group could make the classification easier, as it would provide more dysgraphia cases in each age group. Nevertheless, their results showed the possibility to differentiate between dysgraphia and non-dysgraphia children with ML.

Using the same dataset as Drotar and Dobes (2020), Kunhoth et al.(2023) developed ML models to identify dysgraphia in children. The tablet recorded pen's trajectory, timing, pressure and tilt, capturing both On-Surface and In-Air movements as temporal data. In-Air features capture pauses, pen lifts, and atypical movement patterns that may indicate handwriting difficulties. From these time series, the kinematic, dynamic, temporal, and spatial features were extracted and combined into feature vectors representing each child's handwriting. Results showed that temporal and kinematic features were most important for distinguishing typical from dysgraphia handwriting. The extracted features was used to train several ML algorithms where the adaptive boost model performed best with an accuracy of 80.8%, precision of 83.3%, recall of 78.5%, and a F1-score of 80.1%. In-Air features enhance performance by capturing pauses and atypical movements not visible On-Surface.

Asselborn et al. (2020) advanced this line of research by developing a model that incorporates the motor process into dysgraphia assessment, rather than only on the static handwriting outcomes. In their study, 448 children, including 58 dysgraphia children aged 5-12 were recruited. The children were writing the first five sentences from the BHK test on an iPad with an Apple Pencil. The features extracted were classified in four categories: Static, Kinematic, Pressure, and Tilt. The data was analyzed using unsupervised methods, where Principal Components Analysis and K-mean clustering were utilized to identify underlying patterns in the data. Instead of a binary classification, the study divides the severity into five levels. The model achieved a Recall of 91% and a specificity of 90%. It is critical to mention that this study only included children with severe dysgraphia in the disease group. One of the interesting findings in this study was that the kinematic and pressure features were the most informative tested features.

3.2 Impact of the Writing Medium

In previous studies by Gerth et al.(2016) , Alamargot et al.(2015), No and Choi(2021), it is shown that tablet as writing medium can pose several challenges. Children tend to make more errors, produce less

legible handwriting, and struggle with precision and motor control due to the reduced friction of the screen surface. While adults compensate by adjusting pressure and letter size, children often experience lower writing quality and accuracy. Although writing speed is usually higher on tablets which comes at the cost of reduced motor control. Based on these findings, paper is generally considered the more suitable medium for developing and assessing early writing skills.

To examine how the writing medium affects the writing accuracy, Gerth et al. (2016) investigated performance differences between writing on paper and tablet among both children and adults across various writing tasks. On the tablet, a Lenovo pen was used, and for writing on paper, an Intuous Inking Pen was used. Their results showed that children made significantly more errors when using a tablet compared to paper during graphomotor exercises, which assess fine motor control through repetitive writing patterns. In contrast, adults exhibited a higher error rate on paper than on tablets for the same tasks. In visual motor exercises, defined as tasks requiring the copying of visually presented geometric forms, both younger and older children performed better on paper than on tablets, whereas adults achieved perfect scores regardless of medium. These findings indicate that tablets place higher demands on motor coordination for beginning writers, which may explain the increased error rates among children.

3.2.1 Reduced Handwriting Legibility

Regarding handwriting legibility, Gerth et al. (2016) found that adults produced more readable text on paper than on tablets, while children demonstrated a similar trend, although the difference was not statistically significant. Another study by Alamargot et al. (2015) also found reduced letter legibility for children when comparing tablet and paper surfaces. During tests of second-grade and ninth-grade students' handwriting skills, the participants were instructed to write the alphabet and their full name on both a tablet using a plastic-tipped stylus and on paper using the same pen equipped with a ballpoint ink tip. Writing movements were recorded to analyze handwriting legibility, speed, and motor control. The study found a significant decrease in letter legibility and an increase in pause duration when writing on the tablet, for both age groups. It was also observed that the ninth grade students compensated by increasing the letter size and pressure when writing on the tablet.

3.2.2 Writing Speed and Motor Control

Gerth et al. (2016) also found that handwriting speed was generally higher on tablets across all age groups. This increase was attributed to the reduced friction of the tablet surface, which results in diminished control and precision. The same study reported that first- and second-grade students experienced difficulty adapting to the tablet's surface, resulting in lower writing quality and accuracy. In contrast, adults were better able to compensate for the reduced friction by adjusting writing pressure

and letter size. The study claims that writing on a tablet places greater demands on motor control and may pose challenges for beginning writers, making paper a more suitable medium for assessing early writing development. Additional evidence of medium dependent motor demands was provided in a study by No and Choi (2021), which reported that 6 year old preschoolers’ writing skills vary depending on the writing medium. The children wrote larger, faster, and with less pressure when writing on the tablet screens than on paper. These results further support the conclusion that tablet surfaces systematically alter children’s handwriting movements and speed.

3.3 Sensor Equipped Pen Assessment with Raw Time Series

Based on the findings of Gerth et al. (2016), Alamargot et al. (2015), and No and Choi (2021), it is evident that tablet surfaces systematically influence children’s writing behavior, leading to faster but less controlled movements and reduced legibility. These studies suggest that movement and speed measurements should be derived from alternative sources rather than the tablet surface itself. Overall, the evidence indicates that tablets promote higher writing speed at the cost of precision, particularly among younger children, whereas paper remains a more suitable medium for assessing early writing development due to its support for greater control, fewer errors, and clearer handwriting.

Several writing tools have been developed to choose the optimal writing surface. Technologies that have shown promising results for temporal handwriting monitoring are the *SensoGrip Smart Pen*, *Diagnostic Pen*, and the *Smart Ink Pen (SIP)*.

The *SensoGrip Smart Pen* is a sensor equipped pen that registers handwriting dynamics. The pen is developed in the study by Bublin et al. (2023) [5]. Using the pen, it is possible to write on a more natural writing surface, such as paper. The pen has integrated force-, accelerometer-, and gyroscope sensors, which facilitate the measurement of pressure, velocity, acceleration, angle, and pauses between strokes. Bublin et al. (2023) applied deep learning models to predict dysgraphia. The study did not provide a binary output as dysgraphia or no dysgraphia, but the scores of the German version of the Systematic Screening for Motor Handwriting Difficulties (SEMS), which provides scores between 0 and 12, despite their data only presents scores between 0 and 8. The developed model with the most success was the Long Short-Term Memory (LSTM) model, which achieved an accuracy of 99.8% and a F1 score of 97.78%, but on a small dataset, which only includes 22 children with only 1 child in each group with the scores 1, 7, and 8. The study also investigated the effect of including age and gender variables, and found that gender and age are significant factors for accurate SEMS score prediction. The study opens the possibilities for writing on paper to reduce the surface impact on the writing quality while still acquiring informative dynamic features of the writing movements.

The *Diagnostic Pen* was developed to facilitate early diagnosis of Parkinson’s disease by converting handwriting movements into electrical signals in the study by Chen et al. (2025). The diagnostic pen has a soft magnetoelastic tip, which contains magnetic particles. By using the pen for writing, the mechanical pressure and motion deform the tip, altering the magnetic field. These changes are captured by a conductive coil wrapped around the pen, which converts magnetic flux variation into electrical signals. Furthermore, the pen uses supermagnetic nanomagnets ink that responds dynamically to magnetic fields during writing.

The *SIP* was developed within the European MoveCare Project [24] which aimed to detect early signs of cognitive and physical decline in independent elderly adults living at home. Lunardini et al. (2021) describe how the SIP was designed to monitor age-related changes in handwriting and tremor during daily life. The SIP allows the user to write on a normal piece of paper instead of a tablet by recording pressure and movement data. For this study, data acquired with the SIP were utilized to develop a more quantitative approach to detect dysgraphia. The SIP are further described in Chapter 7.

3.4 Summary of Studies

A summary of the highlighted studies is presented in Table 3.1, including their methods, participant characteristics, and key results.

Study	Methods	Participants	Results
Dimauro et al. (2020)	Automated BHK assessment using image-processing to extract 13 handwriting parameters. Classification of dysgraphia performed with supervised ML models trained.	104 children including 12 with dysgraphia, grades 2–5	Accuracy=96% Recall=83% Specificity=98%
Drotar & Dobes (2020)	Non-linear classification using digitizing-tablet handwriting data. More than 1,000 kinematic and temporal features were extracted from position, velocity, acceleration, pressure, and pen-lifts	120 children including 57 with dysgraphia, aged 8–15 years	Accuracy=79.5% Specificity=76.5% Recall=79.7%
Asselborn et al. (2020)	Tablet-based handwriting acquisition with ML: raw temporal data were collected and transformed into 53 features across static, kinematic, pressure, and tilt. PCA for dimensionality reduction, followed by k-means clustering to group children by handwriting quality.	448 including 58 with dysgraphia, aged 5–12 years	Recall=91% Specificity=90%
Kushki et al. (2011)	Sensor equipped pen measuring grip force, surface pressure, and handwriting kinematics during a 10-minute copying task. Writing was segmented into five 2-minute intervals, and kinematic changes over time were analyzed using hierarchical linear models.	105 fourth grade children	Higher grip and normal forces over time, reflecting fatigue compensation and reduced fluency.
Kunhoth et al. (2023)	Tablet-based handwriting acquisition capturing both On-Surface and In-Air movements, combined feature extraction over kinematic, temporal, dynamic, spatial features. Best performing ML classification were achieved by Adaptive Boost.	120 children including 57 with dysgraphia, aged 8–15 years	Accuracy=80.8% Precision=83.3% Recall=78.5% F1-score=80.1%
Bublin et al. (2023)	SensoGrip smart pen recorded continuous pressure, grip, tilt, and motion data while children wrote on paper. LSTM network used to predict SEMS handwriting scores from raw sensor time series.	22 children, aged 7–9 years	Accuracy=99.8% F1-score=97.78%

Table 3.1: Summary of studies focused on quantitative assessment of typical and dysgraphia handwriting, including data acquisition methods, participants, and key performance metrics.

4. Raw Time Series Classification Techniques

Important features in raw time series data were reported in the study by Bublin et al.(2023). In this study, less computationally expensive time series classification methods were investigated to assess the risk of dysgraphia.

In a review by Middlehurst et al. (2024) [27], different time series classification methods are compared. The review highlights that MultiROCKET-HYDRA (MR-HYDRA) and HIVE-COTE v2 (HC2) demonstrate significantly higher performance than other time series classification methods. HC2 achieves the highest overall accuracy and performs best on balanced datasets, but it requires substantially more computation time. In contrast, MR-HYDRA attains nearly the same accuracy with a much faster runtime, shows the best performance on imbalanced datasets and performs well on larger datasets. QUANT is a strong interval based method, providing competitive accuracy with high efficiency which makes it suitable for fast exploratory analysis. The review shows that hybrid algorithms which combine convolutional or shapelet based features with dictionary based approaches remains highly effective. Additionally, incorporating techniques such as dilation and transformed series, including first-order differences, further improves accuracy. These improvements are particularly evident in the methods highlighted in the review, such as ROCKET and HC2. While HC2 requires much more computational time, this study focused on MR-HYDRA and QUANT.

4.1 Convolution Based Classification Techniques

Convolution classifiers apply kernels to extract discriminative features from time series data was addressed by Middlehurst et al. (2024). The convolution operators slides each kernel across the time series, producing an activation map that represent the transformed signal and provides a compact summary of its extracted features. Pooling operations are then applied to each activation map to extract a single feature per kernel, and the resulting features are concatenated into a feature vector. Typically, a ridge classifier is trained on this vector to perform classification.

4.1.1 ROCKET Methods

Even the most accurate time series classification methods are often computationally heavy, slow to train even on small datasets, and limited scalability [8, 32]. State-of-the-art classification accuracy can be achieved in a fraction of the time by transforming time series data with random convolutional kernels and training a linear classifier on the resulting features where the complex training is avoiding

as in convolutional neural network [8]. A method developed for the purpose is RandOm Convolutional KErnel Transform (ROCKET) [8]. The ROCKET methods provided by Dempster et al. (2020) is a fast method for feature extraction of time series, the diagram of the ROCKET pipeline is illustrated in Figure 4.1. Described by Dempster et al. (2021) ROCKET transforms each time series data by using 10,000 random convolutional kernels with varying parameter such as random length, weight, bias, dilation and padding. Each kernel produces two features, maximum pooling value and proportion of positive values (PPV), which are used to train a linear classifier, typically ridge regression. Since the kernel weights are fixed, computation is inexpensive, making it feasible to apply a large number of kernels at low computational cost. The problem with the ROCKET method is that it still relies on random kernels, which makes it inefficient and non-deterministic.

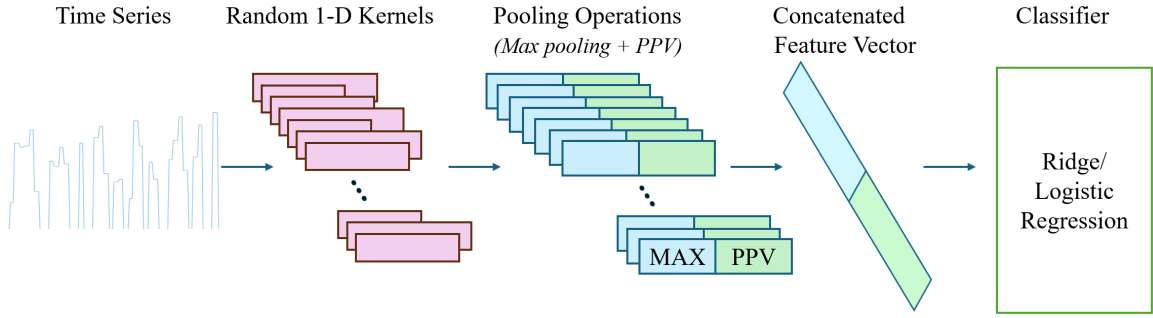


Figure 4.1: Visualization of the ROCKET method pipeline, where Random 1 Dimensional(1-D) kernels are applied to the Time Series signal, followed by Pooling operators of Max Pooling(MAX) and Proportion of Positive Values(PPV). Lastly, a concatenation of the feature vectors is performed before classification with a Ridge Classifier.

To improve ROCKET, the MINImally RandOm Convolutional KErnel Transform (MINIROCKET) was developed by Dempster et al. (2021) [9], which is 75 times faster on larger datasets. The MINIROCKET preserves the key elements of the ROCKET, which are dilation and PPV, while using 84 fixed kernels, making it almost deterministic and much faster. It scales efficiently to large datasets and is therefore recommended over ROCKET. However, MINIROCKET is limited by using a single pooling operator and applying convolutions only to the raw series, which reduces feature diversity and could lead to lower classification accuracy [32].

In a further development of the MINIROCKET, the MultiROCKET by Tan et al. (2022) was developed with multiple pooling operators, and transformation were applied. The extension of MINIROCKET of applying 84 kernels to both the original time series and its first-order difference, using separate dilations and biases for each. MultiROCKET extends by adding three additional pooling operators to increase feature diversity and discriminative power, producing about 50,000 features per series. It achieves higher accuracy than MINIROCKET while maintaining fast training compared to ensemble methods, though MINIROCKET remains faster in computation. Overall, MultiROCKET improves MINIROCKET by introducing first-order differencing and additional pooling operators, but this en-

hancement substantially increases the computational expense compared to MINIROCKET [32].

Another time series classification method is HYbrid Dictionary–Rocket Architecture (HYDRA) by Dempster et al. (2023a), combining the approach of competing convolutional kernels from ROCKET with aspects of conventional dictionary methods for time series classification. Dictionary methods extract symbolic patterns from time series through transformations, quantization, and symbolization, classifying based on pattern frequency. HYDRA transforms the input time series into dictionary features by counting which convolutional kernel within each group achieves the highest response at each time point, capturing competitive pattern matching. It limits kernels per dilation to 512, keeping the total feature count below 10,000, and uses both hard and soft counting to capture pattern response. HYDRA achieve higher accuracy compared with other dictionary methods, while combined with respectively ROCKET and MINIROCKET the accuracy is comparable with MultiROCKET.

The ROCKET, HYDRA, and MR-HYDRA approaches was tested in the study by Aderinola et al. (2025) [1]. The purpose of the study was to develop a realistic, effective, and precise method for registering falls based on real fall data and modern time series classification techniques. The used fall data was acquired with Inertial Measurement Unit (IMU) sensors placed around the center of mass, preferred for fall detection. The simulated fall data were performed by 38 healthy participants. The data was given as input to ROCKET, HYDRA, and MR-HYDRA models, where MR-HYDRA performed best on the simulated fall data ($F1 \geq 96.7$).

As reported by Dempster et al. (2023a), the combination of MultiROCKET and HYDRA achieved the best accuracy compared to the other ROCKET variants combined with HYDRA. By concatenating the feature sets produced by HYDRA and MultiROCKET, an enhanced representation is created that leverages the strength of both methods. Afterwards, a linear classifier is trained on the combined feature space. Integrating MultiROCKET with HYDRA significantly improves accuracy compared to using either method individually. The combination performs almost as well as HC2 but is much faster in runtime. Overall, HC2 remains the most accurate method, but MR-HYDRA provides a more balanced trade-off between accuracy and computational efficiency. Dempster et al. (2023a) demonstrated that accuracy of MR-HYDRA are slightly higher than MINIROCKET-HYDRA on larger datasets while their performance is comparable on smaller datasets. However, MINIROCKET-HYDRA is computationally more efficient than MR-HYDRA, as MINIROCKET requires far fewer convolutional kernels and therefore has a substantially faster feature extraction and training process.

4.2 Interval Based Classifications Techniques

The Interval based classifier technique is described in the study by Dempster et al. (2023b) [11]. Interval-based methods commonly look at a set of intervals extracted from the input time series, and the transformation of the input time series, such as the first difference or discrete Fourier transform. Afterwards, descriptive statistics such as mean and variance for the values in each interval are computed. Lastly, the computed features are used to train the classifier, which typically is an ensemble of decision trees. Interval-based methods typically vary in different sets of transformations applied to the input time series, the interval characteristics, the use of intervals, the feature selection, and the choice of classifier head.

4.2.1 QUANT

Dempster et al. (2023b) introduce the QUANT, which is an interval-based method for time series classification. It follows the general structure of interval-based classification methods and builds on four main components: the set of input representations, the set of intervals, the features, and the classifier.

To capture different characteristics of the time series, multiple representations of the input data were used. In addition to the original time series, the first difference, the discrete Fourier transform, and the second difference were calculated. These representations provide information on both the time- and frequency-domain. The QUANT divides the time series into fixed, dyadic intervals $2^0, 2^1, \dots, 2^{d-1}$ with the depth denoted d , see Figure 4.2. For depth greater than one, an additional set of intervals is shifted by half the interval length. This was addressed to improve the coverage around the interval edges.

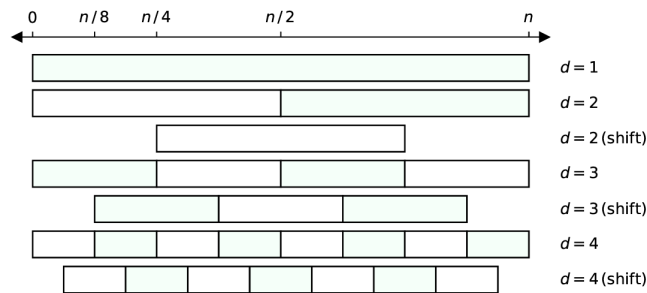


Figure 4.2: Set of intervals for a depth of $d = 4$, including shifted intervals for $d > 1$.

After dividing the time series into intervals, QUANT extracts representative features from each interval. Instead of extracting traditionally descriptive statistics, such as mean and variance, QUANT uses quantile based features derived from the sorted values within each interval. The sorted values are subsampled as quantiles to approximate the empirical distribution. These quantiles provide a compact

yet informative representation of the distribution within each interval, and reduce the dimensionality of the feature space. To improve robustness to level shift, quantiles are computed from the raw values and from mean-corrected values. This was done by subtracting the mean from every second quantile. The extracted quantile based features are used to train a classifier of *Extremely Randomized Trees* [19].

The review by Middelhurst et al. (2024) had compared the QUANT method with other state-of-the-art methods on the 142 UCR datasets and found that the QUANT was more accurate than existing interval methods, although actual differences in accuracy are small. However, Quant is clearly less accurate than the most accurate methods: MR-HYDRA, and HC2. The study further report that a large advantage for the QUANT is the significantly reduced training time of only 15 minutes compared to one to two hours for MR-HYDRA, and multiple weeks for HC2.

5. Problem Delimitation

No standard procedure for diagnosing dysgraphia is recommended, but the quantitative tests are often focused on the final product, where the dynamic aspects are ignored [7, 15, 4]. Writing skills are often developed in the early school grades, and diagnosing dysgraphia during this period is difficult. Therefore, it is often diagnosed later [7], but to reduce the academic and psychosocial consequences of the disorder, ongoing support is essential [7, 4]. The key characteristics of CD are described in Asselborn et al.(2018) as poor motor control, where Rosenblum et al.(2004) further report higher neuro motor noise. As highlighted by Asselborn et al.(2018) and Asselborn et al. (2020), tablets have opened new possibilities for accessing those temporal dynamics during the handwriting tasks. Drotar and Drobes (2020) utilized this approach to extract temporal and kinematics features for use in their ML model, which showed the ability to differentiate dysgraphia and non-dysgraphia in children. Using the same dataset, Kunhoth et al. (2023) slightly improved the performance by including In-Air features, indicating important information about dysgraphia that can not be captured in On-Surface features. The model developed by Asselborn et al. (2020) that also incorporated motor process achieved an improved performance, where they found the kinematic and pressure features most informative.

Previous studies showed that writing performance differs significantly between paper and tablets across age groups, where children produce more errors, less legible text, and struggle to adapt, while adults compensate more effectively [18, 2, 29]. Findings suggest tablets are unsuitable for assessing early writing development, and alternative measures of movement and speed are needed without the bias of tablet surfaces. Several writing tools have been developed to measure movement and speed [5, 6, 25]. Bublin et al. (2023) developed the SensoGrip Smart Pen and a deep learning LSTM model to detect dysgraphia, directly on the raw time series data acquired with the pen. They found that time series captures additional information than engineered features. Building on Bublin et al. (2023), less computationally expensive time series classification techniques are investigated in this study to detect the risk of dysgraphia. The model will perform binary classification to prevent unacceptable small groups in term of age or dysgraphia severity level. The purpose of the model was to detect the risk of dysgraphia using raw time series data collected with a sensor equipped ink pen that allows writing on normal paper, thereby eliminating bias introduced by tablet surfaces. This led to the following problem statement:

How can a time series classifier be developed to detect risk and no risk of dysgraphia using raw time series data collected with a sensor equipped ink pen?

Part II

Problem Solving

6. Solution Strategy

The methodological framework outlines the solution strategy applied in this study, covering the process from data understanding to model implementation and evaluation. The strategy is structured around three main components corresponding to the chapters in Part II, Problem Solving: describing and analyzing the handwriting datasets, developing domain adaptation to reduce device differences, and implementing and presenting the results of the chosen time series classifier.

Initially, the data description is presented in Chapter 7, including details about the study population and the structure of the collected datasets. The datasets include handwriting data collected from both children and adults. Due to the different force sensors used in the prototypes, possible variations in pressure sensitivity and signal scaling between the prototypes must be reduced before combining the data for time series classification. Therefore, the data preparation includes developing a domain adaptation algorithm to ensure comparable pressure representations across the two prototypes. Data from two similar adult groups are used to develop the domain adaptation algorithm, the iterative development process is described in Chapter 8 with its corresponding results. The application of the domain adaptation algorithm, together with the complete preprocessing procedure, is further detailed in Section 8.8. In Chapter 4, the time series classifiers are described. The implementation of these time series classifiers and the evaluation protocol used to assess the performance of the time series classifiers are presented in Chapter 9. In Figure 6.1, an overview of the data flow throughout the Problem Solving is illustrated.

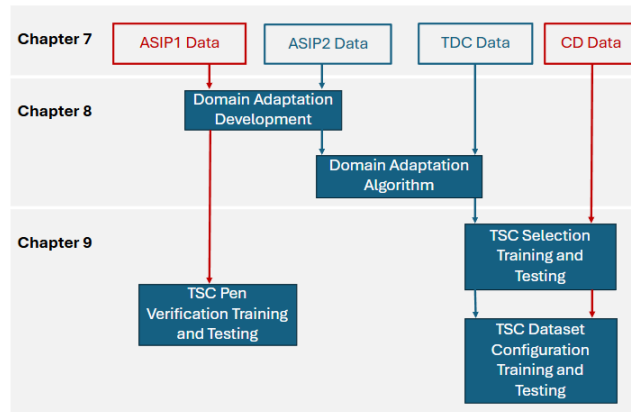


Figure 6.1: Flow diagram showing the data flow throughout the Problem Solving and where the corresponding methods are described in Chapter 7-9. The handwriting data were captured using a smart ink pen (SIP), available in two versions: SIP1 and SIP2. Blue-edged boxes indicate data acquired with SIP1, while red-edged boxes indicate data acquired with SIP2. The diagram includes datasets from adults using SIP1 (ASIP1) and adult using SIP2 (ASIP2), and children diagnosed with dysgraphia (CD), as well as data from typically developing children (TDC). The datasets are described in Section 7 which are used in the Time Series Classifier (TSC).

7. Data Description

In this chapter, the data used for the study are described. All the data was provided by Politecnico di Milano and recorded through two versions of the SIP: SIP1 and SIP2. The pens are further described in Section 7.1. Two handwriting datasets were used, one with data recorded on children for time series classification to detect the risk of dysgraphia, described in Section 7.2, and one with data recorded on adults, used for Domain Adaptation Development and Time Series Classifier Pen Verification, described in Section 7.3

7.1 The Data Acquisition Devices: SIP1 and SIP2

The SIP1 is a typical industrial ink pen that has been modified to record time series data of the dynamic motor processes involved in handwriting, the following presented details are presented in Lunardini et al.(2021). The SIP1 can be seen in Figure 7.1a, and its components in Figure 7.1b. Inside the protective 3D printed plastic case, the SIP1 has three PCBs. The PCB 1 is the core PCB and includes a 32-bit CPU to acquire, filter, and transmit the signal. Furthermore, it facilitates Bluetooth Low Energy(BLE) connectivity and a 1 MB flash memory to store the writing data. PCB 2 has a rechargeable Li-ion battery with a coaxial power connector accessible from the pen cap. PCB 3 is placed in the load cell holder and is responsible for the pre-amplification circuit for the load cell signal. The inertial signal is acquired through the Inertial Measurement Unit (IMU), consisting of an accelerometer measuring the acceleration, a gyroscope measuring the angular velocity, and a load cell measuring the writing force on the tip of the SIP1. The SIP1 is self-operated through movement or a BLE request. Therefore, no activation button for the end user is needed. To inform the user of its state, an LED is visible to indicate the operating mode and the battery state of the pen.

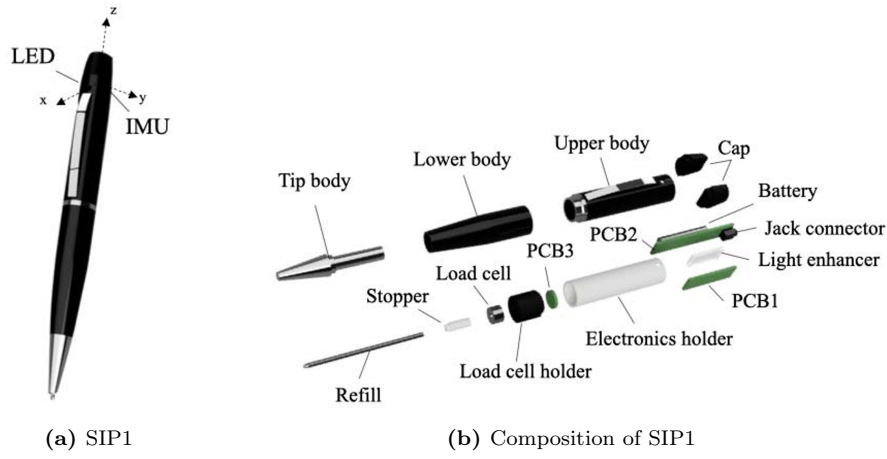


Figure 7.1: a) The SIP1, with the dimensions: height 147 mm, maximum diameter: 14,65 mm and weight: 48 g. and b) the components of the SIP1 and their composition in the pen [25].

The SIP1 had two types of data transfer. 1) *Online data transmission* is defined as the process that occurs when the SIP1 is paired to an external device, through BLE. The microcontroller reads the data from the IMU and the load cell and transmits it in real-time through the BLE. 2) *Onboard Storage* is defined as the process that occurs when the pen is used and no external device is paired, the sensor data are saved on the flash memory and downloaded afterward. Data storage stops if the pen remains stationary for 15 seconds.

SIP2 was a further development of SIP1, it aimed to optimize the cost and usability of SIP1. The main improvements of SIP2 were the force sensor HSFPAR003A (Piezoelectric), a twist-lock mechanism, and in the IMU a magnetometer was added. The force sensor can be seen in Figure 7.2b. The Piezoelectric is much cheaper and smaller than the Load Cell, which makes it easier to fit in the pen, and more sensitive than the Load Cell force sensor.

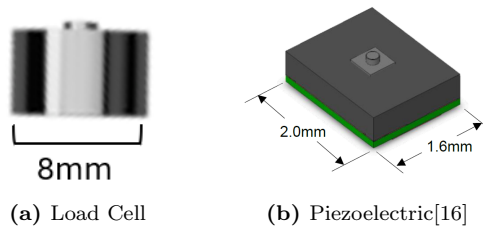


Figure 7.2: a) The Load Cell is the force sensor used in SIP1, and b) Piezoelectric force sensor used in SIP2 [16].

Besides the accelerometer and gyroscope, the SIP2 includes a magnetometer within its IMU, enabling orientation tracking relative to the Earth's magnetic fields. The magnetometer may be disturbed by external magnetic fields, which can occur when the pen is placed near other electronic devices or charging stations. Furthermore, the SIP2 introduces an extra layer of control through a twist-lock mechanism. This ensures that the pen does not start recording data immediately when it is picked up or moved. Instead, the user must twist the SIP2 to extend the tip before writing is possible. The SIP2

is activated only when both conditions are met: movements are detected, and the tip is extended. This additional step is meant to reduce the amount of recorded data under non-writing activities, such as handling, transportation, or walking with the pen.

7.2 Dysgraphia Prediction Dataset

The Dysgraphia Prediction Dataset was used for training and testing the time series classifiers to predict the risk and no risk of dysgraphia based on time series data. The dataset was divided into two subsets: handwriting tests performed by TDC, and handwriting tests performed by CD. The children in the TDC data subset ranged from 1st to 8th grade, where grade refers to school class level, corresponding to 6 to 14 years old, while the CD data subset included children from 8 to 14 years old. In Table 7.2a, the data distribution is shown. Each writing test consists of multiple subtests, described in subsection 7.2.1. The CD and TDC data subset had performed the Sentence writing test, including both the MS, VS, MC, and the VC subtests. In the TDC data subset, Grade 1 T1-T5 are the same group of children tested twice a year at five time points; Grade 1 T1 and Grade 1 T2 belong to 1st grade, Grade 1 T3 and Grade 1 T4 belong to 2nd grade, and Grade 1 T5 belongs to 3rd grade. Additionally, Grade 2 T1-T2 is also divided into two time points of testing, and both belong to the 2nd grade. Lastly, Grades 4-8 belong to their corresponding grade level. All grade levels in TDC performed the BVSCO tests on both capital and cursive letters, except for Grade 1 T1 and Grade 2 T1, which performed the tests with their preferred writing style, either capital or cursive letter. The 1st grade and 2nd grade children performed the Sentence tests in capital letters, where the 3rd to 8th grade children performed the Sentence tests in both capital and cursive letters. The data overview of the subgroups and their corresponding performed tests is illustrated in Figure 7.3.

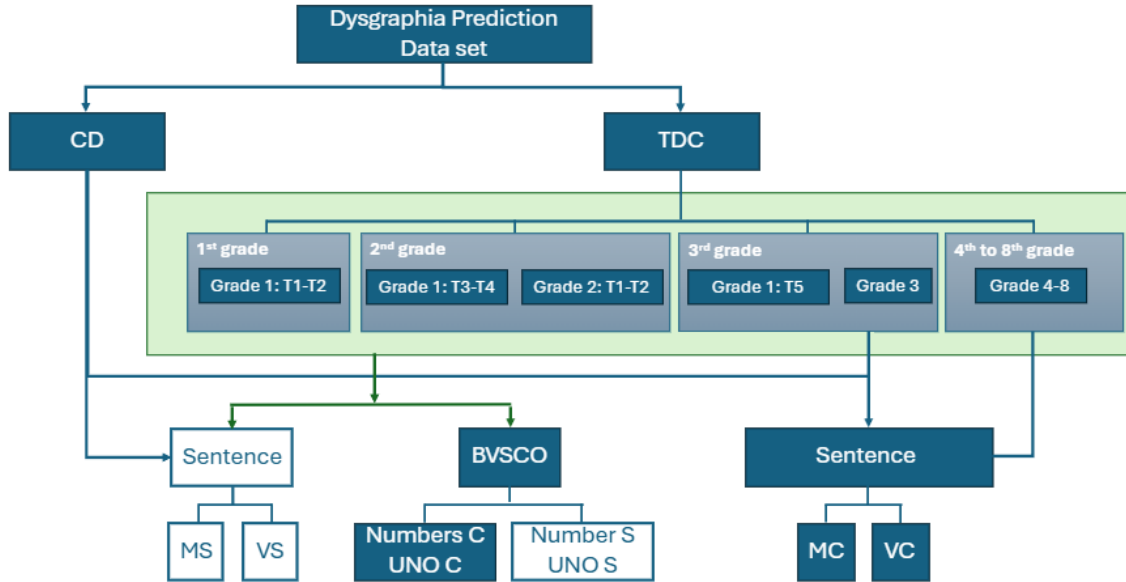


Figure 7.3: Data overview of the Dysgraphia Prediction Dataset including data from children diagnosed with dysgraphia (CD) and typically developing children (TDC) with corresponding grades and time points (T) in selected grades. The Sentence writing tests are divided into fast writing (V) and legible writing (M). All writing tests are performed in Capital Letters (S) and Cursive (C). The included writing tests are illustrated with blue boxes where the excluded are illustrated with white boxes.

During training of the time series classifier, the data were labeled as either **Risk** or **NoRisk**. All CD were labeled with **Risk**, and the TDC were labeled with either **Risk** or **NoRisk**, manually labeled based on their tests results. In the data from CD, only the age of the children were noted and not the grade whereas it was provided for TDC. To ensure comparability, CD participants were assigned fictive grade levels based on typical age ranges derived from the TDC dataset. The midpoint between each grade was found between the median ages of each grade and used as the decision boundary. Children in the CD without noted age were placed in grade 0. This conversion allows evaluation of predictions across grade levels. The conversion from age to grade for the CD dataset is shown in Table 7.1.

Age(years)	Grade	Amount
0 - 7.20	1	0
7.20 - 8.14	2	2
8.14 - 9.15	3	10
9.15 - 10.18	4	16
10.18 - 11.16	5	15
11.16 - 12.18	6	6
12.18 - 13.15	7	5
≥ 13.15	7	9
NaN	0	32

Table 7.1: Age til grade-level conversion table. The age boundaries are decided on the Typical Developed Children(TDC) age distribution on each grade-level. The amount column only shows the amount of Children with Dysgraphia (CD) included with the cursive writing style.

Since dysgraphia cannot yet be reliably diagnosed by evaluating capital letter handwriting, and only the TDC cursive writing test includes the labels, this study considers only the tests performed in

cursive writing style. In Figure 7.3, the excluded writing test are shown as white boxes. For cases in which a test lacked a label, even after excluding the capital writing style, it is investigated whether the subject had performed multiple tests. If the subject contributed multiple tests and all of them shared the same label, either **Risk** or **NoRisk**, the unlabeled test was assigned the same label. If the subject’s other tests showed inconsistent labels or no label at all, the unlabeled test was excluded. Likewise, if the subject contributed a single unlabeled test, the test was excluded. The number of data files included and the data distribution are presented in Table 7.2b.

7.2.1 Writing Tests

The writing tests performed by the children were divided into two main tests: Sentence and BVSCO. In each test, several subtests were executed.

- Sentence: In this test, the child must copy sentences with different focus depending on the subtest.
 - MS: Copy sentences as legibly as possible in capital letters.
 - VS: Copy sentences as fast as possible in capital letters.
 - MC: Copy sentences as legibly as possible in cursive letters.
 - VC: Copy sentences as fast as possible in cursive letters.
- BVSCO: In this test, the child must write numbers in letters or ‘uno’ in different letter styles as fast as possible for 1 minute.
 - Numbers S: Write the numbers one to nine in capital letters.
 - UNO S: Write ‘*UNO*’ repeatedly in capital letters.
 - Numbers C: Write the numbers one to nine in cursive letters.
 - UNO C: Write ‘*uno*’ repeatedly in cursive letters.

7.3 Domain Adaptation Dataset

A Domain Adaptation Dataset was introduced, to address the version related differences between the SIP prototypes, as these differences in pressure scale and sensitivity became critical because the CD wrote using the SIP1 whereas the TDC used the SIP2. Without preprocessing, time series classifiers may learn device-specific characteristics instead of dysgraphia-related patterns. Therefore the pen differences must be reduced.

The Domain Adaptation dataset consists of two groups of adults without writing disorders: one group wrote with SIP1, and the other with SIP2. These subsets are referred to as ASIP1 and ASIP2, respectively. This dataset contributed to the development of the the Domain Adaptation Algorithm. Both ASIP1 and ASIP2 included two writing exercises: List and Text. In the List and Text tasks,

participants used the pen to produce grocery-list-like free writing exercises. All exercises contained capital, cursive and mixed writing styles. Figure 7.4 shows an overview of the dataset with the writing tasks and writing style. In Table 7.2c the distribution of the dataset for development of the domain adaptation algorithm is shown. The data distribution of the dataset used to verify that the time series classifier does not learn pen differences is presented in Table 7.2d.

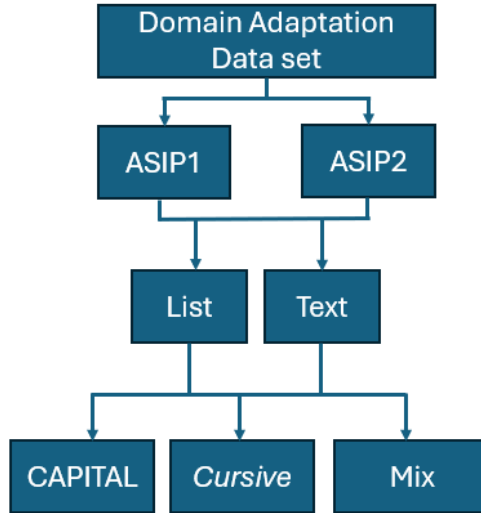


Figure 7.4: Data overview of the Domain Adaptation Dataset of adults performing with SIP1 and SIP2 including List and Text writing tasks along with their corresponding writing styles.

7.4 Ethical Approvals

The TDC subset originates from two ethically approved study protocols, conducted at the Politecnico di Milano, approved by the institutional Ethical Committee under the protocol numbers 28/2022 and 52/2024. The CD subset derives from the ethically approved study protocol number 07.23/02/2022 by *IRCCS Fondazione Don Carlo Gnocchi*.

The ASIP1 subset originates from two ethically approved study protocols, conducted at the Politecnico di Milano, approved by the institutional Ethical Committee under protocol numbers 04/2021 and 10/2018. The ASIP2 subset likewise derives from two ethically approved study protocols: one approved under protocol number 29/06/2021 by the *Comité de Ética de la Investigación de Badajoz*, and the other approved under protocol number 2574CE(September 2021) by *IRCCS Maugeri*.

7.5 Table Summary of Datasets

Subset	Files	Risk/NoRisk/unk.	Male/Female/unk.	Age	Writing Style
<i>a) Dysgraphia prediction full subsets</i>					
CD	145	145 / 0 / 0	113 / 32 / 0	10.67±1.75	Capital, Cursive
TDC	7608	805 / 3890 / 2913	3658 / 3851 / 99	9.32±4.28	Capital, Cursive
<i>b) Filtered Dysgraphia prediction subsets used for time series classification</i>					
CD	95	95 / 0 / 0	72 / 23 / 0	10.74±1.76	Cursive
TDC	2662	729 / 1933 / 0	1228 / 1434 / 0	9.61±1.87	Cursive
<i>c) Domain algorithm development subsets</i>					
ASIP1	322	- / - / -	138 / 176 / 8	53.49±21.85	Capital, Cursive, Mix
ASIP2	716	- / - / -	292 / 424 / 0	54.28±21.28	Capital, Cursive, Mix
<i>d) Filtered subsets used to verify that the time series classifier does not learn pen differences</i>					
ASIP1	166	- / - / -	58 / 103 / 5	57.28± 22.60	Cursive
ASIP2	467	- / - / -	161 / 306 / 0	58.47± 20.94	Cursive

Table 7.2: Overview of the distribution of the datasets used in Dysgraphia Prediction time series classification and Domain Adaptation Development, including the number of participants, participant demographics, and writing style.

a) Full children with dysgraphia(CD) and typical developing children(TDC) subsets with all available files.

b) Filtered subsets used for time series classification after applying all exclusion criteria and transferring unlabeled files to Risk or NoRisk label.

c) Domain adaptation development subsets, including adult data acquired with SIP1(ASIP1) and adult data acquired with SIP2(ASIP2).

d) Filtered Domain Adaptation subsets used to verify that the time series classification do not learn pen specific differences.

Note: Participants without a notated age were not included in the age statistics.

8. Domain Adaptation Development

The data acquired for this study were recorded using two prototypes of the SIP pen, SIP1 and SIP2. Due to the sensor differences described in Section 7.1, the recorded data showed varying characteristics in the pressure signals depending on the used prototype. This could lead to an unacceptable source of error when training a time series classification to detect risk and no risk of dysgraphia in children, as the CD wrote with SIP1 and the TDC wrote with SIP2. The following chapter focuses on the investigation of the signal differences across the two prototypes. In Section 8.1, it is described which tests and analyzes have been executed to verify if the SIP2 signal processing has become comparable with the SIP1 pressure signal. In Section 8.2, it was investigated whether the differences could be minimized with simple techniques such as linear scaling and rounding to the nearest integer. Afterwards, a new approach was tested in Section 8.4 to see if the quantile mapping could make the SIP2 pressure signal comparable to the SIP1 pressure signal. In Section 8.5, it was investigated whether excluding test groups could improve the comparability of the datasets. The accelerometer and the gyroscope signals are investigated in Section 8.6 to ensure that there are no practical differences between the two pens on these signals. Lastly, the summary of the domain adaptation development are outlined in Section 8.7 where the final applied domain adaptation steps are presented in Section 8.8.

To minimize the risk that the time series classifiers learns to distinguish between the pen prototypes rather than detect risk of dysgraphia related patterns, the differences in the pressure signals between the two pens must be minimized. To achieve this, data were collected from adults, including two groups, one group that wrote using SIP1 and another group that wrote using SIP2 while performing the same types of tasks.

8.1 Domain Discriminability Test

The domain discriminability test was conducted to measure and compare the pressure data from both pens with the aim of minimizing potential device-specific differences.

A Support Vector Machine (SVM) with a linear kernel was used to classify the data according to the pen prototype. The dataset was stratified by class and randomly split into a training set (80%) and a test set (20%). To assess the robustness and generalization of the SVM model, a k-fold cross-validation (k=5) was performed on the training data, using four folds for training and one for validation in each iteration. For each fold, the model's performance metrics were computed. The mean and standard

deviation across the folds were used to summarize the model performance stability. After cross-validation, a final SVM model was trained on the full training dataset and then tested on the unseen test set, reporting the same set of performance metrics. The *Accuracy*, *Precision*, *Recall*, and *F1-score* were calculated as defined in Appendix 12, where a performance close to 50% would show that the classifier failed to distinguish the pressure data from the two pen prototypes. Therefore, a threshold of 60% was set as the limit for when the domain adaptation was accepted.

Since the pressure data were not normally distributed, Cliff’s δ , a non-parametric effect size, were used to compare two independent groups, the Cliff’s δ value was introduced by Miessel and Yao (2024)[26]. It represents the probability that a randomly selected observation from one group is larger than one from the other, minus the reverse probability, ranging from -1 to $+1$. Values near 0 indicate overlapping distributions, while values closer to ± 1 indicate stronger dominance. Approximate interpretation:

- $|\delta| < 0.15 =$ negligible
- $0.15 \leq |\delta| < 0.33 =$ small
- $0.33 \leq |\delta| < 0.47 =$ medium
- $0.47 \leq |\delta| =$ large effect

Besides the domain discriminability test with the SVM model, each feature was analyzed using box plot, with the corresponding derived δ value used to quantify the difference between pen prototypes, see Appendix 13 for the box plots.

The features evaluated in the domain discriminability test were mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF), and amplitude range, where computed per stroke and then averaged across all strokes. The CV was computed as the ratio between STD and mean, capturing relative pressure variability, while the NCF was calculated as the number of local changes in force within each stroke, reflecting how often the applied pressure fluctuates during writing.

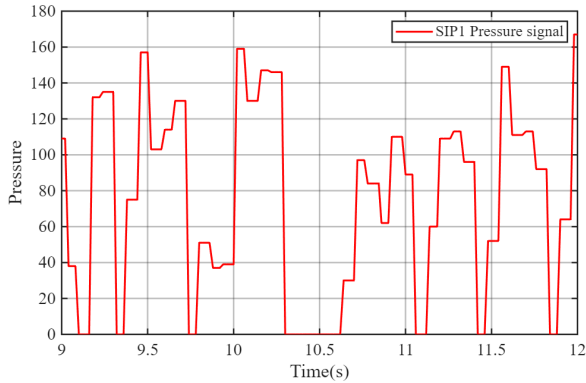
Additionally, feature importance was estimated based on the normalized absolute magnitude of the SVM feature weights, the calculation of the feature importance is described in Appendix 12. This allowed interpretation of which features contributed most to the classification process.

An over-representation of SIP2 data files led to a critical imbalance in the class distribution between the two pen prototypes. To address this, random under-sampling of the SIP2 data files was performed to match the number of SIP1 files. This ensured a balanced 50/50 class distribution in both the train and test sets, thereby preventing bias in the classifier’s learning process.

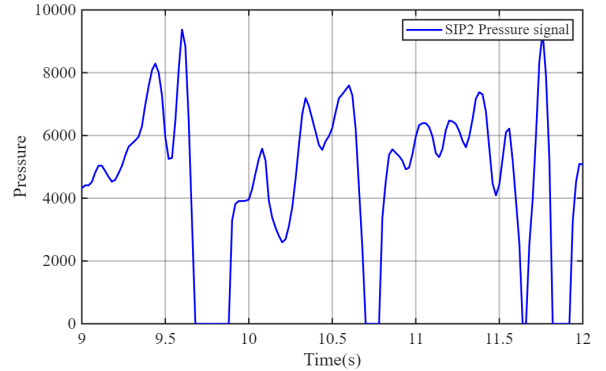
The following sections outline the methods used to reduce potential device-specific differences and achieve comparable pressure data across the pens.

8.2 Comparison of SIP1 and SIP2 Pressure Signals

The pressure signal from the two SIP prototypes had a large difference in pressure amplitude range, which is visualized in Figure 8.1. The pressure amplitude for the SIP1 signal ranges approximately from 0 to 180, whereas the pressure amplitude for the SIP2 signal ranges from 0 to 10,000. The difference in amplitude ranges may cause the time series classifier to classify based on pen-specific properties rather than dysgraphia-related patterns. Therefore, it was investigated whether amplitude scaling could make the SIP1 and SIP2 comparable.



(a) 3 seconds of the SIP1 pressure signal with an amplitude range of 0-180.



(b) 3 seconds of the SIP2 pressure signal with an amplitude range of 0-10,000.

Figure 8.1: Visualization of the pressure amplitude range of a) SIP1 and b) SIP2 pressure signal.

In Table 8.1, the δ values summarize the differences between SIP1 and SIP2 across pressure based features. The results show consistently large negative δ values for mean, median, STD, and amplitude range, indicating substantial discrepancies between the two prototypes before domain adaptation for these features. The δ values are defined as the difference between SIP1 and SIP2, calculated as $\text{SIP1} - \text{SIP2}$. Consequently, negative δ values indicate higher feature values measured by SIP2 compared to SIP1 which is consistent with the amplitude of the pressure signal is substantially higher for SIP2 compared to SIP1 as seen in Figure 8.1.

Feature	δ values
Mean	-1.00
Median	-1.00
STD	-1.00
CV	0.24
NCF	-0.83
Amplitude Range	-1.00

Table 8.1: The Cliff δ values derived from pressure signal features when comparing SIP1 and SIP2, including the features mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF), and amplitude range.

The impact of the pressure amplitude is further supported in Figure 8.2, which presents a bar chart of feature importance in distinguishing between SIP1 and SIP2. The features with the highest impact are mean, STD, median, and amplitude range, highlighting the influence of amplitude related pressure values in differentiating the devices.

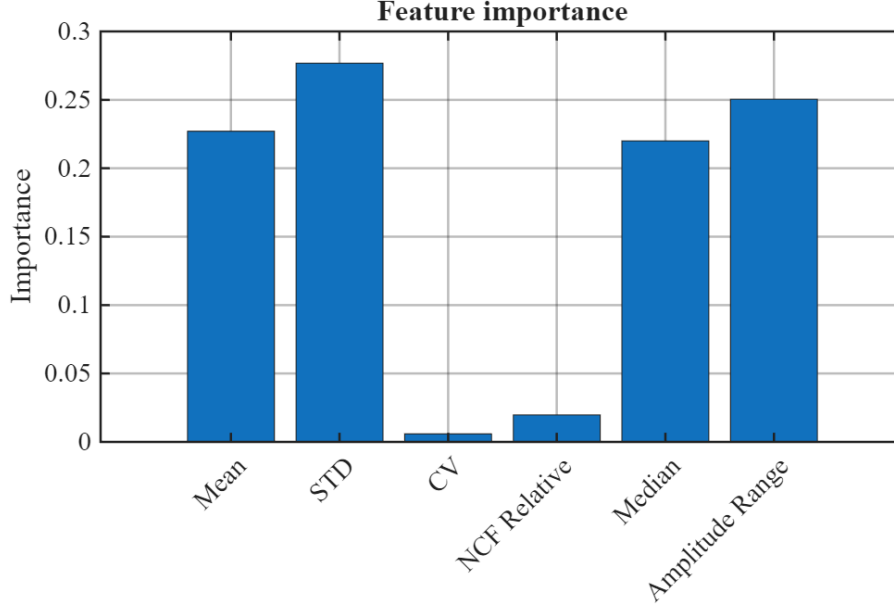


Figure 8.2: Bar chart showing the importance of the features for the SVM classifier of Mean, standard deviation (STD), coefficient of variation of force in the stroke (CV), number of changes in force (NCF) Relative, Median, and Amplitude Range features in distinguishing between SIP1 and SIP2.

8.3 Amplitude Scaling and Rounding

To investigate whether the amplitude differences were linearly comparable, the analysis evaluated whether a simple linear transformation could harmonize the signals or if more advanced preprocessing would be required for the classifier. All of the pressure data were concatenated into long vectors, one for SIP1 and one for SIP2. Extreme outliers above the percentile of 9.99 and zeros were removed. Afterward, they were split into quantiles in 1000 steps between 0 and 1. On the quantiles for both pens, a linear model fit with zero intercept was made, the model can be seen in Equation (8.1). The linear fit performed a good regression fit ($R^2 = 0.996$) with a low error ($RMSE = 3.235$).

$$f(x) = 0.026 \cdot x, R^2 = 0.996, RMSE = 3.235 \quad (8.1)$$

A Bland-Altman plot was used to evaluate the linear scaling in Equation 8.1. The Bland-Altman plot residuals showed a small systematic deviation as shown in Figure 8.3, but with relatively small error bars and therefore not considered to affect the comparability between the prototypes.

Bland-Altman: SIP2 Scaled, zero-intercept regression

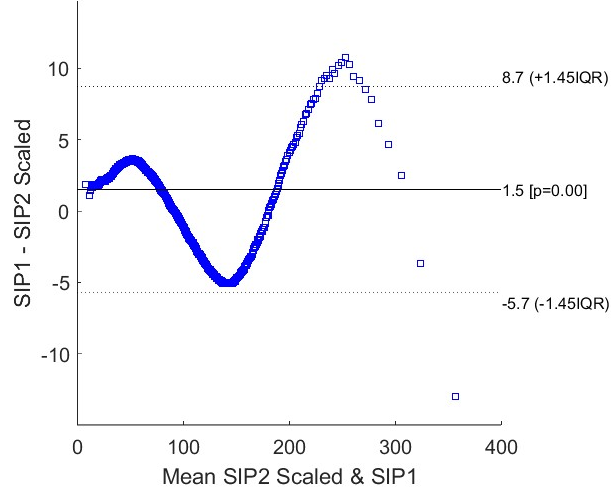


Figure 8.3: Bland-Altman plot for the quantile distribution for SIP1 and SIP2 Scaled.

8.3.1 Integer Rounding

By investigating the sensitivity resolution according to the signal amplitude for SIP1 and SIP2 in Figure 8.4a, the SIP1 signal shows clearly integer steps in sensitivity, where the SIP2 signal contains a finer sensitivity resolution. In the left side of both plots in Figure 8.4, there is a steep line of close dots seen for the small amplitudes, which appears when the force jumps from zero, no contact with the paper, to a higher value indicating paper contact. The line appears because the SIP1 and SIP2 pressure signals are shifted with a fixed decimal number throughout the whole signal, probably applied to the signal when removing the baseline drift. To meet this, the scaled SIP2 pressure signals are rounded to the nearest integer by using the MATLAB `round` function, the result can be seen in Figure 8.4b.

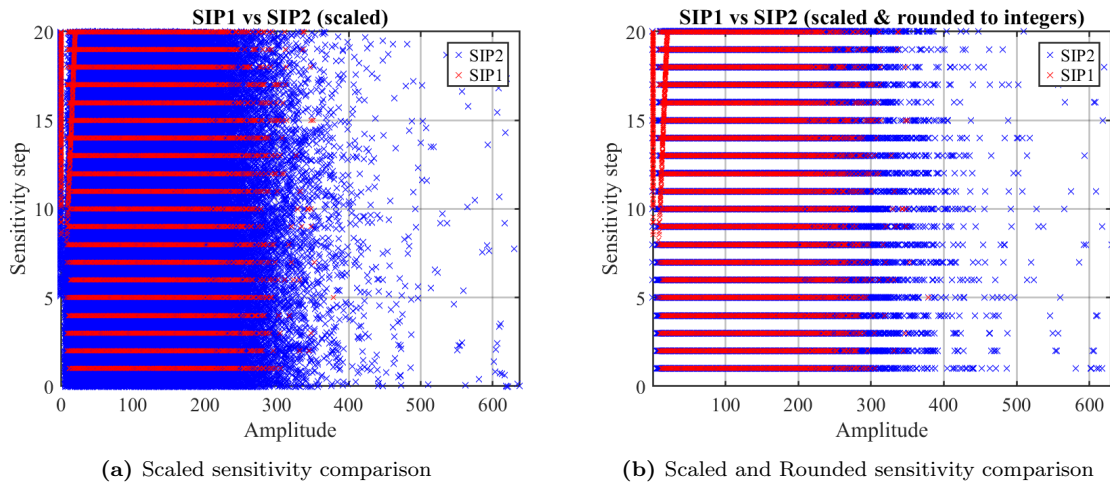


Figure 8.4: Scatter plot that shows the sensitivity steps in amplitude for all concatenated SIP1 signals and all concatenated SIP2. a) Shows the SIP2 Scaled signal with the linear model compared to SIP1. b) Shows the SIP2 Scaled signal with the linear model and rounded to the nearest integer, compared to SIP1 signal.

Furthermore, a histogram of the amplitude distribution of pressure between SIP1 and SIP2 Scaled and Rounded was made to investigate systematic amplitude outage or peaks. The histogram in Figure 8.5 shows the amplitude distribution followed a normal distribution. In general, there is a larger amount of the SIP2 pressure signal amplitudes than the SIP1, which can be seen in the generally higher bin boxes for the SIP2 signal.

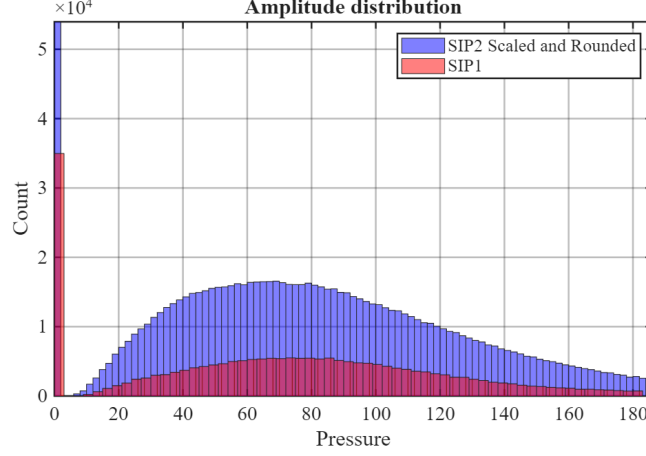
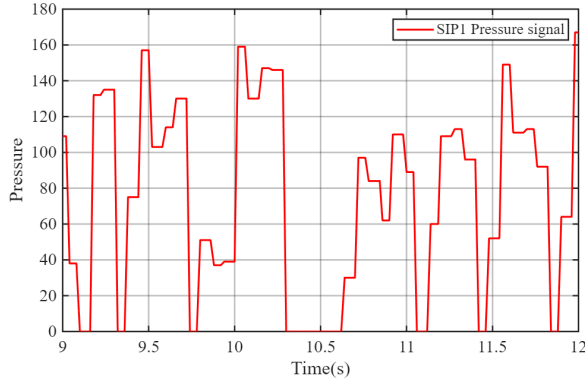
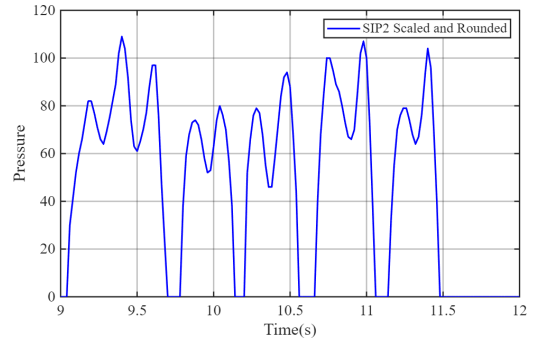


Figure 8.5: Histogram of the amplitude distribution of pressure for all datasets in SIP1 and SIP2 Scaled and Rounded with maximum amplitude set to the 95th percentile.

3 seconds of the SIP1 pressure signal is shown in Figure 8.6a, similar 3 seconds of the SIP2 Scaled and Rounded pressure signal is shown in Figure 8.6b. By comparing the two outtakes of the pressure signals, it is clear that the SIP1 signal still contains different characteristics than the SIP2 signal, such as longer plateau phases and more edge signal changes.



(a) 3 seconds of the SIP1 Pressure signal.



(b) 3 seconds of the SIP2 Scaled and Rounded pressure signal with an amplitude range similar to SIP1.

Figure 8.6: Pressure signal plots of 3 seconds acquired with the SIP1 and SIP2. a) Shows the 3 seconds of the SIP1 pressure signal, and b) shows 3 seconds of the SIP2 Scaled and Rounded pressure signal.

8.3.2 Discriminability Test of SIP1 and SIP2 Scaled and Rounded

To evaluate the effect of the applied procedure on the SIP2 pressure signals, δ values are derived to compare SIP1 features with the features of SIP2 Scaled and Rounded. Furthermore, a domain discriminability test was applied using a SVM classifier.

In Table 8.2, the δ values summarize the differences between SIP1 and SIP2 Scaled and Rounded across pressure based features. The δ values range from 0.02 to 0.05 for mean, median, STD, and amplitude range which corresponds to a negligible effect size. In contrast, CV shows $\delta=0.24$ indicating small effect while the largest observed effect is found in NCF with δ value of 0.75 indicating large effect which reflects its sensitivity to dynamic characteristics. The high negative value indicates that the SIP2 is more sensitivity compared to SIP1. NCF appears to be more sensitive to signal variations that are not affected by the amplitude scaling, which may explain the difference in NCF between SIP1 and SIP2 Scaled and Rounded.

Feature	δ values
Mean	-0.03
Median	-0.04
STD	0.05
CV	0.24
NCF	-0.75
Amplitude Range	-0.02

Table 8.2: The Cliff δ values derived from pressure signal features when comparing SIP1 and SIP2 Scaled and Rounded, including the features mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF), and amplitude range.

After scaling SIP2, the amplitude ranges become more similar, but the sensitivity resolution between the pressure signals remains different as seen in Figure 8.6 where the SIP2 still appears more smooth than the SIP1 signal. This is reflected in the SVM classifier performance where training results in Table 8.3 show a clear difference in classifier performance before and after scaling and rounding. Before scaling and rounding of SIP2, the SVM classifier achieved perfect performance metrics (100%), indicating that the SVM classifier could easily distinguish between SIP1 and SIP2 pressure data. The performance metrics decrease slightly after scaling and rounding with accuracy of 81.8 (± 1.2), precision of 86.8 (± 5.1), recall of 75.5 (± 6.3), and F1-score of 80.5 (± 2.2) on the training in Table 8.3, indicating that SIP1 and SIP2 become more distinct once amplitude related differences are minimized by using scaling and rounding on the SIP2 signal. This suggests that scaling and rounding reduces the amplitude-based bias between SIP1 and SIP2, allowing the classifier to focus more on sensor-specific responses such as NCF sensitivity.

The test results in Table 8.3 show further decrease in performance with an accuracy of 82.0%, precision of 84.7%, recall of 78.1%, and F1-score of 81.3%, indicating that SIP1 and SIP2 become more comparable, but highlight that there remains a difference in the sensitivity resolution.

SVM Classifier test	Accuracy	Precision	Recall	F1-score
<i>Training</i>				
SIP2 (n=516)	100(± 0.0)	100(± 0.0)	100(± 0.0)	100(± 0.0)
SIP2 Scaled and Rounded (n=516)	81.8(± 1.2)	86.8(± 5.1)	75.5(± 6.3)	80.5(± 2.2)
<i>Test</i>				
SIP2 (n=128)	100.0	100.0	100.0	100.0
SIP2 Scaled and Rounded (n=128)	82.0	84.7	78.1	81.3

Table 8.3: SVM Classifier test to distinguish between SIP1 and SIP2 before and after scaling. N reference to the number of data files used for the test.

By comparing the feature importance before Scaling and Rounding in Figure 8.7a with after applying Scaling and Rounding on the SIP2 pressure signal in Figure 8.7b, the feature importance showing that before applying Scaling and Rounding the Mean, STD and Median and Amplitude range was the most important features, all highly depending on the amplitude of the signal. After applying Scaling and Rounding, NCF becomes the most important feature, whereas amplitude-related features have their contribution reduced.

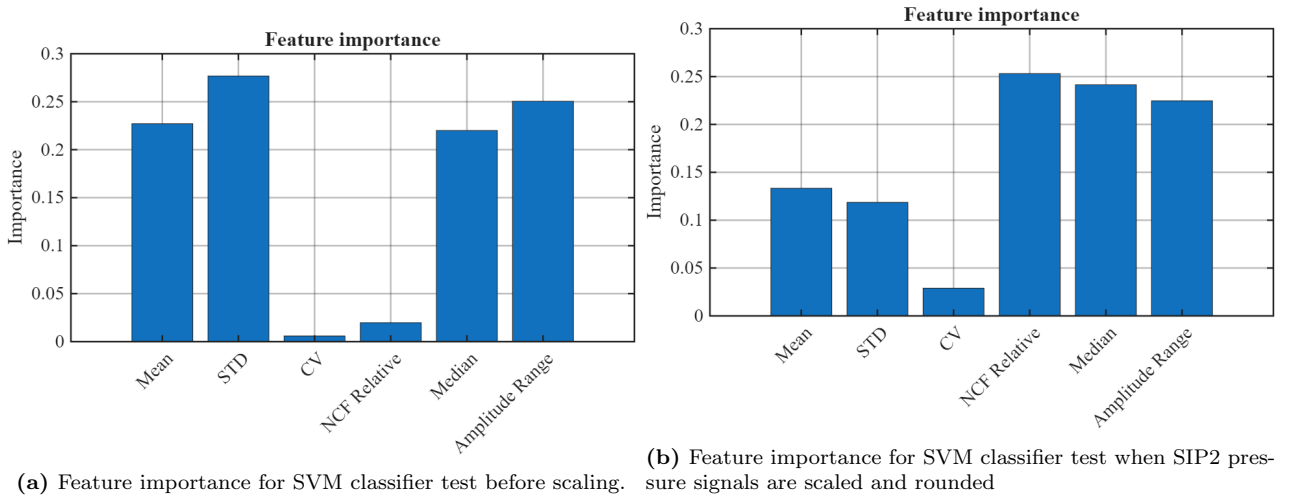


Figure 8.7: Bar chart showing the feature importance for the SVM classifier of Mean, standard deviation (STD), coefficient of force in the stroke(CV), number of changes in force(NCF) Relative, Median, and Amplitude Range features in distinguishing between SIP1 and SIP2 before and after Scaling and Rounding of the SIP2 pressure.

8.4 Stepwise Quantile Mapping

The amplitude scaling method was insufficient to reduce the sensitivity differences between SIP1 and SIP2, since it adjusts amplitude differences but does not affect sensitivity-related variations. The statistical bias method *quantile mapping* was applied to adjust SIP2 to be comparable to SIP1. To obtain the discrete steps as in SIP1, the *stepwise quantile mapping* method was utilized.

8.4.1 Stepwise Quantile Mapping Method

The stepwise quantile mapping method adjusts the amplitude and distribution properties of the SIP2 pressure signal to the SIP1 pressure signal while preserving the original order and timing of the values.

The purpose of quantile mapping is to compare the quantiles of the input signal, SIP2, with those of a reference signal, SIP1, and subsequently adjust the input values of SIP2. The data is divided into *numsteps* quantile bands where *numsteps* specifies the number of bands to split the data into. The quantile boundaries are defined by generating equally spaced values between 0 and 1, ensuring that the data are split into *numsteps* equally populated quantile bands. For each quantile band, all data points belonging to that range are identified and each quantile band is then assigned a constant value that reflects its relative position within the overall distribution. Gain values are estimated from the sensitivity ratio between SIP1 and SIP2, which are used to progressively adjust the SIP2 pressure distribution to match the characteristics of SIP1. In this way, every quantile is replaced by a representative plateau value that is related to its original location in the data distribution. By using quantile mapping, the influence of outliers is reduced.

Different *numstep* sizes were investigated during Bayesian Optimization to identify the value that resulted in the lowest classification performance of accuracy, as a lower performance indicates that the two pen prototypes are more comparable and therefore more difficult to distinguish. In the Bayesian Optimization, the optimizer evaluates 30 iterations in the range from zero to the maximal number of unique amplitude values of SIP1, with a k-fold of 5 in training to achieve the optimal *numstep* for the test of the quantile mapped SIP2 signal. Bayesian Optimization investigates new points, both points near low objective values and points in areas with high model uncertain. After performing the Bayesian Optimization in Figure 8.8, the optimal *numstep* was 22, where the lowest accuracy was achieved. The SIP2 signal is divided into 22 equally populated quantile bands.

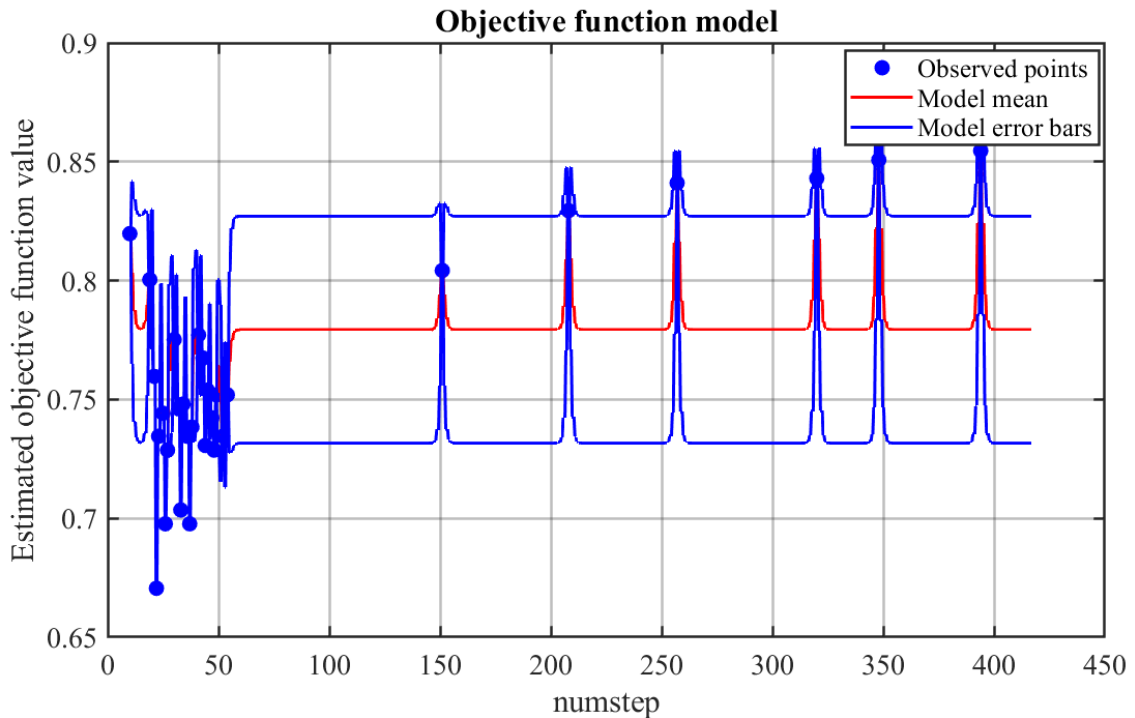
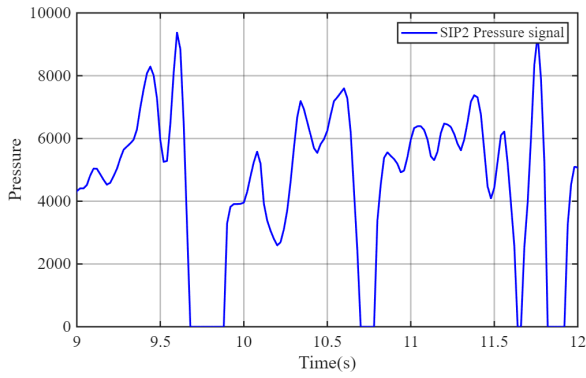
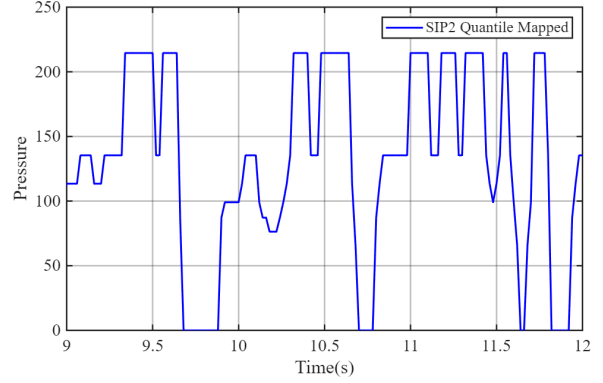


Figure 8.8: The result of Bayesian Optimization guesses, where *numstep*=22 was found as the best number of quantile bands, according to not being able to distinguish the ASIP1 and the ASIP2.

In Figure 8.9, the pressure signal of SIP2 before and after quantile mapping with *numstep* of 22 is shown. Quantile mapping keeps the data's relative position but assigns each value to the corresponding quantile in the target pressure distribution, stretching upper quantiles and compressing lower ones. As a result, some peaks from the original signal become relatively higher in the transformed signal, while intermediate values are compressed which can be seen in Figure 8.9b where the nine largest peaks become more emphasized than in the original signal in Figure 8.9a.



(a) 3 seconds of the SIP2 pressure signal before quantile mapping.



(b) 3 seconds of the SIP2 pressure signal after quantile mapping with *numstep* of 22.

Figure 8.9: SIP2 pressure signal before and after quantile mapping, where a) shows 3 seconds of the SIP2 pressure with no pre-processing, and b) shows the same 3 seconds of the SIP2 pressure signal after applying quantile mapping with *numstep* of 22.

Furthermore, a histogram of the amplitude distribution of the pressure between SIP1 and SIP2 Quantile Mapped was made to investigate systematic amplitude outage or peaks. The histogram in Figure 8.10 shows the amplitude distribution follow a normal distribution. In the histogram of the SIP2 Quantile Mapped data, the pressure values clearly fall into the rough amplitude collections. This shows that the quantile mapping compresses the pressure data while preserving the relative distribution between bands.

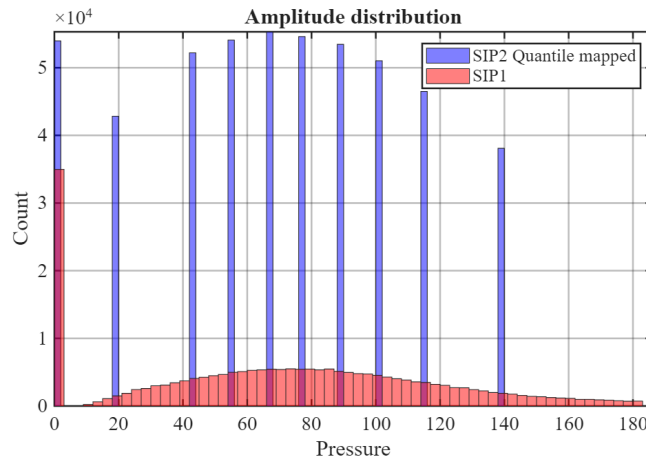


Figure 8.10: Histogram of the amplitude distribution of pressure for all datasets in SIP1 and SIP2 Quantile Mapped with maximum amplitude set to the 95th percentile.

8.4.2 Discriminability Test for SIP1 and SIP2 Quantile Mapped

To investigate the effect of the stepwise quantile mapping applied to the SIP2 signal, δ values are derived to compare SIP1 features with the features of SIP2 Quantile Mapped. Furthermore, a domain discriminability test was applied using a SVM classifier.

In Table 8.4, the δ values demonstrates that the quantile mapping effectively aligns amplitude-related characteristics of SIP2 with SIP1, as seen for the mean, median, STD, and amplitude range features, where the δ values are between -0.12 and 0.06 indicating negligible difference in the effect size. For the feature CV, the δ value of 0.46 indicates the largest effect size. This is substantially higher than for SIP2 Scaled and Rounded where CV showed a smaller effect size with a δ value of 0.24. Additionally, the effect size of NCF is reduced to a δ value of -0.21, representing small effect compared to the SIP2 Scaled and Rounded, where the NCF exhibited a large effect size with a δ of -0.75. This reducing in NCF indicates that the quantile mapping effectively reduces the sensitivity differences between SIP1 and SIP2.

Feature	δ values
Mean	-0.12
Median	-0.10
STD	0.06
CV	0.46
NCF	-0.21
Amplitude Range	-0.07

Table 8.4: The Cliff δ values derived from pressure signal features when comparing SIP1 and SIP2 Quantile Mapped, including the features mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF), and amplitude range.

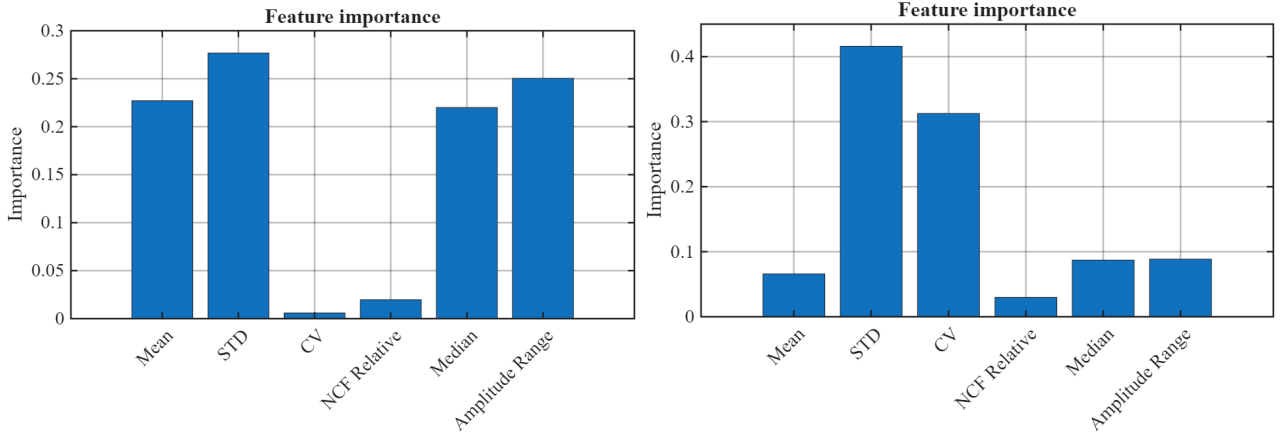
To evaluate the effectiveness of the stepwise quantile mapping, an SVM classifier was trained and tested on the ASIP1 and ASIP2 in Table 7.2c using $numsteps = 22$. The training results in Table 8.5 show that the SVM classifier achieved an accuracy of 80.8 % (± 5.7), precision of 77.5 % (± 6.0), recall of 87.2 % (± 5.2), and F1-score of 82.0 % (± 5.2), indicating an improvement compared to the amplitude scaling and rounding approach as seen in Table 8.5.

The SVM classifier test results of all performance metrics in Table 8.5 show a lower performance compared to the SVM classifier test of SIP2 Scaled and Rounded, with an accuracy of 57.8 %, precision of 60.9 %, recall of 43.8 %, and F1-score of 50.9 %. This performance is close to a random classification, indicating that the SVM classifier was no longer able to reliably distinguish between SIP1 and SIP2 after the stepwise quantile mapping in the test results.

SVM Classifier test	Accuracy	Precision	Recall	F1-score
<i>Training</i>				
SIP2 Scaled and Rounded (n=516)	81.8(± 1.2)	86.8(± 5.1)	75.5(± 6.3)	80.5(± 2.2)
<i>numsteps</i> = 22 (n=516)	80.8(± 5.7)	77.5(± 6.0)	87.2(± 5.2)	82.0(± 5.2)
<i>Test</i>				
SIP2 Scaled and Rounded (n=128)	82.0	84.7	78.1	81.3
<i>numsteps</i> = 22 (n=128)	57.8	60.9	43.8	50.9

Table 8.5: Result of the SVM Classifier test by using *numsteps* = 22 in quantile mapping. N reference to the number of data files used for the test.

The bar chart in Figure 8.11b shows the importance of the different features after applying quantile mapping to SIP2. Before quantile mapping of SIP2 in figure 8.11a, amplitude related features were the most dominant features which matches the overall different amplitude range between the two prototypes. After quantile mapping in Figure 8.11b, amplitude related features importance are reduced.



(a) Feature importance for SVM classifier test before quantile mapping of SIP2. (b) Feature importance for SVM classifier test after quantile mapping of SIP2.

Figure 8.11: Bar chart showing the feature importance for the SVM classifier of Mean, standard deviation (STD), coefficient of force in the stroke(CV), number of changes in force(NCF) Relative, Median, and Amplitude Range features in distinguishing between SIP1 and SIP2 before and after applying quantile mapping of the SIP2 pressure signal.

8.5 More Similar Pen Tests and Test Groups

To investigate whether the observed difference in effect size of δ values is influenced by specific factors due to the test groups and not the pens, the impact of excluding the writing style of capital letters and subjects over 70 years was examined. Capital letters were excluded from the analysis because they typically involve more pen lifts due to the presence of multiple strokes, whereas cursive letters are more continuous and produce smoother, uninterrupted movements, which is expected to increase the NCF. Subjects over 70 years were excluded, as age-related declines in handwriting motor function are known to accelerate after this age [25]. The effect of excluding both test groups simultaneously was further investigated.

The feature NCF in Table 8.2 appears to show the largest difference between SIP1 and SIP2 Scaled and Rounded with a δ value of -0.75 among all examined features where the CV feature has a δ value of 0.24 indicating small effect size. After quantile mapping of SIP2, the largest effect size appears in the feature CV in Table 8.4 with a δ value of -0.46 among all examined features where the δ value of NCF is -0.21. The impact of the features CV and NCF was investigated across the different test groups in both the SIP2 Scaled and Rounded and SIP2 Quantile Mapped while these feature indicates minimum small effect size.

8.5.1 Data Exclusion Experiments of the SIP2 Scaled and Rounded Dataset

In this section, the effects of excluding capital letters, participants over 70 years, and the simultaneous exclusion are examined for reducing the effect size of δ in the CV and NCF features in SIP2 Scaled and Rounded dataset. In Table 8.6, the δ values demonstrates the effect of the exclusion process. Before any exclusion, δ was -0.75 , and although excluding capital letters reduced the δ value to -0.69 , the effect size remained large. Excluding subjects over 70 years did not substantially alter the result, and the combined exclusion of capital letters and subjects over 70 years resulted in an even larger difference with a δ of -0.84 , confirming NCF as a dominant feature for distinguishing the two pen prototypes. In contrast, the CV feature exhibited smaller δ values overall. Initially, δ was 0.24, with the most pronounced reduction observed when capital letters were excluded with a δ value of 0.11. Excluding subjects over 70 years increased the δ value, while applying both exclusions resulted in a moderate reduction to a δ value of 0.15. Consistent with the NCF results, excluding capital letters had the strongest impact on reducing the difference between the pens.

Feature	δ values			
	Incl. All	Excl. CL	Excl. >70 years	Excl. CL + >70 years
CV	0.24	0.11	0.26	0.15
NCF	-0.75	-0.69	-0.88	-0.84

Table 8.6: The Cliff δ values derived from pressure signal features when comparing SIP1 and SIP2 Scaled and Rounded before (Incl. All) and after excluding (Excl.) Capital Letters (CL), participants over 70 years(>70 years), and the simultaneous exclusion, including the features coefficient of variation (CV) and number of changes in force (NCF).

Table 8.7 illustrates how the different exclusion groups influence the SVM classifier’s performance in distinguishing between SIP1 and SIP2 Scaled and Rounded data. Exclusion of capital letters slightly reduce the performance but exclusion of over 70 years or both slightly improves performance. Overall, the difference in results before and after exclusion are small. The decrease in performance after exclusion capital letters indicates that the writing style influence the performance to distinguish between SIP1 and SIP2 Scaled and Rounded.

Exclusion group	Accuracy	Precision	Recall	F1-score
<i>Training</i>				
SIP2 scaled (n=516)	81.8(± 1.2)	86.8(± 5.1)	75.5(± 6.3)	80.5(± 2.2)
CL excluded (n=302)	82.5(± 6.9)	85.8(± 4.3)	77.6(± 10.9)	81.3(± 8.0)
Over70 excluded (n=378)	88.4(± 3.1)	89.5(± 4.4)	87.3(± 6.1)	88.2(± 3.1)
CL + Over70 excluded (n=198)	84.9(± 5.3)	84.9(± 6.7)	86.0(± 13.4)	84.7(± 6.4)
<i>Test</i>				
SIP2 scaled (n=128)	82.0	84.7	78.1	81.3
CL excluded (n=76)	75.0	75.7	73.7	74.7
Over70 excluded (n=94)	88.3	90.9	85.1	87.9
CL + Over70 excluded (n=50)	84.0	84.0	84.0	84.0

Table 8.7: SVM Classifier test to distinguish between SIP1 and SIP2 Scaled and Rounded with and without test with Capital Letters (CL) and subjects over 70 years. N reference to the number of data files used for the test.

The bar chart in Figure 8.12 illustrates the importance of the different features after exclusion of capital letters, subjects over 70 years and simultaneous. NCF remains as the most important features in exclusion of subjects over 70 years and both test groups, followed by median, amplitude range, mean and STD as for the SIP2 Scaled and Rounded without exclusion while CV contributes with minimal impact. For the capital letters, amplitude range appears as the most important feature, followed by NCF, STD and median while CV and mean contributes with minimal impact in this case. The bar chart of feature importance in the exclusion of capital letters indicates that the exclusion minimize the impact of NCF in the SVM classifier test. These results of the exclusion process indicates that the exclusion of capital letter slightly improve the comparability of SIP1 and SIP2 Scaled and Rounded based on the NCF and CV features.

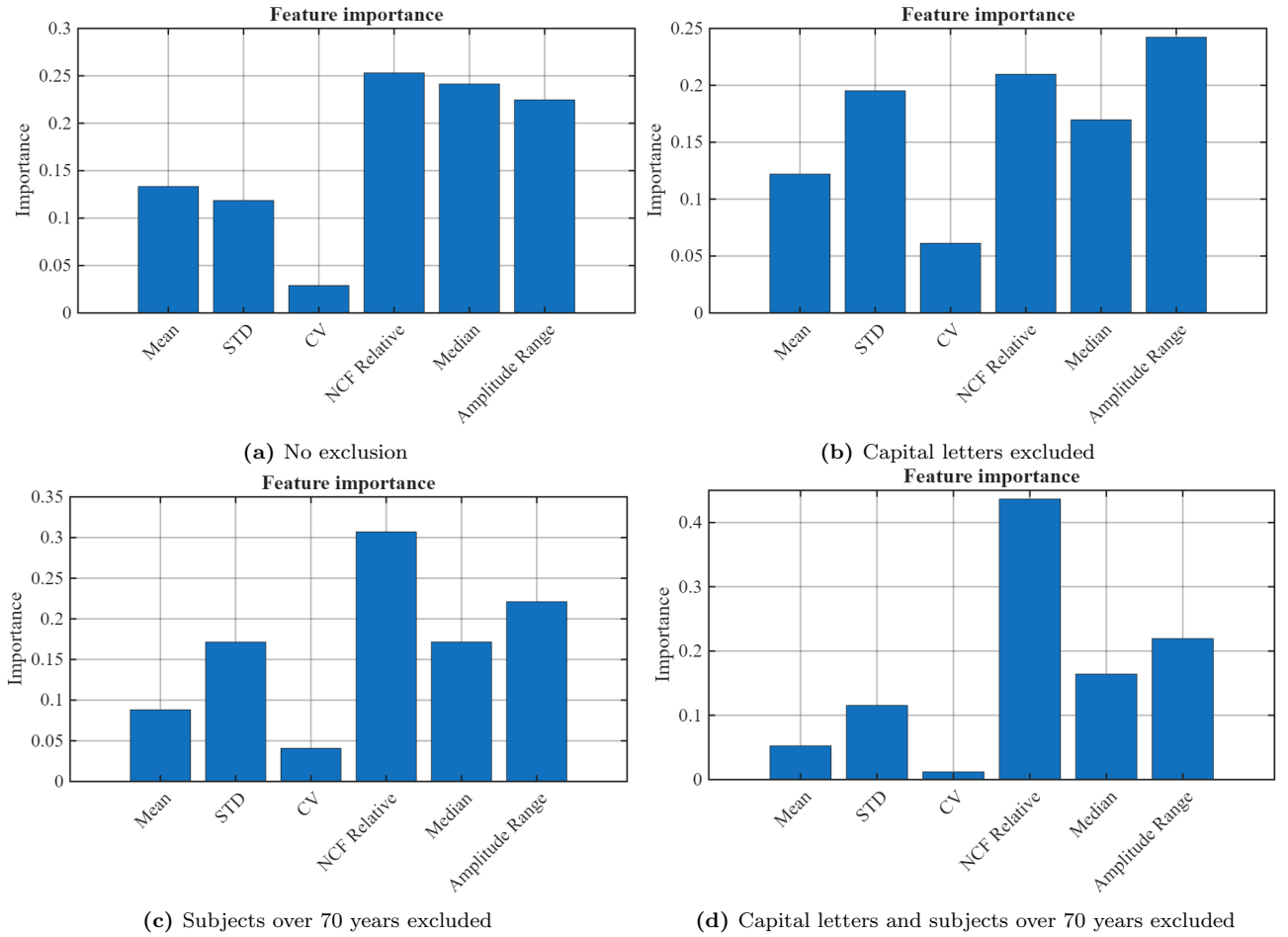


Figure 8.12: Bar charts showing the feature importance for the SVM classifier to distinguishing between SIP1 and SIP2 after Scaling and Rounding of the SIP2 pressure. The features are Mean, standard deviation (STD), coefficient of force in the stroke(CV), number of changes in force(NCF) Relative, Median, and Amplitude Range. The four bar chart shows the feature importance when a) No groups were excluded, b) capital letters were excluded, c) subjects over 70 years were excluded, and d) both capital letters and subjects over 70 years were excluded.

8.5.2 Data Exclusion Experiments of the SIP2 Quantile Mapped Dataset

In this section, the effects of excluding capital letters, participants over 70 years, and simultaneous are examined for the CV and NCF features after quantile mapping SIP2. In Table 8.8, the δ values demonstrates the effect of the exclusion process. For SIP2 Quantile Mapped data, the NCF feature showed a small initial difference between SIP1 and SIP2 Quantile Mapped with a δ value of -0.21 . Excluding capital letters reduced the δ value to -0.01 , indicating a negligible difference, whereas excluding subjects over 70 years increased the difference with a δ value of -0.39 . When both exclusion criteria were applied, the δ value remained negligible with a δ value of 0.01 . This confirms that excluding capital letters substantially reduces the distinguishability between SIP1 and SIP2 for NCF after quantile mapping, thereby increasing device comparability. A similar pattern was observed for the CV feature. Before any exclusion, δ was -0.46 , with the most pronounced reduction achieved by excluding capital letters with a δ values of 0.15 . Excluding subjects over 70 years resulted in a minor reduction to a δ value of 0.41 , while applying both exclusions increased the difference to a δ value of 0.58 . Consistent with previous analyses, excluding capital letters has the strongest impact on reducing

δ values, indicating its primary role in decreasing differences between the pen versions.

Feature	δ values			
	Incl. All	Excl. CL	Excl. >70 years	Excl. CL + >70 years
CV	0.46	0.15	0.41	0.58
NCF	-0.21	-0.01	-0.39	0.01

Table 8.8: The Cliff δ values derived from pressure signal features when comparing SIP1 and SIP2 Quantile Mapped before (Incl. All) and after excluding (Excl.) Capital Letters (CL), participants over 70 years (>70 years), and the simultaneous exclusion, including the features coefficient of variation (CV) and number of changes in force (NCF).

Table 8.9 illustrates how the exclusion criteria applied influence the SVM classifier’s performance in distinguishing between SIP1 and SIP2 Quantile Mapped. By excluding the capital letters, the overall performance is substantially decreased where the accuracy are 50% in the test, corresponding to random guessing for a binary classification task. The performance by excluding subjects over 70 years is substantially increased on the contrary, indicating that the exclusion does not make the prototypes more comparable. When both criteria were excluded, the precision, recall, and F1-score all dropped to 0, as the model failed to predict any true positives. This also affected the calculation of accuracy. The exclusion of capital letters appears to reduce the model’s ability to distinguish between SIP1 and SIP2 after quantile mapping, because this exclusion increases the comparability of the pressure data across the two devices.

Exclusion group	Accuracy	Precision	Recall	F1-score
<i>Training</i>				
Quantile Mapped (n=516)	80.8($\pm$ 5.7)	77.5($\pm$ 6.0)	87.2($\pm$ 5.2)	82.0($\pm$ 5.2)
CL excluded (n=302)	71.2($\pm$ 6.1)	73.5($\pm$ 9.3)	68.2($\pm$ 8.1)	70.3($\pm$ 5.9)
Over70 excluded (n=378)	82.8($\pm$ 4.7)	80.8($\pm$ 2.6)	86.8($\pm$ 8.5)	83.3($\pm$ 5.0)
CL + Over70 excluded (n=198)	82.3($\pm$ 6.5)	76.2($\pm$ 8.1)	95.9($\pm$ 4.2)	84.6($\pm$ 4.7)
<i>Test</i>				
Quantile mapped (n=128)	57.8	60.9	43.8	50.9
CL excluded (n=76)	50.0	50.0	31.6	38.7
Over70 excluded (n=94)	68.1	68.1	68.1	68.1
CL + Over70 excluded (n=50)	38.0	0.0	0.0	0.0

Table 8.9: SVM Classifier test to distinguish between SIP1 and SIP2 Quantile Mapped with and without test with Capital Letters (CL), subjects over 70 years, and simultaneously. N reference to the number of data files used for the test.

The bar chart in Figure 8.13 illustrates the importance of the different features after exclusion of capital letters, subjects over 70 years and both test groups. The exclusion of capital letters results in reducing the impact of CV in the classifier test. In both the exclusion of subjects over 70 years and the simultaneous exclusion, the impact of CV remains high in according to the feature importance of the classifier test. Similar to the findings for SIP2 Scaled and Rounded, the SIP2 Quantile Mapped results also show that excluding capital letters improves comparability by reducing the importance of both CV and NCF.

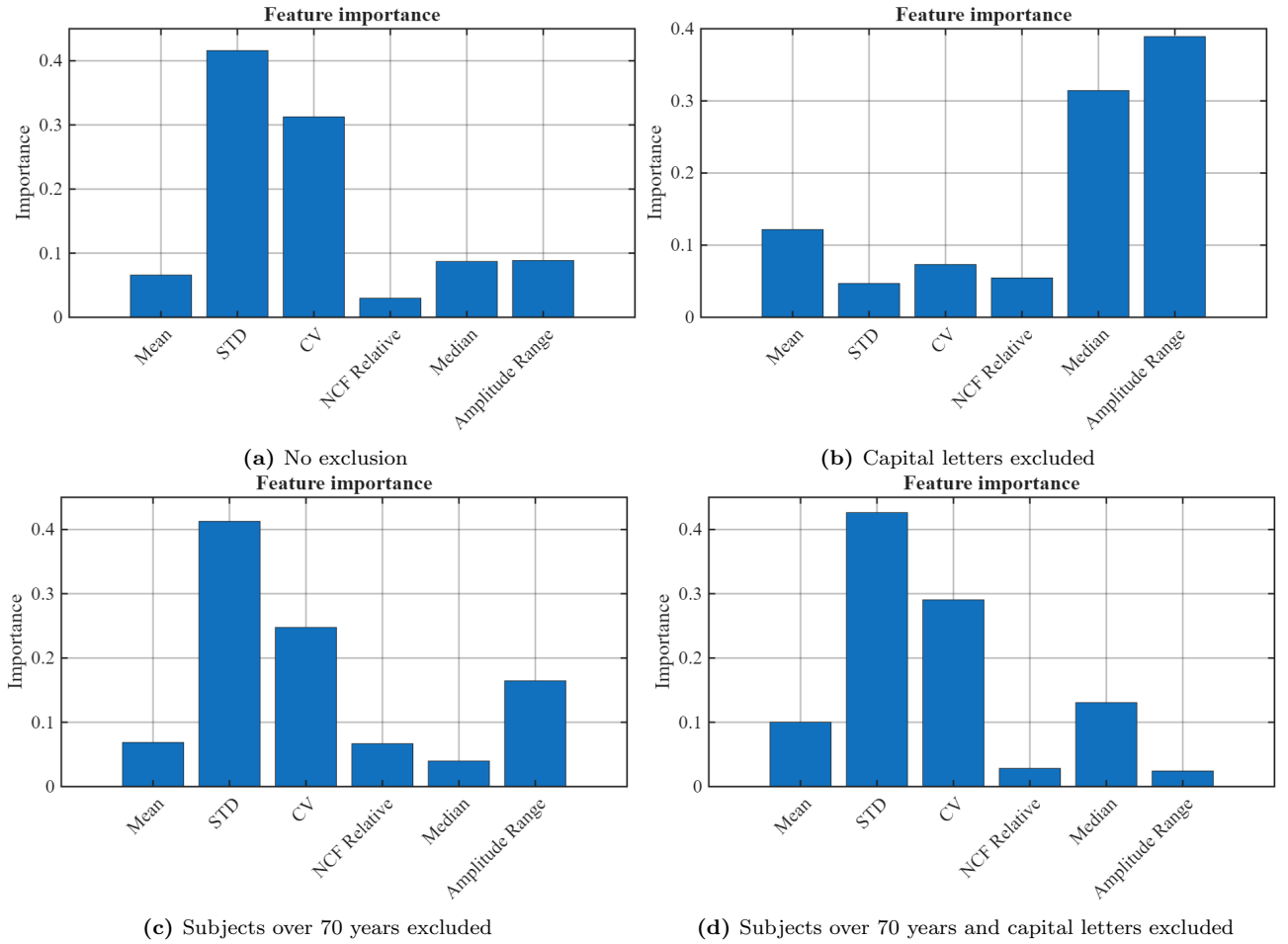
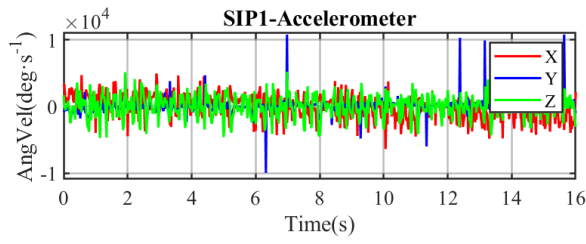


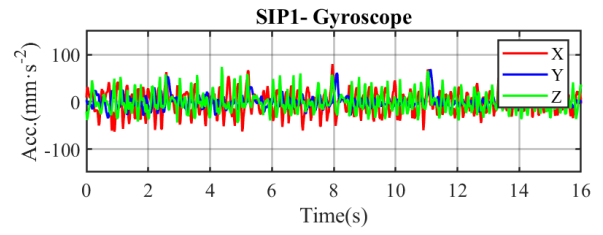
Figure 8.13: Bar charts showing the feature importance for the SVM classifier in distinguishing between SIP1 and SIP2 after quantile mapping the SIP2 pressure. The features are Mean, standard deviation (STD), coefficient of force in the stroke(CV), number of changes in force(NCF) Relative, Median, and Amplitude Range. The four bar chart shows the feature importance when a) No groups were excluded, b) capital letters were excluded, c) subjects over 70 years were excluded, and d) both capital letters and subjects over 70 years were excluded.

8.6 Accelerometer and Gyroscope Data Across the Pens

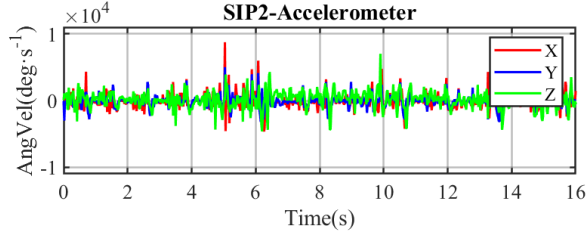
To compare the accelerometer and gyroscope data across the two prototypes, the data was split and balanced as described in Section 8.1. The X and Y axes in the IMU of SIP2 are flipped compared to the X and Y axes in SIP1. This applies to both the accelerometer and the gyroscope data. Therefore, flipping of the Y and X axes of SIP2 was applied before the analysis. An example of the accelerometer and gyroscope data in the X, Y, and Z directions for SIP1 is shown in Figure 8.14a and 8.14b, and for SIP2 after flipping the axes in Figure 8.14c and 8.14d.



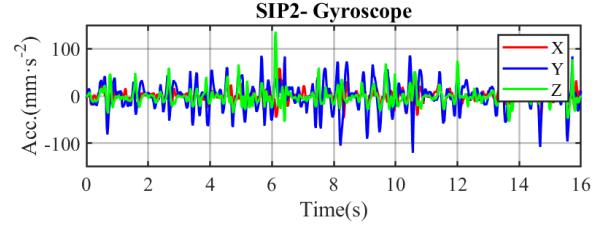
(a) 16 seconds of the SIP1 accelerometer signals X, Y, and Z.



(b) 16 seconds of the SIP1 Gyroscope signals X, Y, and Z.



(c) 16 seconds of the SIP2 accelerometer signals X, Y, and Z.



(d) 16 seconds of the SIP2 Gyroscope signals X, Y, and Z.

Figure 8.14: Accelerometer and Gyroscope signal examples of the SIP1 and the SIP2, where the X and Y axes of SIP2 were flipped to match the axis direction in SIP1.

In the accelerometer and gyroscope data for both SIP1 and SIP2, DC offset was removed, and the signal was rectified, to highlight the variability differences between the signals. For each prototype, both mean, median, and STD with corresponding δ values were examined to observe if larger differences occurred in the features between the pens.

To quantify practical differences in Table 8.10, the δ value was calculated for each axis and feature. All δ values are close to 0 indicating negligible effect where several δ values are indicating a small effect size with δ values close to -0.20 . The negative values indicate that SIP2 tends to have slightly higher measurements than SIP1, but the magnitude of these differences is minimal. Overall, the negligible to small δ values suggests that the two prototypes produce highly comparable sensor data. Therefore, SIP1 and SIP2 can be considered comparable in terms of the measured IMU features.

Feature	δ values		
	x	y	z
<i>Accelerometer</i>			
Mean rectified	-0.04	-0.16	-0.01
Median rectified	-0.01	-0.17	0.01
STD rectified	-0.11	-0.11	-0.06
STD	-0.07	-0.13	-0.02
<i>Gyroscope</i>			
Mean rectified	-0.18	-0.13	-0.15
Median rectified	-0.20	-0.19	-0.17
STD rectified	-0.09	-0.01	-0.08
STD	-0.13	-0.03	-0.10

Table 8.10: The Cliff δ values derived from accelerometer and gyroscope signal features when comparing SIP1 and SIP2, including the features mean, median, and standard deviation (STD) where the features are derived from the axis X, Y and Z.

The SVM classifier test results in Table 8.11 show an accuracy of 55.5% with precision, recall, and F1 scores all close to 60%. These values are only slightly above random chance for a binary classification task, indicating that the classifier struggles to distinguish between SIP1 and SIP2 based on the extracted features. This low discriminative performance aligns with the negligible to small Cliff’s δ values, confirming that the sensor data from the two prototypes is highly similar and does not exhibit strong systematic differences. The bar chart of the feature importance in Figure 8.15 highlights which IMU features contributed most to the classification. The feature importance values range from approximately 0.01 to 0.11, which is a relatively narrow span. This indicates that the SVM classifier does not rely heavily on any single feature, but rather on a broad set of features with comparable contributions. The distribution reinforces the observation that the prototypes exhibit similar sensor characteristics.

SVM Classifier test	Accuracy	Precision	Recall	F1-score
<i>Training</i>				
(n=516)	60.5($\pm$ 1.9)	61.7($\pm$ 2.3)	55.0($\pm$ 5.2)	58.1($\pm$ 3.2)
<i>Test</i>				
(n=128)	55.5	55.9	51.6	53.7

Table 8.11: SVM Classifier test of IMU data. N reference to the number of data files used for the test.

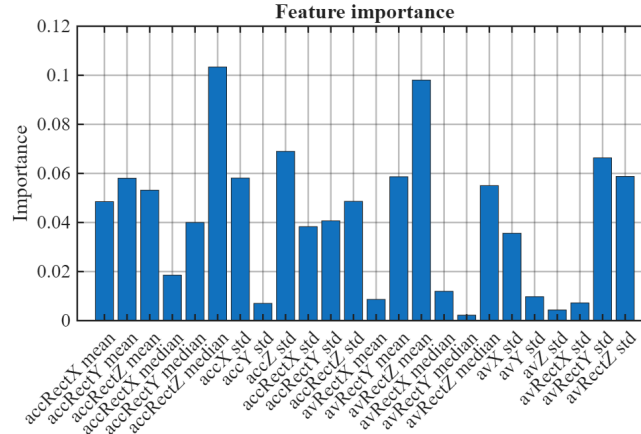


Figure 8.15: Bar chart showing the importance of the features for the SVM classifier of accelerometer (acc) and gyroscope (av) where mean and median are calculated from rectified (RECT) signals, while standard deviation (STD) is calculated for both rectified (RECT) and non-rectified signals for each X, Y, and Z axis.

8.7 Domain Adaptation Development Summary

Initially, amplitude scaling and simple rounding of SIP2 aligned the general amplitude ranges between the two prototypes but failed to account for the sensitivity differences, as reflected by the variation in the NCF and CV features and the high classification performance of the SVM classifier. Furthermore, the SVM classifier performance test results after applying stepwise quantile mapping to SIP2 approached random classification, indicating that the model was no longer able to reliably distinguish between SIP1 and SIP2. This suggests that the pressure data from the two prototypes became more

comparable after quantile mapping. The NCF feature appears to support this outcome by minimizing the impact of sensitivity related differences, suggesting that the sensitivity between the two prototypes becomes more comparable as well.

The promising results from the quantile mapping SVM classifier test confirm that the pressure data from the two prototypes can be used as a feature for adding additional information about the handwriting patterns in the time series classification models for the detection of the risk of dysgraphia. Keeping in mind that a more complex ML model than the SVM classifier may still be able to distinguish between the two pen prototypes rather than the dysgraphia characteristics. The final model must be verified with data acquired with both pen prototypes containing similar subject groups as used for developing the domain adaptation algorithm, to ensure that the pen device differences do not affect the performance.

The evaluation of the exclusion test groups showed that excluding capital letters led to a further decrease in classifier performance. This reduction suggests that removing capital letters increases the comparability between SIP1 and SIP2. According to the accelerometer and the gyroscope, the SVM classifier struggled to distinguish the pen types, it is assumed that the IMU data are comparable. These results indicate that the accelerometer and gyroscope signals are comparable across the prototypes, and they can therefore be used directly as input to the time series classifier without requiring further preprocessing.

8.8 The Applied Domain Adaptation Algorithm

The applied domain adaptation methods aimed to reduce the difference between SIP1 and SIP2 pressure recording feature spaces. To make the accelerometer and the gyroscope axis comparable across the pens the X and Y axis on the SIP2 are flipped. Based on the SVM classifier results, it is expected that applying quantile mapping to the SIP2 pressure signal makes the pressure data comparable to SIP1. The preprocessing of the SIP2 pressure signal, therefore, consists of applying quantile mapping, ensuring that the pressure data becomes comparable to SIP1 and can be reliably used as input to the time series classifier.

The final domain adaptation algorithm applying the signal processing to the data acquired with SIP2:

1. Pressure signal was Quantile Mapped
2. Accelerometer and gyroscope signals X and Y axis were flipped

9. Time Series Classifiers

This chapter describes the implementation of the time series classifiers used to classify **Risk** or **NoRisk** of dysgraphia by using raw time series data recorded from both SIPs. The chapter outlines the preparing of data of splitting the data before model testing. The implemented classifiers including MINIROCKET-HYDRA, MR-HYDRA, and QUANT were further described in Chapter 4. Finally, the pen verification, evaluation methods, and classification results of the tested classifiers are presented.

9.1 Datasets for Training and Evaluating the Time Series Classifiers

Data used for this chapter was preprocessed in MATLAB as described in Section 8.8 and subsequently imported into Python for time series classification. Each recording contained the grade, age, writing style, pen prototype, time series data from the accelerometer and the gyroscope, in the X, Y, Z axes, and the time series pressure data.

As described in Chapter 7, the dataset consisting of 95 samples from CD, labeled with **Risk**, and 2662 samples from TDC. Among the TDC, 729 samples showed a risk of dysgraphia and were labeled **Risk**, while 1933 samples showed no risk and were labeled with **NoRisk** as shown in table 7.2b.

The data were randomly divided into a training set of 80% and a test set of 20%. In Table 9.1, the distribution of **Risk** and **NoRisk** children are visualized according to the grade as described in Section 7.2. The distribution of grades was imbalanced because the number of children in the study from grade 2 to grade 5 were larger. As well as the distribution of **Risk** and **NoRisk** were imbalanced in the majority of the grades. To ensure representation of all grade levels in both train and test, the data were stratified by grade levels and class labeling which is illustrated in table 9.1. For each grade, 80% of the **Risk** samples and 80% of the **NoRisk** samples were assigned to the training set. Additionally, 20% of the **Risk** samples and 20% of the **NoRisk** samples from each grade were assigned to the test set.

			<i>Total</i>		<i>Training</i>		<i>Test</i>		
Grade			Risk	NoRisk	Total	Risk	NoRisk	Risk	NoRisk
CD without age			32	0	32	26	0	6	0
	Grade 1		110	100	210	88	80	22	20
	Grade 2		112	347	459	90	278	22	69
	Grade 3		174	480	654	139	384	35	96
	Grade 4		134	249	383	107	199	27	50
	Grade 5		129	318	447	103	254	26	64
	Grade 6		53	210	263	42	168	11	42
	Grade 7		37	140	177	30	112	7	28
	Grade 8		43	89	132	34	71	9	18

Table 9.1: The distribution of **Risk** and **NoRisk** according to the school grade level, including train and test partitions.

9.2 Feature Selection

The models were trained and tested on gyroscope and accelerometer data with and without the inclusion of pressure data to test whether the pressure data could improve the performance. While dysgraphia symptoms could be reflected in the surface pressure, as mentioned in Section 2.4. This is further supported in Chapter 3 where Asselborn et al. (2020) identified pressure as one of the most informative features. Similar studies, such as Drotar & Dobes (2020), Kunhoth et al. (2023), and Kunhoth et al. (2023) have also used pressure to distinguish normal from dysgraphic handwriting. The models were both tested with and without the pressure data to follow up on this hypothesis.

The test results presented in Section 8.4 show minor differences in the pressure signals remain across the pens, although these differences are not sufficient for the SVM classifier to separate them clearly. To fully eliminate pen related differences, the models were also tested totally without the data recorded with SIP1, corresponding to the exclusion of test recording from the CD.

As described in Section 2.1, the typical writing development process rapidly improves from the 1st to 3rd grade which can indicate how far the children are in their writing development process. If a systematic pattern is observed in detection performance across grades, the effect of including grade level as an additional feature will be investigated.

In Table 8.11, the SVM classifier results for gyroscope and accelerometer were stable between training and testing, where the results indicate that the SVM classifier could not distinguish between the pen versions. Therefore, the gyroscope and accelerometer data were included in all tested models. The time series classifiers were therefore trained and tested under 6 conditions:

- TDC_baseline: Accelerometer and gyroscope data from TDC.
- TDC_P: Accelerometer and gyroscope data from TDC including pressure data.
- TDC_P_G: Accelerometer and gyroscope data from TDC including pressure data and school

grade level.

- TDC_CD: Accelerometer and gyroscope data from TDC and CD.
- TDC_CD_P: Accelerometer and gyroscope data from TDC and CD including pressure data.
- TDC_CD_P_G: Accelerometer and gyroscope data from TDC and CD including pressure data and school grade level.

Data from TDC was acquired with SIP2, where data from CD was acquired with SIP1.

9.3 Implementation of the Time Series Classifiers

This section describes the implementation of the models used in the study’s classification. The model development of the three models was based on the implementations provided in the AEON library [28]. In the tested models, the `random_state` parameter is set to 42 to ensure reproducibility. The models are tested by using the `class_weight` parameter to address the imbalance in the dataset and to investigate its effect on model performance.

9.3.1 Implementation of MINIROCKET-HYDRA

To develop the combined MINIROCKET-HYDRA model, features extraction was performed using both the MINIROCKET and HYDRA transformers. First, the MINIROCKET transformer from AEON library [28] was fitted on the training data in order to learn its convolutional kernel parameters and subsequently transform the time series into a MINIROCKET feature representation. In this study, MINIROCKET was used with default settings where the most relevant parameters include `n_kernels` and `max_dilations_per_kernel`. The `n_kernels` parameter defines the number of random convolutional kernels set to 10,000, where the number of fixed kernels was set to 84. In a similar way, the HYDRA transformer from AEON library [28] was fitted on the same training data to learn its kernel grouping structure and produce the corresponding HYDRA feature set. In this study, the HYDRA transformer was used by adjusting `n_kernels` and `n_groups` parameter. Although higher accuracy is typically obtained with larger HYDRA kernel configurations, such as 64 groups of 8 kernels which substantially increases the computational cost because HYDRA must evaluate a much larger set of competing kernels at every time point. [10] After both transformations, the resulting MINIROCKET and HYDRA feature matrices were concatenated into a single combined feature set. A RidgeClassifierCV was then used to train the classification model on the combined feature set. The RidgeClassifierCV is a Ridge classification model with build in cross validation. The default Leave-One-Out Cross-Validation was used, with the alpha values (-3, 3, 10) proposed by Dempster et al. (2021) and Dempster et al. (2023) when training the individually MINIROCKET and HYDRA models.

For model testing, the trained classifier was applied the test data. The MINIROCKET and HYDRA transformers were used to transform the test data into feature representations using the same learned parameters as in the training. These test features were then passed to the trained ridge classifier to generate predictions.

9.3.2 Implementation of MultiROCKET-HYDRA

The MR-HYDRA was implemented using the MultiRocketHydraClassifier from AEON library [28]. The MultiRocketHydraClassifier is a combination of the MultiROCKET and Hydra transformer where the feature outputs are concatenated and trained in a linear classifier. In this study, the model was used with the default setting. As described in Section 4.1.1, MultiROCKET is an extension of the MINIROCKET. Accordingly, the MR-HYDRA model was implemented in the same way as the MINIROCKET-HYDRA model but incorporates the additional feature representations introduced in MultiROCKET. Specifically, MultiROCKET applies a kernel set to both the raw time series and the first-order difference series by using separate dilation sets for each representation. Additionally, MultiROCKET includes three additional pooling operators, which provide a more discriminative feature space.

9.3.3 Implementation of QUANT

The QUANT model was implemented using the QUANTClassifier from AEON library [28]. QUANTClassifier performs interval-based quantization of time series where the classifier exposes several parameters controlling how intervals and quantized features are generated. In this study, the model was used with default settings where the most relevant parameters include `interval_depth`, `quantile_divisor`, and `estimator`. The `interval_depth` parameter determines how deep the interval extraction process proceeds. Starting from the full time series, QUANT extract 2^{depth} intervals at each level, and features from all levels are concatenated to form the final representation. With the default values of `interval_depth` = 6, QUANT performs interval extraction across seven levels from 0 to 6, resulting in a total of 127 intervals whose features are concatenated into the final representation. The `quantile_divisor` parameter determines how many quantiles are computed within each level which is calculated as in equation 9.1.

$$1 + \frac{\text{interval_length} - 1}{\text{quantile_divisor}} \quad (9.1)$$

With the default of `quantile_divisor` = 4, the number of quantiles in an interval is reduced by dividing the interval length with four, meaning that longer intervals produce more quantiles while shorter intervals will produce fewer. This default setting defines the level of detail in the quantised features. The `estimator` parameter specifies the classifier used on the extracted features, where the default is *Extremely Randomized Trees* with 200 trees.

9.4 Verification of Pen Difference

The verification process was performed using the ASIP1 and ASIP2 datasets, det datasets are showed in Table 7.2d, on which the domain adaptation algorithm was originally developed on. To evaluate whether the differences between SIP1 and SIP2 were reduced after preprocessing and domain adaptation of SIP2, the adult data were used to train and test on the selected classifier in distinguishing between the prototypes. This step was essential because the selected classifier was the same classifier used to train and test the detection of **Risk** and **NoRisk** for dysgraphia in the children’s dataset, which includes CD data from SIP1 and TDC data from SIP2. Thus, the verification ensures that any remaining pen related differences were eliminated in the exact model applied to the children’s data, preventing classification performance from being confounded by pen differences. For the verification, the selected model was trained and tested on accelerometer and gyroscope data from ASIP1 and ASIP2, as well as an additional version where pressure data was included in the training and testing. A distinguishability with an accuracy of 60% was decided as a threshold for how well the model must be able to distinguish the pen version, to confirm the comparability across the pens. The purpose of the verification with only the accelerometer and gyroscope data were to verify that the pen differences after preprocessing of the accelerometer and gyroscope were eliminated and the prototypes were comparable. Similarly, the purpose of the verification, including pressure data, was to assess whether the domain adaptation and preprocessing removed the pen related differences, thereby verifying the comparability between the ASIP1 and ASIP2 including pressure data.

9.5 Evaluation Method

In this study, Accuracy, Precision, Recall, F1 score, Receiver Operating Characteristic(ROC), and Area Under the Curve(AUC) were selected as evaluation metrics. Although accuracy provides a general indication of model performance, it may be misleading when applied to an imbalanced dataset, as seen in Table 7.2b. Since the class distribution in the tested datasets is 30% **Risk** and 70% **NoRisk**, more emphasis was placed on precision, recall, and F1-score, which better capture the model’s ability to correctly identify minority and majority classes. These metrics are also widely used in the literature [22] and, therefore, relevant to use as comparison metrics. Furthermore, the confusion matrices for each model are presented to support a detailed error analysis. The ROC curves and AUC scores were generated using the predicted class probabilities obtained for the dataset. These metrics provide insight into how well the model distinguish the **Risk** and **NoRisk** classes in this study. Furthermore, the model performance in each school grade level was investigated to gain insight into whether the model struggles to predict the **Risk** class in different grades.

The evaluation of the performance metrics was used to select the best performed model. In the

selection process, MINIROCKET-HYDRA, MR-HYDRA and QUANT are trained and tested only on the TDC data to avoid potential pen difference with only the accelerometer and gyroscope data included. These baseline models are trained and tested with and without balanced weighted class distribution to investigate the impact of balanced classes due to the imbalanced dataset. The best performing selected time series classifier was afterwards trained and tested with different dataset configurations depending on the pen verification test to investigate the impact of different features related to handwriting development. The dataset configurations are described in Feature Selection in Section 9.2. The overall best performed model configuration was selected based on the feature configuration performance.

The best performed model was chosen based on the highest recall, as recall reflects the model's ability to correctly identify children with handwriting difficulties. In the context of dysgraphia detection, false negatives are particularly undesirable, as failing to identify affected children may lead to unaddressed writing difficulties and subsequent negative consequences. In case of two models achieve the highest recall, the highest F1-score will be used to evaluate the best performing model. The F1-score provides a balanced measure by accounting for both precision and recall, ensuring that the chosen model not only detects most positive cases but also limits the number of false positives. The same success criteria must be obtained to chose the best performed dataset configuration of the model.

The results of model selection based on the baseline models are presented in section 9.6.1 where the results of features configuration are presented in section 9.6.3.

9.6 Classification Results

This section presents the results of the time series classifier testing. In Subsection 9.6.1, the test results of the implemented models described in Chapter 9 are presented. In Subsection 9.6.2, the selected time series classifier was used to verify if the ASIP1 and ASIP2 datasets were comparable. When clarification of the comparability between the pens, the selected time series classifier was trained and tested with combinations of dataset configurations in Subsection 9.6.3. In Appendix 14, the belonging ROC and Confusion Matrix can be found for Subsection 9.6.1, 9.6.2 and 9.6.3.

9.6.1 Time Series Classifier Selection Result

In this section, the performance of the MINIROCKET-HYDRA, MR-HYDRA, and QUANT with and without balanced weights with respect to class distribution was compared and evaluated. The test results were obtained using the TDC_baseline, including only the TDC dataset with accelerometer and gyroscope data. Table 9.2 shows the results of the tested models, with and without balanced weights.

Model	Balanced	Acc.	Prec.	Recall	F1	AUC
MINIROCKET-HYDRA	No	78.79	68.96	41.09	51.50	0.81
MR-HYDRA	No	76.36	60.00	41.09	48.78	0.78
QUANT	No	78.79	70.88	38.35	49.77	0.80
MINIROCKET-HYDRA	Yes	70.54	46.62	52.05	49.19	0.74
MR-HYDRA	Yes	72.04	48.96	48.63	48.79	0.71
QUANT	Yes	78.23	69.23	36.98	48.21	0.80

Table 9.2: Overview of the test results on the trained time series classifiers. All time series classifiers were trained and tested on the TDC_baseline data, including only the TDC dataset with accelerometer and gyroscope data. The shown performance metrics are Accuracy(Acc.), Precision(Prec.), Recall, F1-Score(F1), and Area Under the Curve(AUC).

MINIROCKET-HYDRA was the best performing model according to recall, and was therefore chosen to be the primary model. The Receiver Operating Characteristic(ROC), confusion matrix, and the school grade confusion matrix are shown in Figure 9.1.

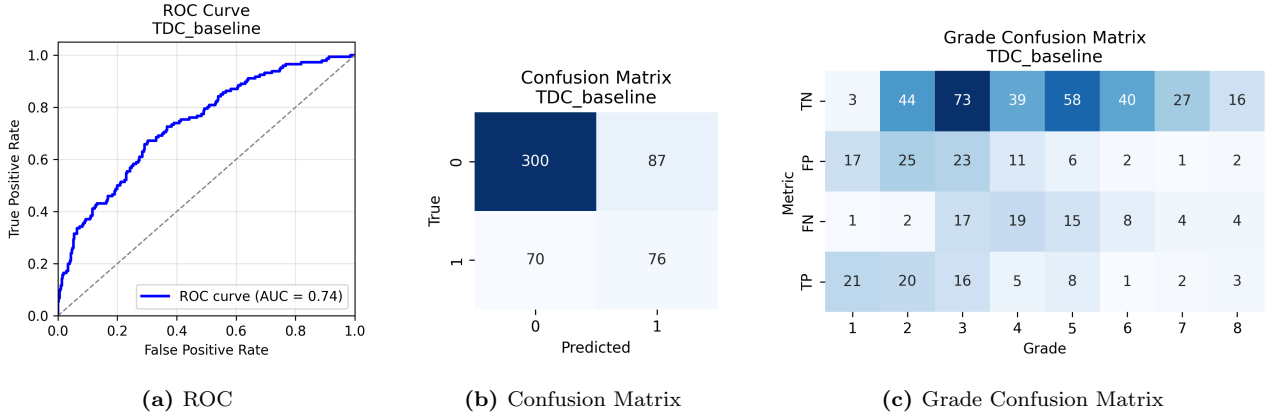


Figure 9.1: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights chosen as the primary model, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each school grade.

The school grade confusion matrix in Figure 9.1c shows that the number of false positive in grade 1-3 was higher than the false positives, and conversely in grade 4-7 where the number of false negative were higher. Due to the systematic grade-related pattern, grade was introduced as a feature.

9.6.2 Time Series Classifier Pen Verification Result

In this section, the SIP_baseline and the SIP_P dataset configurations were used to train and test MINIROCKET-HYDRA with balanced class weights. The dataset configurations are shown on the right side of Table 9.3, while the classification results are presented on the left side in the same table.

Dataset	Performance metrics					Configuration			
	Acc.	Precision	Recall	F1	AUC	ASIP1	ASIP2	AG	Press
SIP_baseline	79.53	60.00	63.64	61.76	0.85	X	X	X	
SIP_P	100	100	100	100	100	X	X	X	X

Table 9.3: The results of the MINIROCKET-HYDRA classifier with balanced class weights trained and tested on the ASIP1 and ASIP2 datasets, including accelerometer and gyroscope data(AG), depending on the dataset, pressure data(Press) was included. The shown performance metrics are Accuracy (Acc.), Precision(Prec.), Recall, F1-Score(F1), and Area Under the Curve(AUC).

For both the SIP_baseline and the SIP_P dataset configuration, the model achieved an accuracy above the predefined threshold of 60%, as seen in Table 9.3. Therefore, the following results on the MINIROCKET-HYDRA with balanced class weights to detect the risk of dysgraphia not includes the CD data, while the CD dataset is acquired with the SIP1. The results are presented in Subsection 9.6.3

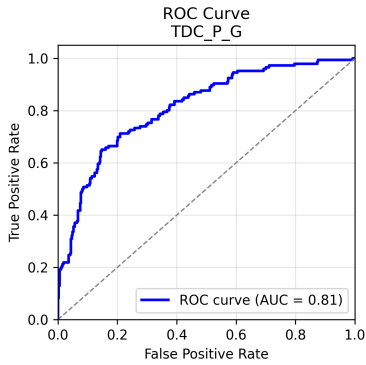
9.6.3 Time Series Classifier Dataset Configuration Results

Dataset configurations with combinations of introducing school grade level and pressure were investigated to determine which combination provides the best performance for MINIROCKET-HYDRA with balanced class weights. Table 9.4 shows the performance of the MINIROCKET-HYDRA model, with balanced weights with respect to the class distribution, across the different feature set configurations.

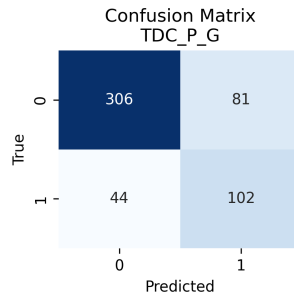
Dataset	Performance metrics					Configuration			
	Acc.	Prec.	Recall	F1	AUC	TDC	AG	Press	Grade
TDC_baseline	70.54	46.62	52.05	49.19	0.74	X	X		
TDC_P	72.04	49.06	54.10	51.46	0.75	X	X	X	
TDC_G	74.67	53.03	65.75	58.71	0.81	X	X		X
TDC_P_G	76.54	55.73	69.86	62.00	0.81	X	X	X	X

Table 9.4: The results of the MINIROCKET-HYDRA classifier with balanced class weights trained and tested on dataset feature configurations with accelerometer and gyroscope data(AG) and combinations of including pressure data(Press) and school grade information(Grade). The shown performance metrics are Accuracy (Acc.), Precision(Prec.), Recall, F1-Score(F1), and Area Under the Curve(AUC).

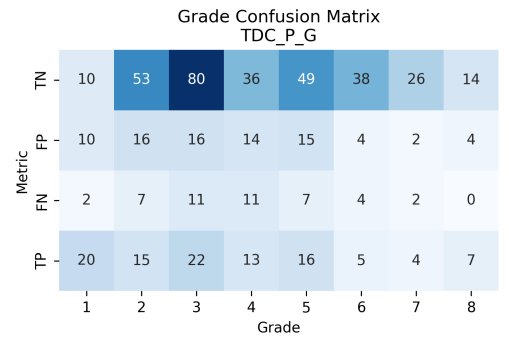
The TDC_P_G showed to be the best tested dataset configuration, according to the Accuracy, Recall, F1-Score, and AUC. The ROC, Confusion Matrix, and Grade Confusion Matrix are shown in Figure 9.2. By including the grade level feature, the grade confusion matrix in Figure 9.2c illustrates that the false positive and false negative are reduced. As well as the overall true positive and true negative are increased in the confusion matrix in Figure 9.2b.



(a) ROC



(b) Confusion Matrix



(c) Grade Confusion Matrix

Figure 9.2: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced class weights, trained and tested on the TDC_P_G dataset configuration, including accelerometer, gyroscope, pressure data and school grade information: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

Part III

Discussion and Conclusion

10. Discussion

Dysgraphia is characterized by difficulties with etc. legibility, writing, and fine motor skills [7], which can result in a negative impact on children’s academic results, daily life, and self-esteem [4, 3, 15, 20]. Tablet-based ML algorithms have been investigated to detect dysgraphia [4, 3, 14, 22, 23], but the surface seems insufficient for this task [18, 2, 29]. Therefore, this study aimed to develop a time series classifier to detect risk and no risk of dysgraphia using raw time series data collected with a sensor equipped ink pen. The best performing time series classifier was the MINIROCKET-HYDRA with balanced class weights and with the dataset configuration TDC.P.G, which was based on TDC data including accelerometer, gyroscope, pressure, and grade as features. It achieved an accuracy of 76.54%, a precision of 55.73%, a recall of 69.86%, a F1-score of 62.00%, and an AUC of 0.81.

Compared to tablet-based studies using engineered features to classify dysgraphia, higher performance is reported. Drotar and Dobes (2020) achieved an accuracy of 79.5% and recall of 79.7% using temporal and kinematic, including pressure features, as in this study. Kunhoth et al.(2023) further improved performance to an accuracy of 80.8%, precision of 83.3%, recall of 78.5% and F1-score of 80.1% by incorporating In-Air features, indicating the importance of capturing continuous temporal dynamics beyond pen surface contact as in raw time series data. While Drotar and Dobes (2020) and Kunhoth et al. (2023) classify engineered features, this study used MINIROCKET-HYDRA to detect the risk of dysgraphia directly from raw time series data, providing an approach without explicit feature engineering. As a result, the performance outcomes are not directly comparable, but demonstrating the potential of raw time series classification. In contrast, Bublin et al. (2023) reported high performance achieving an accuracy of 99.8% and F1-score of 97.78% using raw time series data from SensoGrip smart pen to classify dysgraphia severity rather than binary classification. While these results highlight the potential of raw time series data representations, they were obtained on a small cohort of only 22 children and addressed a different classification objective, limiting direct comparability. Consequently, the reported performance should be interpreted with caution.

10.1 Data Splitting Strategy

Data splitting is important for the validity and comparison with prior work, particularly in this field where the number of datasets is often limited to fewer than 60 subjects in the disease group [12, 14, 4, 22, 5]. When comparing the results of this study with Bublin et al.(2023), who trained a deep learning model directly on raw pen sensor time series. To compensate the limited dataset of only 22 children,

they introduced splitting of each handwriting test into 20 time series, which could force the model to focus on small local movement patterns and increase the size of training and test samples. This strategy may influence the extremely positive results by introducing the risk of data leakage, as their methodology allowed the same writing test to appear in both the train and test datasets. This enables the model to recognize the specific tests rather than generalizing to new ones.

In contrast, the dataset in this study was split such that each test file appeared exclusively in either the train or test set. While a child could contribute multiple tests at the same time point, or across grade levels, no individual test file was shared across splits. This design is not expected to cause a leakage problem. Tests conducted at the same time point differ in tasks or task demand on focus, and tests across grade levels are expected to differ substantially due to the rapid handwriting development during these years. Therefore, the model was trained and tested at the handwriting test-level rather than the subject-level, which increased the amount of data available for training. A subject-level split could have prevented the overlap across train and test sets, but this approach may conflict with the goals of maintaining an 80/20 split while also stratifying by grade level. Given the limited amount of data available in certain grade levels, introducing all these constraints simultaneously may have resulted in a highly restricted and non-random split. Therefore, the subject-level split was not introduced, and the data 80/20 train and test split was only stratified by class label and grade level. Thus, handwriting test-level splitting was chosen as the most practical balance between data availability and methodological robustness.

10.2 Compensating for the Stage of Handwriting Development

The study by Drotar & Dobes (2020) included a broad age range from 8 to 15 years, which they hypothesized introduced development-related noise to the feature space. Similar age range is observed in studies by Kunhoth et al. (2023) and Asselborn et al. (2020). The findings in this study support the pronounced hypothesis. As shown in the grade-level confusion matrix Figure 9.1c, when grade is not included as a feature, a clear pattern is seen, children from 1st to 3rd grade produce many false positives (FP). This suggests that their handwriting skills are still in the early stage of development [30, 7], leading the model to classify early-stage handwriting as **Risk**. In contrast, for 4th to 8th grade, the model tends to produce more false negatives(FN) than FP. This indicates that the children’s handwriting becomes more mature and consistent [7], even children at risk of dysgraphia may display more developed motor patterns, making them more difficult for the model to distinguish from typically developing peers.

Adding grade level as a feature changes this pattern, as seen in the grade level confusion matrix Figure 9.2c, where grade level and pressure are introduced as features. Now FN are markedly reduced, and FP

appears more often than FN for 1st to 5th grade, while no clear trend is found from 6th to 8th grade. In 1st grade, the model predicts only three negative out of 21, indicating the difficulty of separating early-stage writing development patterns from dysgraphia patterns. This shows that including grade-level information helps the model focus on dysgraphia-related patterns, but early grades remain challenging, whereas performance improves slightly in the higher grades. Also, in the performance metrics, the positive effect of introducing grade level is reflected. The MINIROCKET-HYDRA with the dataset configuration TDC_P_G including pressure and grade, the recall are increased to 69.86% indicating that the model’s ability to detect actual **Risk** cases increase. To enhance the model performance further the normalization of the writing experience could be improved of either reducing the age or grade level range in the dataset to let the model focus on more specific age or grade level groups, or adding larger amount of subject specially with **Risk** but also subject with **NoRisk** into each grade, to prevent the model to overfit and to keep an acceptable label balance within each grade level.

10.3 Dysgraphia Handwriting Patterns Found in Pressure Data

The measurements of kinematics and pressure are observed as the most informative indicators of writing quality in the study by Asselborn et al. (2020). Their model obtained a high recall of 91%, but includes children with severe dysgraphia in the disease group. The impact of pressure in detecting dysgraphia is also observed in studies by Kunhoth et al. (2023) and Asselborn et al. (2020). Furthermore, reduced motor control is defined as a key characteristic of dysgraphia [4, 31], and pressure reflects this motor control deficit. In this study, pressure data were included where the MINIROCKET-HYDRA achieved a recall of 69.86%. Nevertheless, these results did not reach the level of the results reported in studies by Kunhoth et al. (2023) with recall of 78.8%, Drotar and Dobes (2020) with recall of 79.7%, and Asselborn et al. (2020) with recall of 91% which also included pressure data. This study is based on typically developing children in risk of dysgraphia and not clinically diagnosed with dysgraphia, as in the mentioned studies. However, this study demonstrated that raw time series data indeed contain meaningful information about handwriting development, even when the classification task is more subtle than in earlier studies [22, 3, 14, 12].

10.4 Limitations Related to Domain Adaptation

In the domain adaptation, quantile mapping was applied to reduce sensitivity and amplitude related differences in pressure between SIP1 and SIP2. The Bayesian Optimization procedure identified the *numstep* of 22 quantile band as the configuration that produced the lowest classification performance and reducing the pen difference most effectively. Dividing the amplitude distribution into 22 quantile bands represents a rough discretization of the signal that limits the resolution of the amplitude mapping as seen in Figure 8.10. As a consequence, dynamic variations in the signal may be lost. Furthermore,

the domain adaptation is developed with respect to the SIP1 signals distributional properties, the algorithm has generalizability limitations. The procedure may not transfer effectively to other signal domains for which it was developed for this study. Nevertheless, the quantile mapping established a consistent preprocessing basis for subsequent time series classification in this study. It provided a practical way to reduce pen differences by enabling a comparable input representation for the time series classification.

As described in Section 8.7, the domain adaptation of the SIP2 domain to the SIP1 domain was successfully achieved when restricting to the accelerometer and the gyroscope data, where the performance of accuracy in the domain adaptation discriminability test with the SVM classifier does not exceed 60%, with an accuracy of 55.5%. To evaluate whether the preprocessing actually mitigates pen differences, the MINIROCKET-HYDRA classifier was trained to distinguish between SIP1 and SIP2 using the adult data as used to develop the domain adaptation algorithm. By using only the accelerometer and gyroscope data, the model achieved an accuracy of 79.53%, indicating that the reduced pen difference is not fully eliminated, as the accuracy exceeds 60%. Similar patterns are observed for the domain discriminability test including pressure data where the SVM classifier results does not exceed 60%, with an accuracy of 50.0%. However, performing the verification test including the pressure data increased MINIROCKET-HYDRA performance to 100% across all metrics, showing that domain adaptation does not completely remove pressure differences. While simpler models like SVM classifier struggle to distinguish pens, MINIROCKET-HYDRA can still detect subtle discrepancies, especially when pressure is included, highlighting that domain adaptation does not completely eliminate pen specific differences.

10.5 Limitations Related to Data Availability and Labeling

The exclusion of capital letter written tasks resulted in a decrease in the TDC dataset from 7608 handwriting tests to 2662 handwriting tests. This limits the amount of training samples, and could decrease the accuracy as reported in Dempster et al.(2023a), where the accuracy of the MINIROCKET-HYDRA starts to stabilizing when using approximately $2^{12}(4,096)$ to $2^{17}(131,072)$ number of training examples. This suggests that a larger amount of data would increase the performance of the MINIROCKET-HYDRA.

In Table 9.4, the results indicate that the inclusion of the CD data did not improve the performance in the feature configuration that incorporated grade level. This may be due to the way grade levels were assigned in the CD dataset, which was based on the age ranges of grade levels within the TDC dataset. Consequently, it could lead to some of the children being assigned to an incorrect grade level as their dysgraphia diagnosis may affect the typical educational progression for their age group,

resulting in discrepancies between their assigned and actual grade levels. Additionally, the results without grade level increase slightly, but not as expected. It was expected that the true positive rate would increase more while all CD were assigned with the label **Risk**. The model learns to distinguish between **Risk** and **NoRisk** of dysgraphia where the CD handwriting may differ substantially while they are diagnosed with dysgraphia. Potentially, the model did not learn dysgraphia patterns because of the small proportion of data from children with dysgraphia written patterns. These discrepancies could be caused by differences in handwriting between TDC at **Risk** of dysgraphia and CD, or the pen differences.

10.6 Future Perspective

The decision to adopt the SIP2 domain to the SIP1 domain was based on the expectation that the more detailed pressure signal of SIP2 would better align with the more rough pressure signal of SIP1. It was done this way around to avoid adding fictive information in the SIP1 pressure signal pattern when comparing it to the SIP2 pressure signal pattern. However, it could be interesting to investigate the possibilities to adopt the SIP1 domain to the SIP2 domain. The time series classifier focuses on detecting the risk of dysgraphia, and the result does not improve when including the CD. It could be worth to investigate the MINIROCKET-HYDRA performance on the TDC dataset without applying the domain adaptation. As the domain adaptation algorithm may reduce the dynamic variation information in the signal by restricting the amplitude level to 22 levels.

To assess the comparability between SIP1 and SIP2, a simple SVM model was used for the discriminability test. The relatively simple model limits the risk of capturing sensitivity variation that still might exists between the prototype. By using a more complex model in the domain adaptation development, the differences between the pens could potentially be evaluated more precisely while a more complex model might be able to offer a more sensitive assessment of whether the prototypes truly behave similarly. In this study, the SVM classifier were utilized as a part of the preprocessing process for evaluating the domain adaptation. Since the domain adaptation process required repeated training and testing, a more advanced model would have been substantially more time consuming to train. However, the SVM classifier served as an efficient and sufficiently sensitive baseline for assessing the comparability between the pen prototypes.

In further work on the time series classifier, several additional features could be investigated to potentially improve its performance. As highlighted in the literature [7], written difficulties are more prevalent among boys, including Gender as a feature may therefore enhance model performance. Other potentially relevant features include the child’s dominant hand, which could influence writing dynamics and contribute to distinguishing patterns in the data. For early detection of **Risk** for dysgraphia,

the recorded data should be focused on children aged 6 to 9 years [30, 7], while this is defined as the period where the fundamental handwriting is developed. By focusing on this age range, it would better support early intervention, capturing children at the beginning of their writing development. In this development phase, legibility tests are suitable to evaluate the writing skills of preschool children, where speed assessment develops later [4, 17]. Therefore, it would be relevant to focus on legibility writing tests.

11. Conclusion

The study aimed to develop a time series classifier to detect risk and no risk of dysgraphia using raw time series data collected with a sensor equipped ink pen. The MINIROCKET-HYDRA classifier showed potential to detect risk of dysgraphia in typically developing children by using raw time series data recorded by a sensor equipped ink pen. However, sensor differences in the sensor equipped ink pens were reflected in the raw time series data characteristics and affected the performance. Additionally, the raw time series data appeared to contain relevant information about dysgraphia handwriting patterns, where school grade information improved the performance by normalizing the handwriting stage. Furthermore, pressure data improved the model performance, which confirms hypotheses from earlier studies and contributes to the understanding that pressure variation is a key characteristic in the dysgraphia writing pattern.

Part IV

Appendix

12. Definitions of Classification Performance Metrics

In this appendix, the classification performance metrics used in the study are described and explained. These metrics include accuracy, precision, recall, F1-score, ROC, AUC, and feature importance.

Accuracy:

Accuracy measures the overall proportion of correct predictions, seen in equation 12.1, representing the fraction of correctly classified instances relative to the total number of cases:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (12.1)$$

where TP and TN denote true positives and true negatives, and FP and FN denote false positives and false negatives.

Precision:

Precision quantifies the proportion of correctly predicted positive cases out of all instances predicted as positive, seen in equation 12.2. It indicates how reliable the model is when it predicts a positive outcome:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12.2)$$

Recall:

Recall, also referred to as sensitivity, measures the fraction of actual positive instances that are correctly identified by the model, seen in equation 12.3. It reflects the model's ability to capture all relevant positive cases:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12.3)$$

F1-score:

The F1-score provides a single metric that balances precision and recall, accounting for both false positives and false negatives, seen in equation 12.4. It represents the harmonic mean of precision and recall and is particularly useful when the class distribution is imbalanced:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12.4)$$

Receiver Operating Characteristic (ROC):

The ROC curve illustrates the relationship between the true positive rate (sensitivity) and the false positive rate(1-specificity) across different classification thresholds.

Area Under the Curve (AUC):

The AUC provides a single summary value of the model's ability to distinguish between the classes positive and negative. A score of 1.0 represents perfect classification, whereas a score of 0.5 corresponds to random guessing.

Feature Importance:

In equation 12.5, the feature importance equation is defined:

$$FeatureImportance_i = \frac{|w_i|}{\sum_j |w_j|} \quad (12.5)$$

Here w_i denotes the weight of feature i and the summation $\sum_j |w_j|$ runs over all features $j = 1, \dots, n$ in the model, ensuring that the importance scores are relative and comparable across features.

13. Domain Adaptation Development Box Plots

In the following sections, the box plots of extracted features from the domain adaptation development in Chapter 8 are visualized.

13.1 Before Domain Adaptation of SIP2

The box plot in Figure 13.1 are before domain adaptation consisting of box plots of the extracted features before applying domain adaptation on SIP2 pressure signal compared with SIP1 pressure signal. The extracted features was mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force in the stroke (NCF) and amplitude range. At the box plot the Cliff δ value are also visualized.

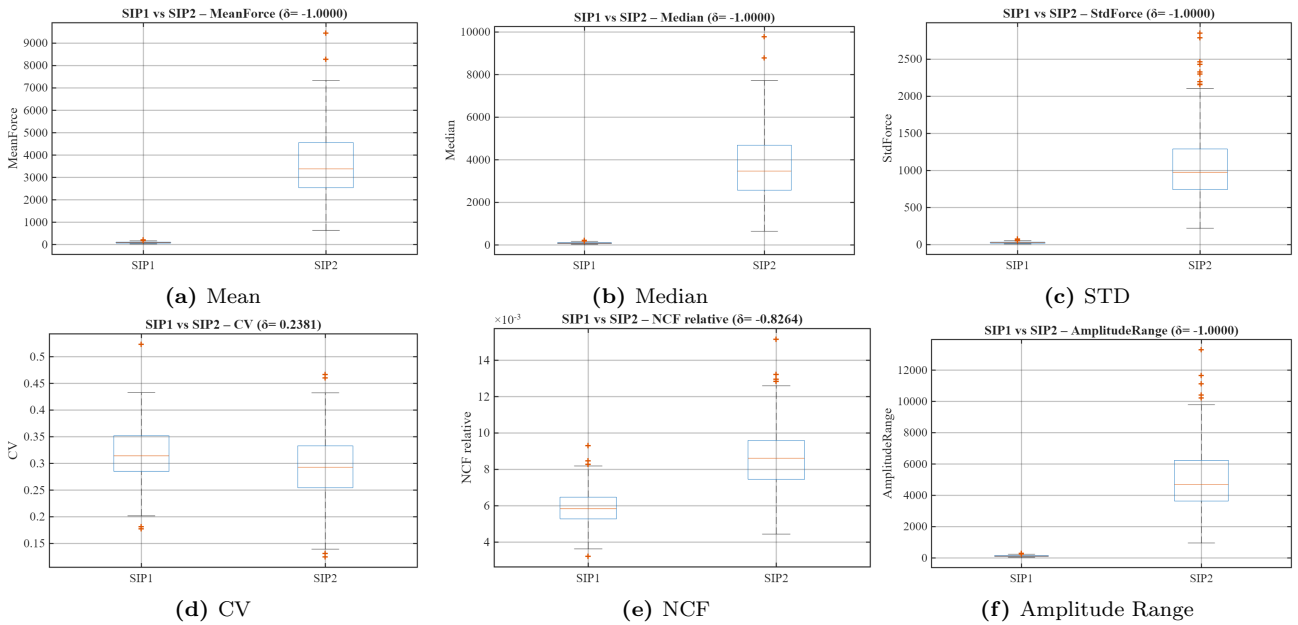


Figure 13.1: Box plots to comparing the features in pressure signal of SIP1 and SIP2. The extracted features was mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF) and amplitude range. The Cliff δ value are visualized on the box plots.

13.2 SIP2 Scaled and Rounded

The box plot in Figure 13.2 are SIP2 Scaled and Rounded pressure signal compared with SIP1 pressure signal consisting of box plots of the extracted features where the domain adaptation approach with scaling and rounding of SIP2 are applied to the pressure signal. The extracted features was mean, median, STD, CV, NCF, and amplitude range. At the box plot the Cliff δ value are also visualized.

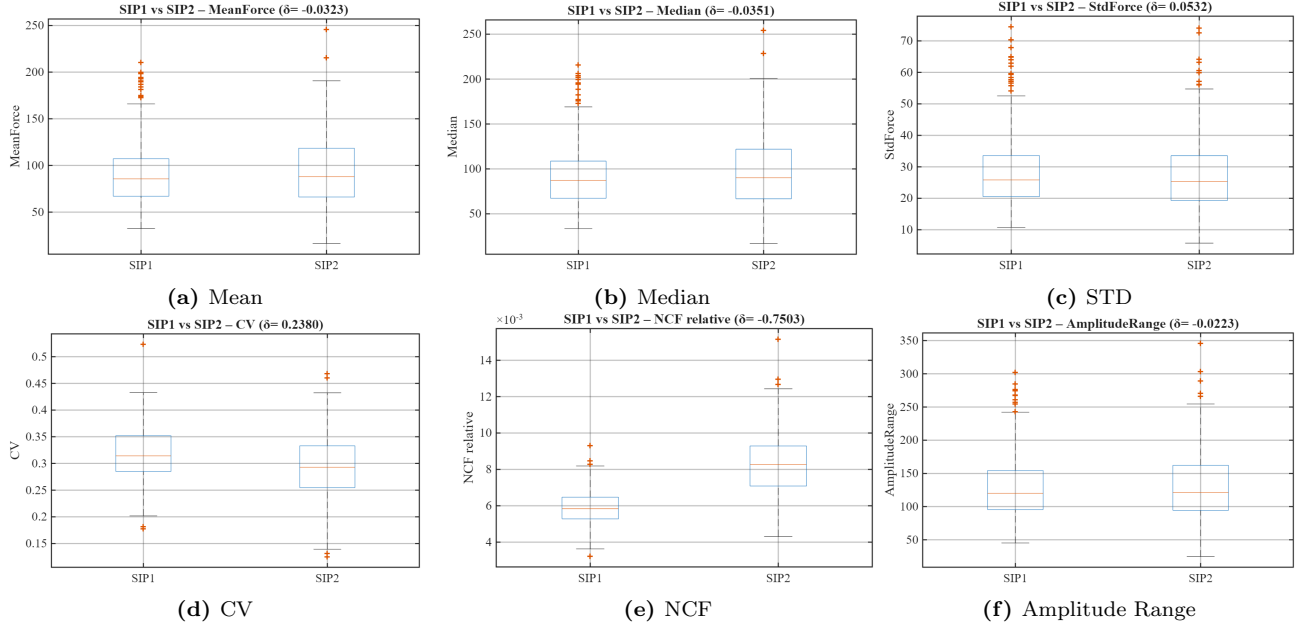


Figure 13.2: Box plots comparing the features in pressure signal of SIP1 and SIP2 Scaled and Rounded. The extracted features were mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF) and amplitude range. The Cliff δ value is visualized on the box plots.

13.2.1 Exclusion Experiments of SIP2 Scaled and Rounded

The box plot in Figure 13.3 are SIP2 Scaled and Rounded pressure signal compared with SIP1 pressure signal consisting of box plots of the CV, and NCF where these features are tested in the exclusion experiments. The exclusion experiments consist of excluding capital letter, subjects over 70 years and simultaneous. At the box plot the δ value is also visualized.

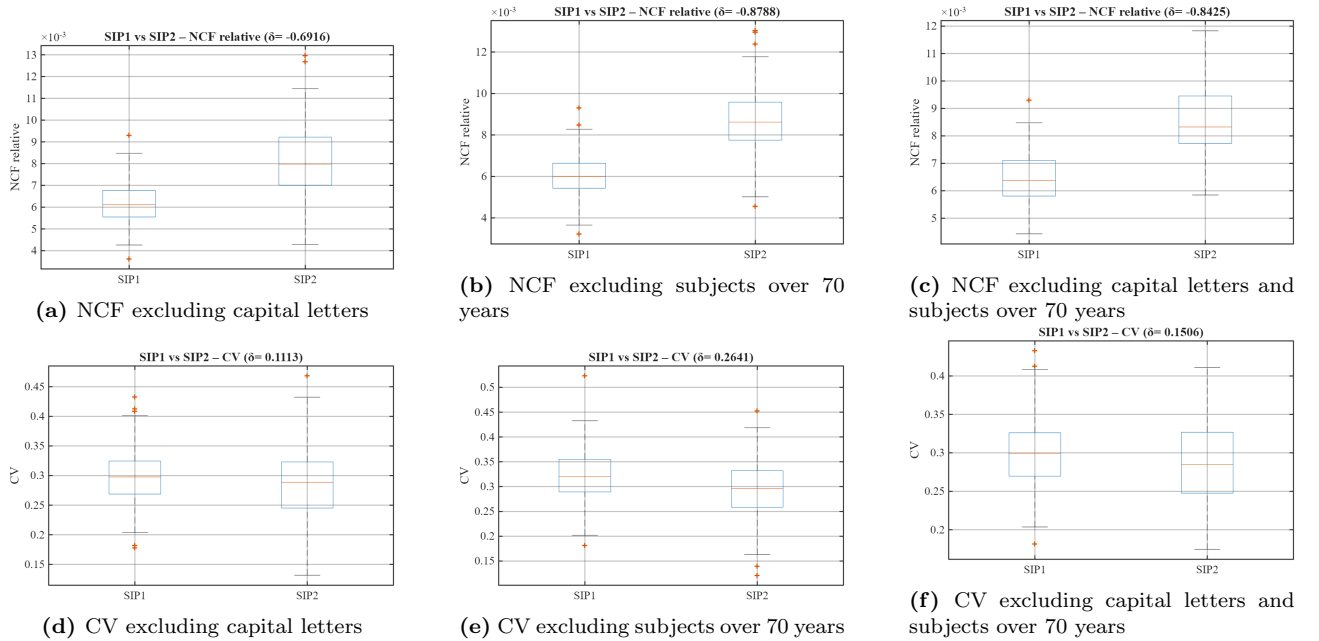


Figure 13.3: Box plots comparing the number of changes in forces (NCF) and coefficient of variation (CV) features in SIP1 pressure signal and SIP2 Scaled and Rounded pressure signal when different test groups are excluded. Excluded test groups are capital letter, subjects over 70 years and simultaneous. The Cliff δ value is visualized on the box plots.

13.3 SIP2 Quantile Mapped

The box plot in Figure 13.4 are SIP2 Quantile Mapped pressure signal compared with SIP1 pressure signal consisting of box plots of the extracted features where the domain adaptation approach with quantile mapping with $numstep=22$ are applied to SIP2 pressure signal. The extracted features was mean, median, STD, CV, NCF, and amplitude range. At the box plot the δ value are also visualized.

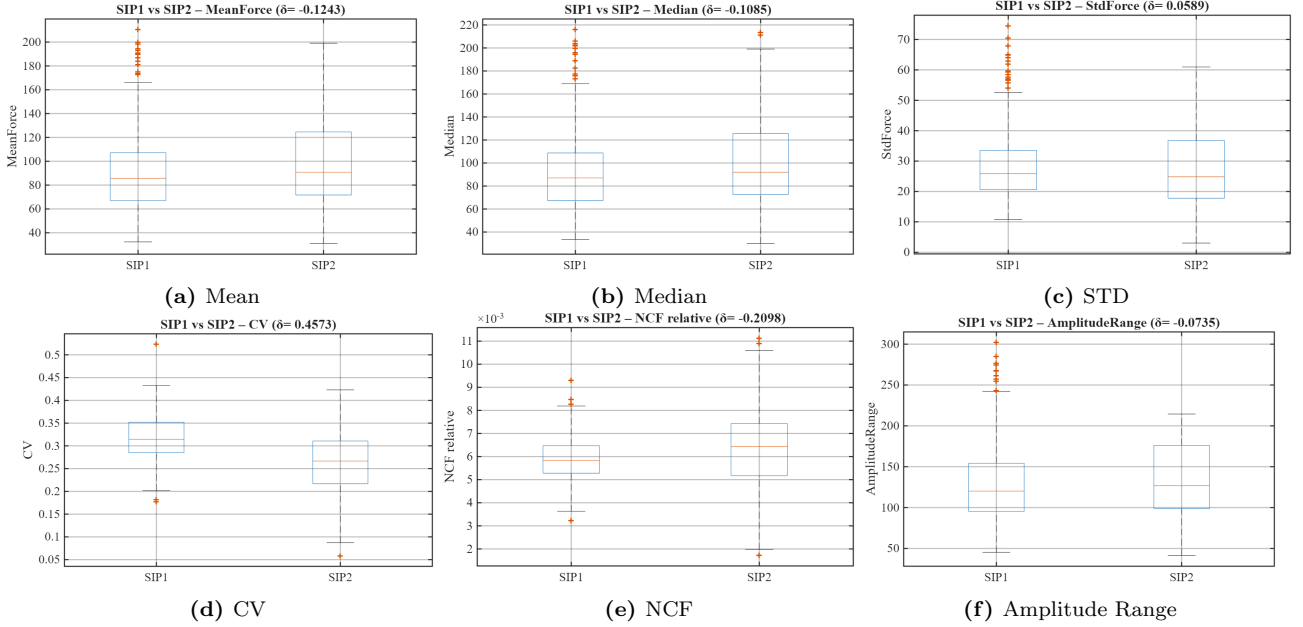


Figure 13.4: Box plot to comparing the features in pressure signal of SIP1 and SIP2 Quantile Mapped using $numstep=22$. The extracted features was mean, median, standard deviation (STD), coefficient of variation (CV), number of changes in force (NCF) and amplitude range. The Cliff δ value are visualized on the box plots.

13.3.1 Exclusion Experiments of SIP2 Quantile Mapped

The box plot in Figure 13.5 are SIP2 Quantile Mapped pressure signal compared with SIP1 pressure signal consisting of box plots of the CV and NCF where these features are tested in the exclusion experiments. The exclusion experiments consists of excluding capital letter, subjects over 70 years and simultaneous. At the box plot the Cliff δ value are also visualized.

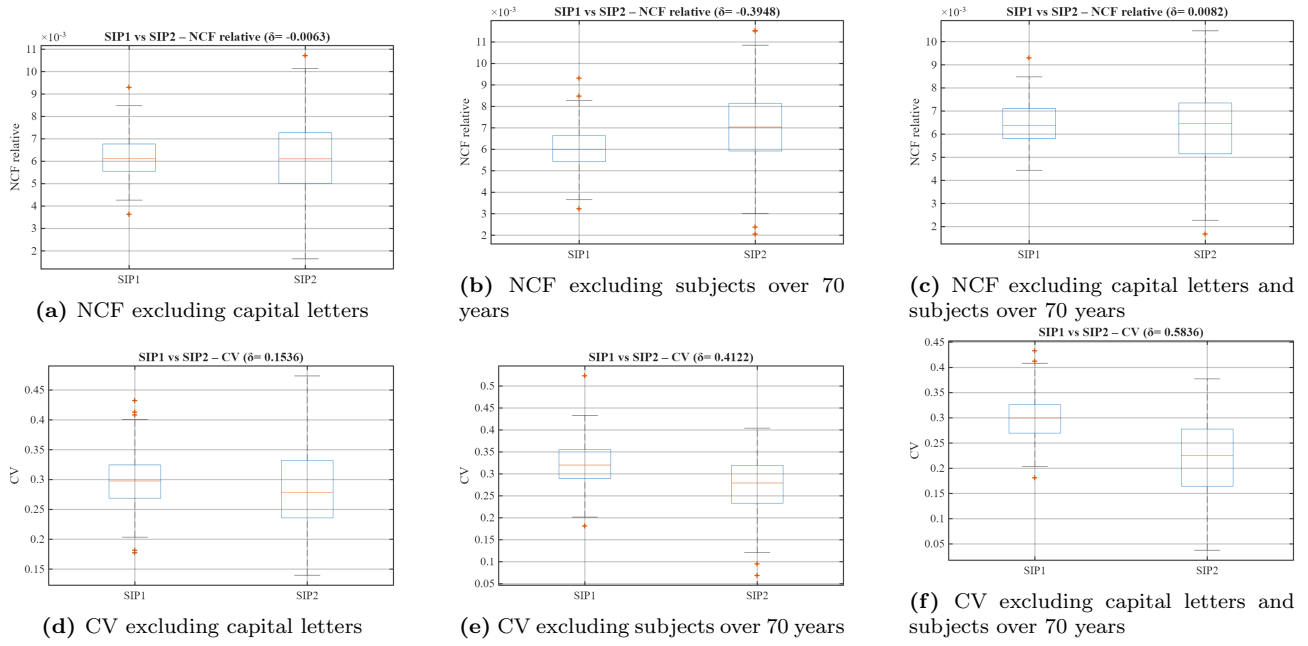


Figure 13.5: Box plots comparing the number of changes in forces (NCF) and coefficient of variation (CV) features in SIP1 pressure and SIP2 Quantile Mapped pressure signal, when different test groups are excluded. Excluded test groups is capital letter, subjects over 70 years and simultaneous. The Cliff δ value are visualized on the box plots.

13.4 Accelerometer and Gyroscope

The box plot in Figure 13.6 are SIP2 compared with SIP1 in the accelerometer and gyroscope features. Mean and median are computed from rectified signals, whereas STD is computed for both rectified and non-rectified signals for each X, Y, and Z axis. At the box plot Cliff δ value are also visualized.

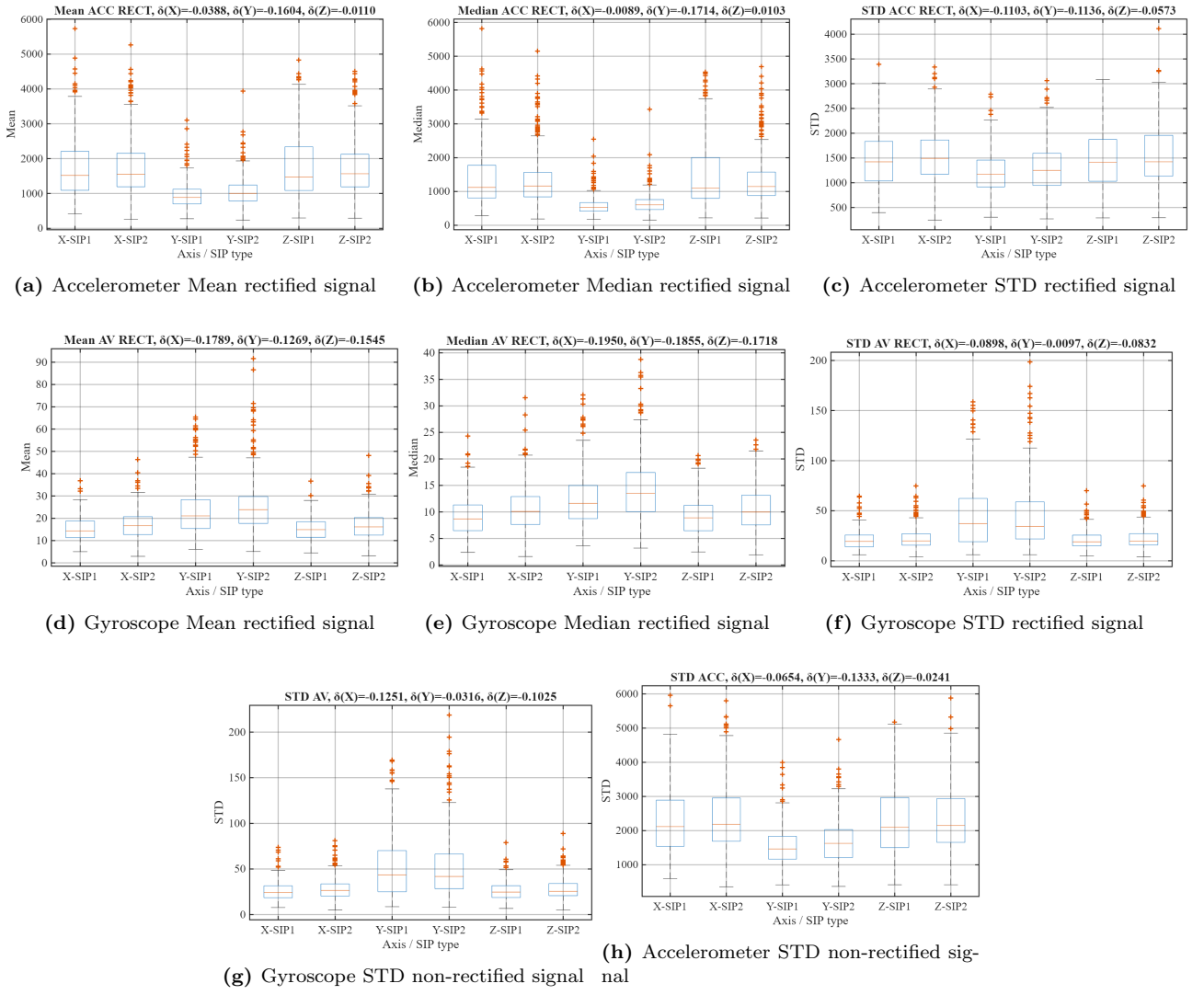


Figure 13.6: Box plot comparing accelerometer and gyroscope features in SIP1 and SIP2 where mean and median are calculated from rectified (RECT) signals, while standard deviation (STD) is calculated for both rectified (RECT) and non-rectified signals for each X, Y, and Z axis. The Cliff δ value are visualized on the box plots.

14. Time Series Classifier Results

In this appendix a performance metric table with all the performance metrics for the section and the Receiver Operating Characteristic(ROC), Confusion Matrix, and the School Grade Confusion Matrix are shown for each test in Section 14.1 Model Selection, Section 14.2 Time Series Classifier Pen Verification, and Section 14.3 Time Series Classifier Dataset Configuration Tests.

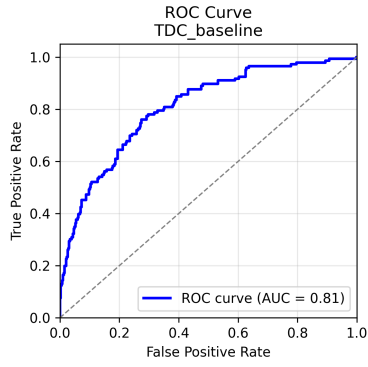
14.1 Model Selection

The test results of selecting the best model to proceed with the analysis are presented in this section. Table 14.1 shows the performance metrics accuracy, precision, recall, F1-score, and AUC for all tested models. The tested models are MINIROCKET-HYDRA, MultiROCKET-HYDRA (MR-HYDRA) and QUANT. The models were tested on the typical developing children (TDC) dataset, only including cursive writing style, with accelerometer and gyroscope features. The models are trained and tested with and without balanced class weights. Below the table the ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown for each test.

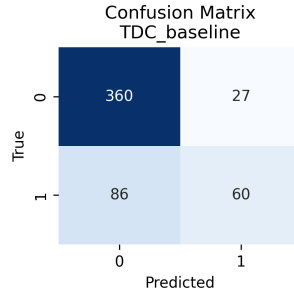
Model	Balanced	Acc.	Prec.	Recall	F1	AUC
MINIROCKET-HYDRA	No	78.79	68.96	41.09	51.50	0.81
MR-HYDRA	No	76.36	60.00	41.09	48.78	0.78
QUANT	No	78.79	70.88	38.35	49.77	0.80
MINIROCKET-HYDRA	Yes	70.54	46.62	52.05	49.19	0.74
MR-HYDRA	Yes	72.04	48.96	48.63	48.79	0.71
QUANT	Yes	78.23	69.23	36.98	48.21	0.80

Table 14.1: Overview of the test results on the trained time series classifiers: MINIROCKET-HYDRA, MultiROCKET-HYDRA (MR-HYDRA) and QUANT. All time series classifiers were trained and tested on the TDC-baseline data, including only the TDC dataset with accelerometer and gyroscope data with and without balanced class weights. The shown performance metrics are Accuracy(Acc.), Precision(Prec.), Recall, F1-Score(F1), and Area Under the Curve(AUC).

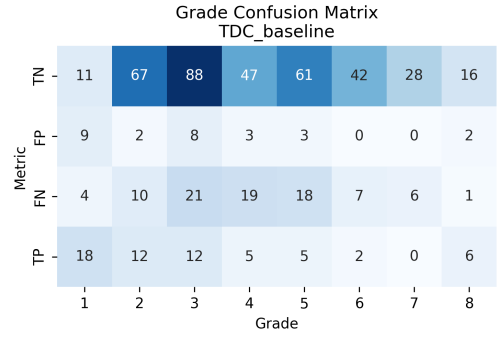
MINIROCKET-HYDRA ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown in Figure 14.1



(a) ROC



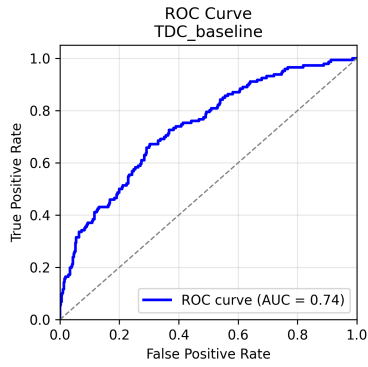
(b) Confusion Matrix



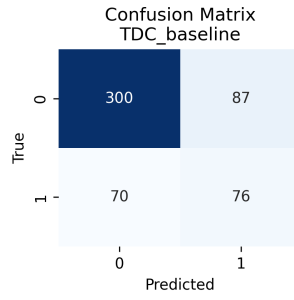
(c) Grade Confusion Matrix

Figure 14.1: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

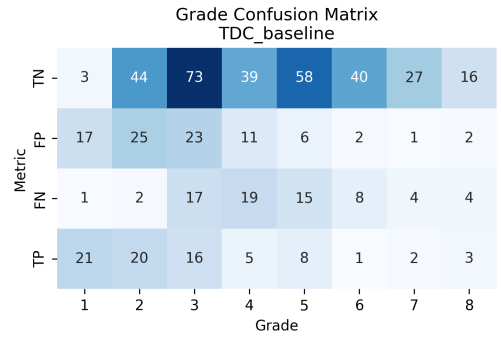
MINIROCKET-HYDRA with balanced class weights ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown in Figure 14.2



(a) ROC



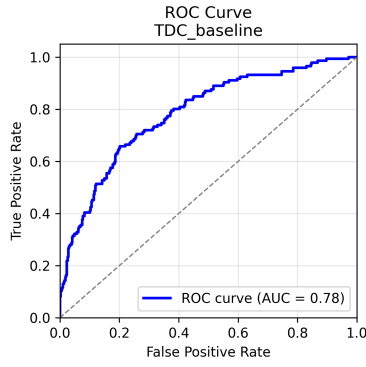
(b) Confusion Matrix



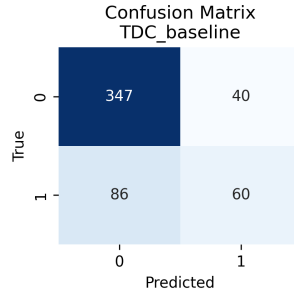
(c) Grade Confusion Matrix

Figure 14.2: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights chosen as the primary model, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

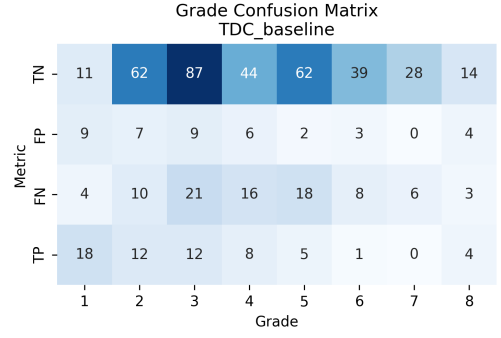
MR-HYDRA ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown in Figure 14.3



(a) ROC



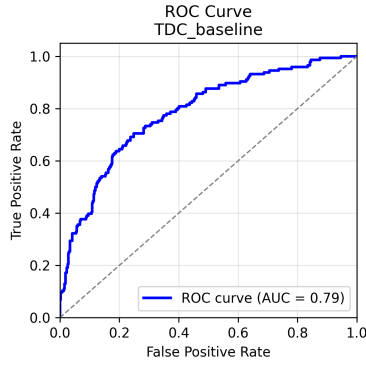
(b) Confusion Matrix



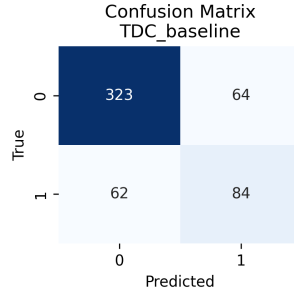
(c) Grade Confusion Matrix

Figure 14.3: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MR-HYDRA, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

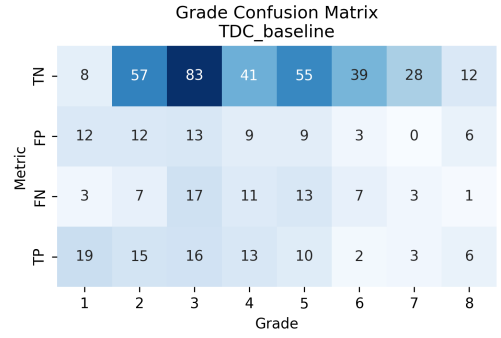
MR-HYDRA with balanced class weights ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown in Figure 14.4



(a) ROC



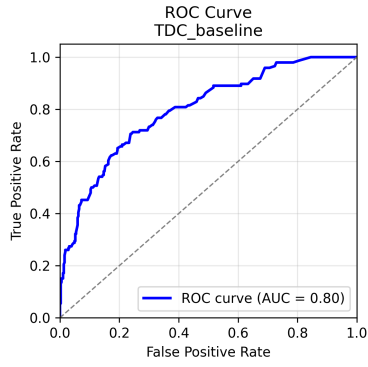
(b) Confusion Matrix



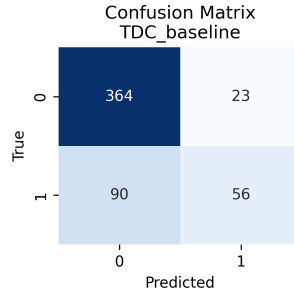
(c) Grade Confusion Matrix

Figure 14.4: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MR-HYDRA with balanced class weights, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

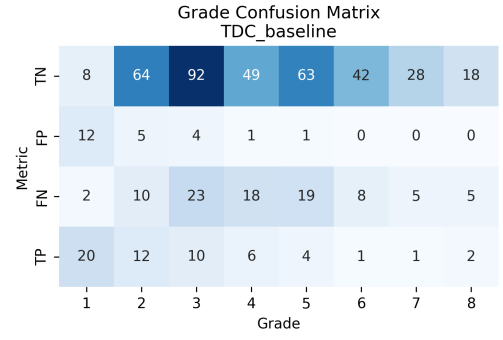
QUANT ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown in Figure 14.5



(a) ROC



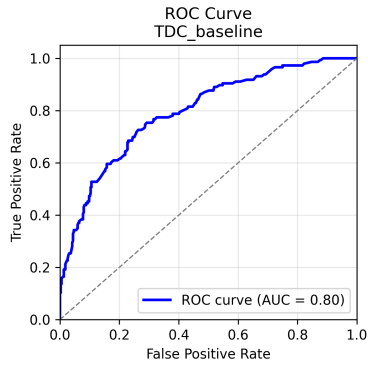
(b) Confusion Matrix



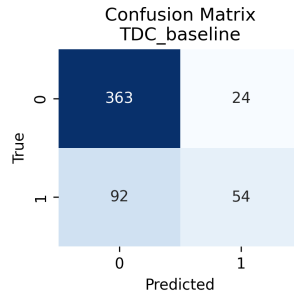
(c) Grade Confusion Matrix

Figure 14.5: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for QUANT, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

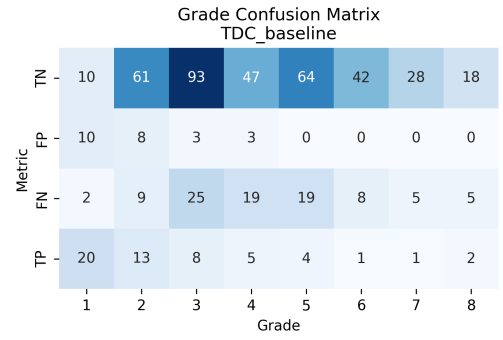
QUANT with balanced class weights ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown in Figure 14.6



(a) ROC



(b) Confusion Matrix



(c) Grade Confusion Matrix

Figure 14.6: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for QUANT with balanced class weights, trained and tested on the TDC_baseline dataset including only TDC dataset with accelerometer and gyroscope: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

14.2 Time Series Classifier Pen Verification

In this section, the test results of the time series classifier pen verification are shown. The performance metrics are collected in Table 14.2. Below the table, ROC and Confusion Matrix are shown for each test.

Dataset	Performance metrics					Configuration			
	Acc.	Precision	Recall	F1	AUC	ASIP1	ASIP2	AG	Press
SIP_baseline	79.53	60.00	63.64	61.76	0.85	X	X	X	
SIP_P	100	100	100	100	100	X	X	X	X

Table 14.2: The results of the MINIROCKET-HYDRA classifier with balanced class weights trained and tested on the ASIP1 and ASIP2 datasets, including accelerometer and gyroscope data(AG), depending on the dataset, pressure data(Press) was included. The shown performance metrics are Accuracy (Acc.), Precision(Prec.), Recall, F1-Score(F1), and Area Under the Curve(AUC).

In Figure 14.7, the test for **SIP_baseline** is performed where the ROC curve and confusion matrix show that the MINIROCKET-HYDRA classifiers are not able to perfectly separate SIP1 from SIP2. This indicates that the pen differences are slightly reduced, as both false positive and false negative cases occur.

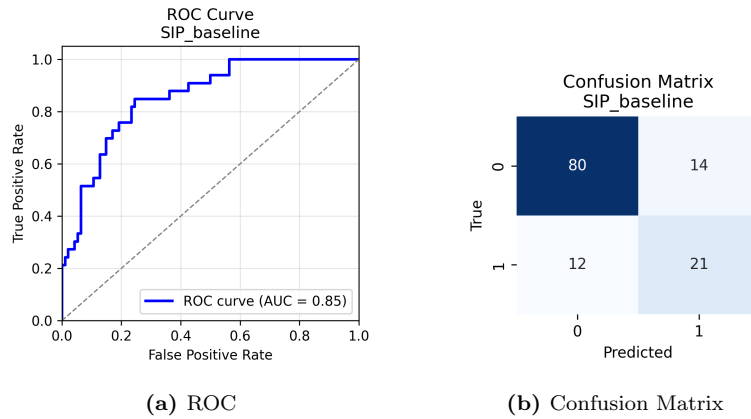


Figure 14.7: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced class weights, trained and tested on the SIP_baseline dataset configuration, including accelerometer and gyroscope data from ASIP1 and ASIP2. a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

In Figure 14.8, the test for **SIP_P** is performed. Where the ROC curve and confusion matrix show that when pressure data is included, the MINIROCKET-HYDRA model could perfectly separate ASIP1 from ASIP2, and the pen differences are not reduced.

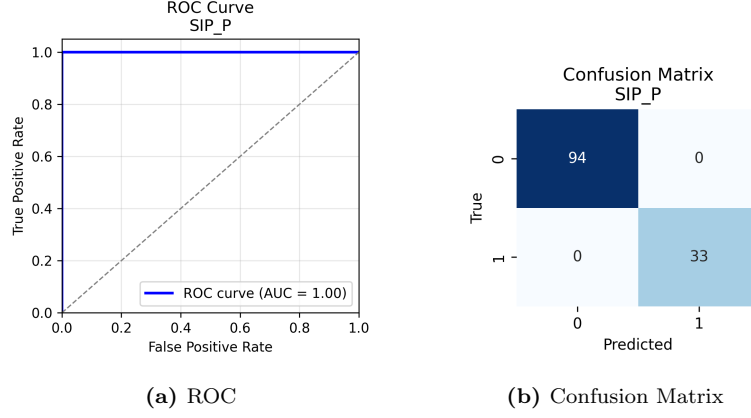


Figure 14.8: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced class weights, trained and tested on the SIP_P dataset configuration, including accelerometer, gyroscope, and pressure data from ASIP1 and ASIP2. a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

14.3 Times Series Classifier Dataset Configuration Tests

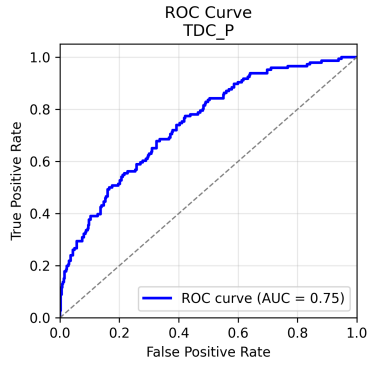
The test results of selecting the dataset configuration are presented in this section. Table 14.3 shows the performance metrics accuracy, precision, recall, F1-score, and AUC for all tested dataset configurations. The dataset configurations were tested with the MINIROCKET-HYDRA balanced class weights classifier. Below the table, the ROC, Confusion Matrix, and the School Grade Confusion Matrix are shown for each test.

Dataset	Performance metrics					Configuration				
	Acc.	Prec.	Recall	F1	AUC	TDC	CD	AG	Press	Grade
TDC_baseline	70.54	46.62	52.05	49.19	74	X		X		
TDC_P	72.04	49.06	54.10	51.46	75	X		X	X	
TDC_G	74.67	53.03	65.75	58.71	81	X		X		X
TDC_P_G	76.54	55.73	69.86	62.00	81	X		X	X	X
TDC_CD	70.47	50.54	55.75	53.02	74	X	X	X		
TDC_CD_P	73.55	55.75	55.75	55.75	76	X	X	X	X	
TDC_CD_G	73.36	54.89	61.21	57.87	78	X	X	X		X
TDC_CD_P_G	75.18	58.43	58.78	58.61	79	X	X	X	X	X

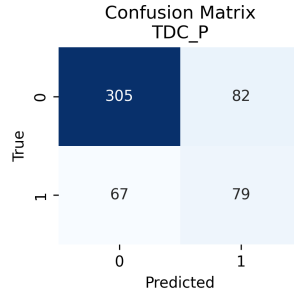
Table 14.3: The results of the MINIROCKET-HYDRA classifier with balanced class weights trained and tested on dataset feature configurations with accelerometer and gyroscope data(AG) and combinations of including pressure data(Press) and school grade information(Grade). The shown performance metrics are Accuracy (Acc.), Precision(Prec.), Recall, F1-Score(F1), and Area Under the Curve(AUC).

TDC_baseline results are the same as MINIROCKET-HYDRA with balanced weights, and its ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.2

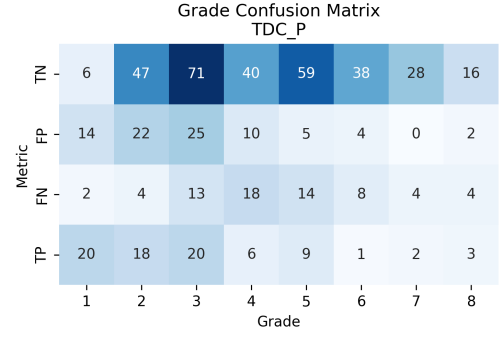
TDC_P ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.9



(a) ROC



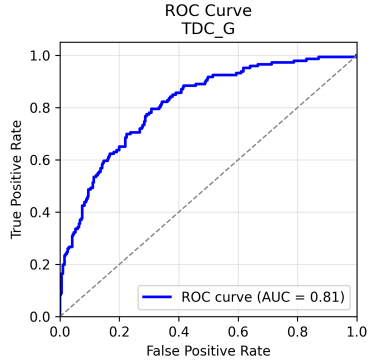
(b) Confusion Matrix



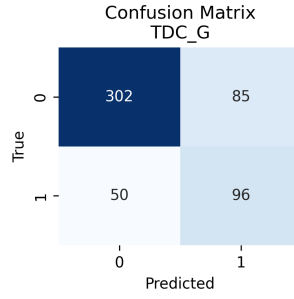
(c) Grade Confusion Matrix

Figure 14.9: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TDC_P dataset configuration including TDC dataset with accelerometer, gyroscope and pressure data: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

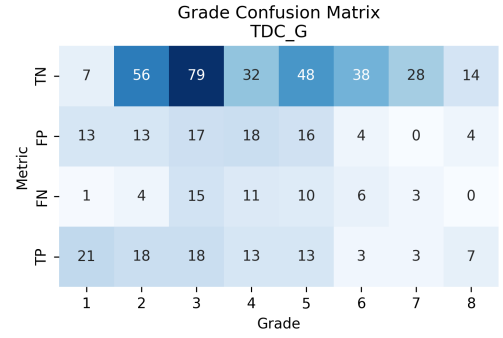
TDC_G ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.10



(a) ROC



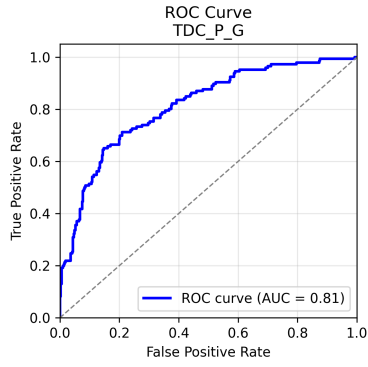
(b) Confusion Matrix



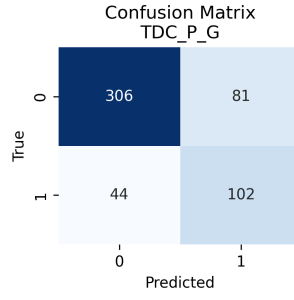
(c) Grade Confusion Matrix

Figure 14.10: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TDC_G dataset configuration including TDC dataset with accelerometer, gyroscope and school grade information data: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

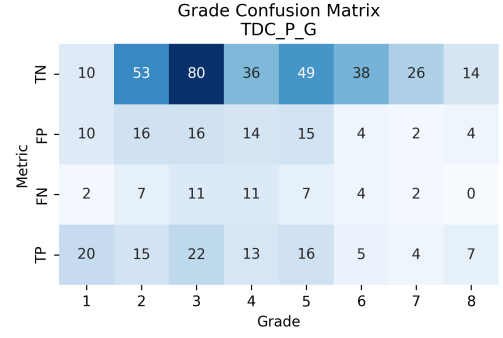
TDC_P_G ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.11



(a) ROC



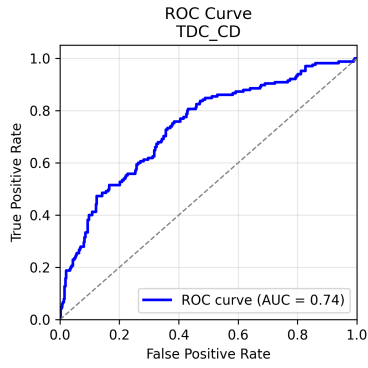
(b) Confusion Matrix



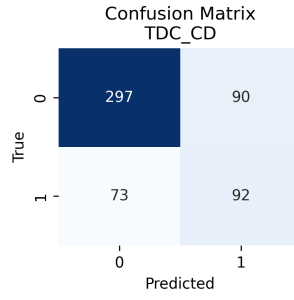
(c) Grade Confusion Matrix

Figure 14.11: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TDC_P_G dataset configuration including TDC dataset with accelerometer, gyroscope, pressure data, and school grade information data: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

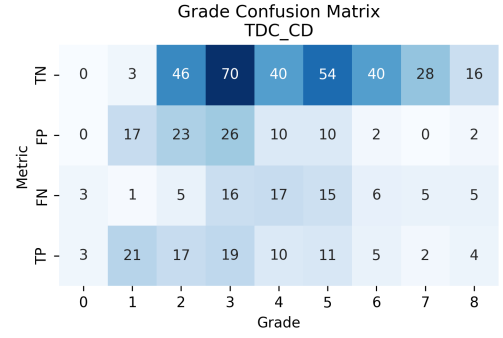
TDC_CD ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.12



(a) ROC



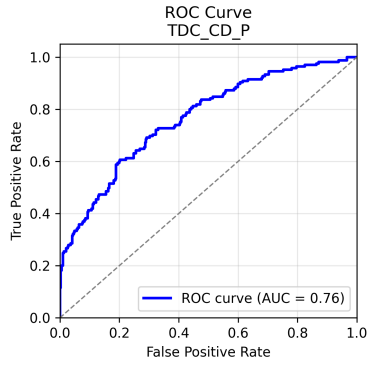
(b) Confusion Matrix



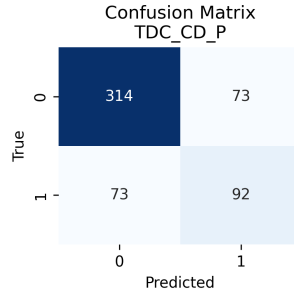
(c) Grade Confusion Matrix

Figure 14.12: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TD_CD dataset configuration including the TDC dataset and the CD dataset with accelerometer and gyroscope data: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

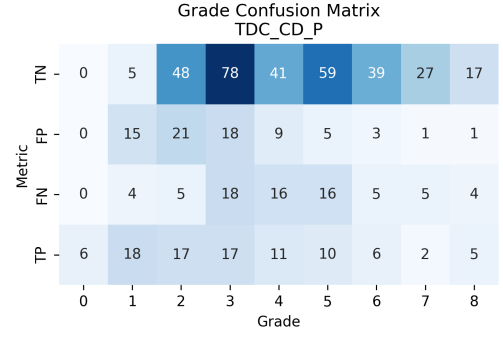
TDC_CD_P ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.13



(a) ROC



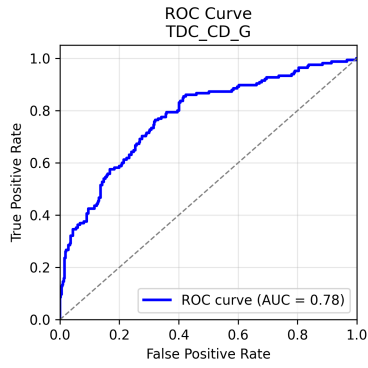
(b) Confusion Matrix



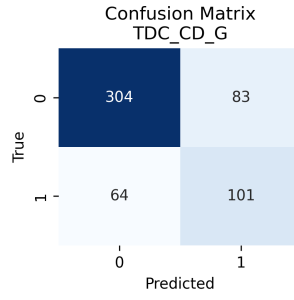
(c) Grade Confusion Matrix

Figure 14.13: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TDC_CD_P dataset configuration including the TDC dataset and the CD dataset with accelerometer, gyroscope, and pressure data: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

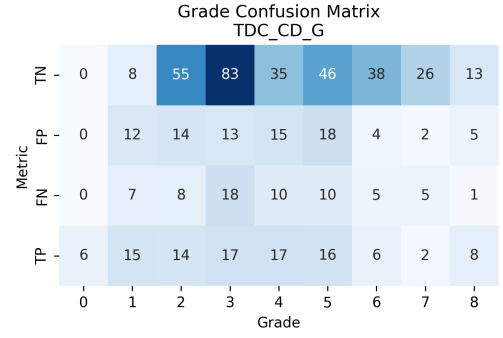
TDC_CD_G ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.14



(a) ROC



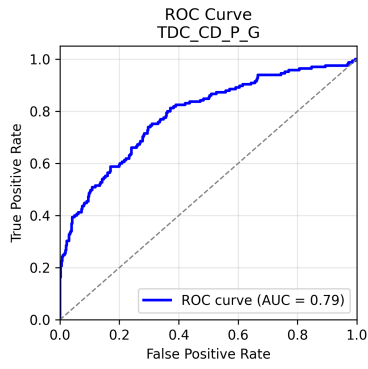
(b) Confusion Matrix



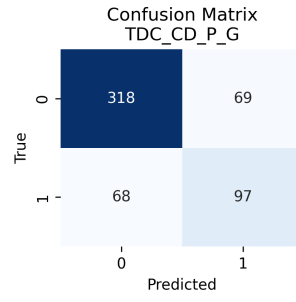
(c) Grade Confusion Matrix

Figure 14.14: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TDC_CD_G dataset configuration including the TDC dataset and the CD dataset with accelerometer, gyroscope, and school grade information: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

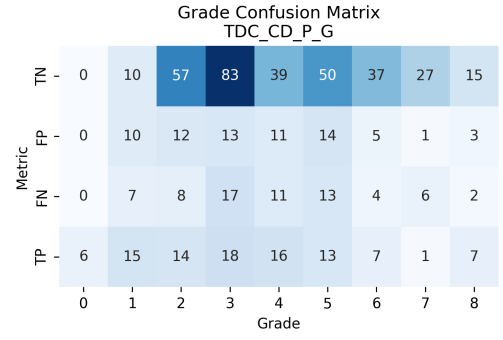
TDC_CD_P_G ROC, Confusion Matrix, and school grade confusion matrix are shown in Figure 14.11



(a) ROC



(b) Confusion Matrix



(c) Grade Confusion Matrix

Figure 14.15: Receiver Operating Characteristic(ROC), Confusion Matrix and School Grade Confusion Matrix shown for MINIROCKET-HYDRA with balanced weights, trained and tested on the TDC_CD_P_G dataset configuration including the TDC dataset and the CD dataset with accelerometer, gyroscope, pressure data, and school grade information: a) ROC, b) Confusion Matrix, c) School Grade Confusion Matrix showing true positive(TP), false negative(FN), false positive(FP), and true negative(TN) for each grade.

15. Collaboration Portfolio

In this appendix, the collaboration portfolio is presented. In Section 15.1, the establishment of the external collaboration is described. This is followed by a description of the collaboration with the supervisors and collaborators throughout this project, in Section 15.1.1. The project management is described in Section 15.2. Lastly, the reflection of the semester is presented in Section 15.3

15.1 Establishment of External Collaboration

As part of the 3rd semester of the Master of Science in Biomedical Engineering in Informatics, students were required to engage in collaborative projects, either domestically or internationally. The group investigated study abroad options by reaching out to researchers at Aalborg University about potential international opportunities. The process resulted in the establishment of a collaboration with Politecnico di Milano, motivated by academic ambitions and interest in the subject. The collaboration enabled the group to work within a research environment while maintaining close interaction with the research group.

The field of the project was biomedical engineering, and the group had the Associate Professor (Assoc. Prof.) Erika G. Spaich from Aalborg University as intern supervisor and Assoc. Prof. Simona Ferrante from Politecnico di Milano as external supervisor. Additionally, the group collaborated with PhD student Simone Toffoli and Junior Researcher Linda Greta Dui, both from Politecnico di Milano, as co-supervisors.

The initial brainstorming for the project proposal was conducted during a Teams meeting at the establishment of the collaboration with Assoc. Prof. Spaich and Assoc. Prof. Ferrante.

15.1.1 Collaboration with supervisors

The group maintained ongoing close communication with the co-supervisors at Politecnico di Milano where progression meetings were held as needed, and established a group chat to facilitate communication and address questions. The meetings were used for discussion about the direction of the project, guidance for the work of the project, and technical guidance according to the data. Additionally, status meetings were held with Assoc. Prof. Ferrante and the co-supervisors where the group presented the current work of the project and the next step followed by a discussion.

During the project, Assoc. Prof. Spaich guided to achieving the learning objectives, reviewed the

written material, and provided constructive feedback, which was subsequently discussed during the online supervisor meetings on Teams.

15.2 Project Management

For the overall project management, the group used a Gantt chart to keep track on milestones in the project as well as the corresponding weeks to work in the milestones. The Gantt chart can be seen in Figure 15.1.

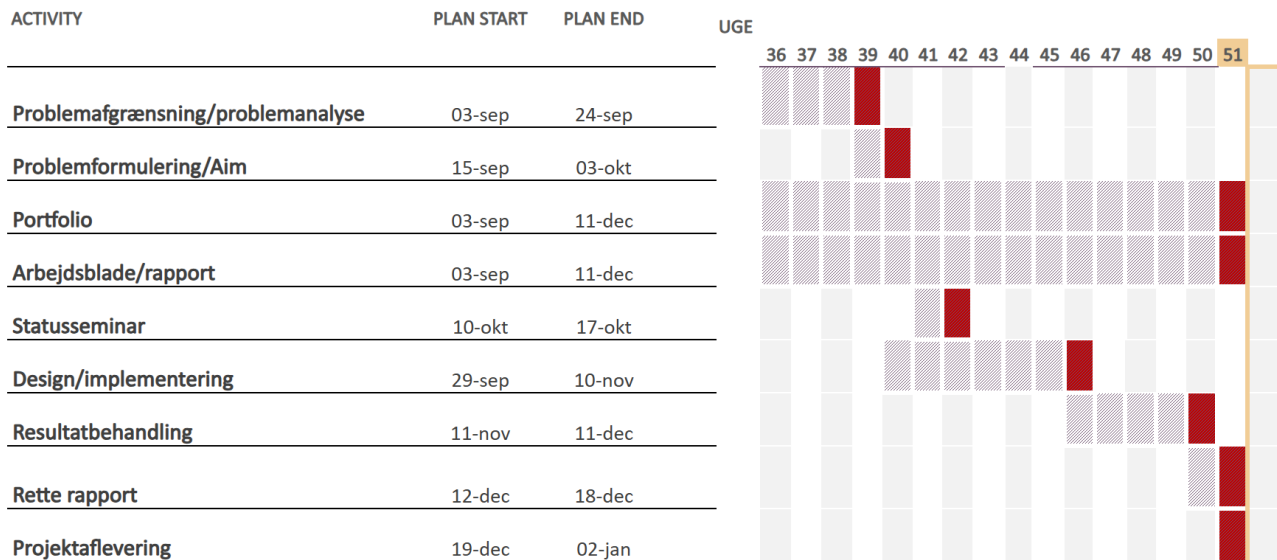


Figure 15.1: The original Gantt chart made at the beginning of the project for time management. Different colors are used to indicate when work should be done. Scratched boxes are work in progress, and red are the milestones for the given activity.

For the last weeks before the deadline of the project submission, a more detailed Gantt chart was made to specify the project management for the final activities to ensure reaching the deadline. The final sprint is specified with milestones of sending material for supervisor and deadlines to finish the specific parts of the project. The final sprint Gantt chart is visualized in Figure 15.2.

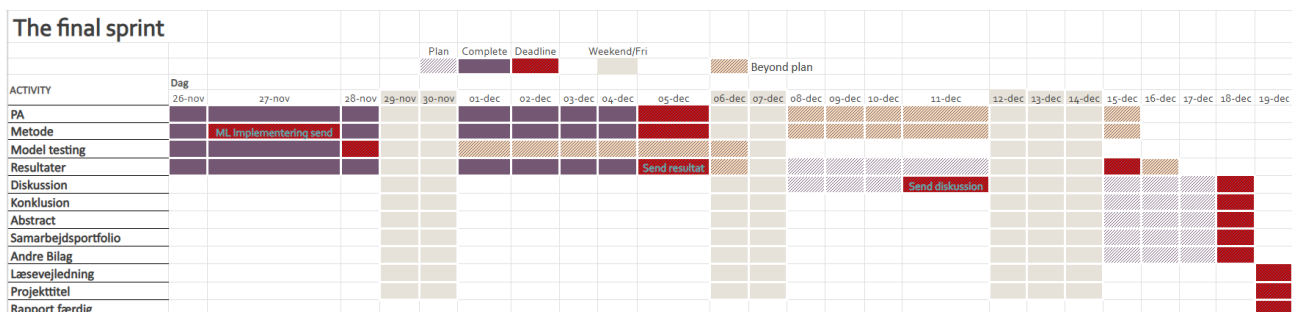


Figure 15.2: The final sprint Gantt chart representing the final work in the project management. Different colors are used to indicate when work should be done where scratched boxes are work in progress, and red are the milestones for the given activity.

For the daily work, an agenda was made, which defined the work the group wanted to do throughout

the day. The agenda was made the day before or at the beginning of the day, to ensure all group members agreed with the work plan for the day.

Besides the agenda, the group created a work plan, which gave an overview of tasks yet to be completed. The purpose of this list was to divide the main task into smaller, more manageable tasks. Whenever new tasks came up, the list was updated. Each group member could mark the progress of completing the task, where a color was used to mark the task when in progress, and another when done. The work plan contributed to creating an overview of the tasks that have not been done yet in the project, and collected the group's expectations of which tasks had to be made. Furthermore, the work plan facilitated a more fluent workflow where each group member could start a new task without involving the rest of the group.

Another management tool used in the group were status meeting which were held each week to discuss goals for the upcoming week and evaluate the previous week. This includes evaluating the effort and the achieved goals. The meeting also includes a time management aspect, where the group members discuss and plan the upcoming week as detail as possible and note it in the agenda for the days in the upcoming week. Furthermore, the meeting facilitated updates about any private matters that may affect the work effort. A template for the status meeting can be seen in Table 15.1.

Subject	Date: xx-xx-2025
Status on project	...
Project goals for the week	...
Project goals for next week	...
Evaluation of time schedule	...
Planning of next week	...
Well-being	...

Table 15.1: Template for the Status meetings

15.3 General Reflection

During the study abroad, the group decided to live together. In contrast to other semesters, this new daily routine of living as roommates required establishing clear boundaries between work hours and personal time as well as creating spaces where team members could have time alone. A shared calender were used to plan weekends together as well as visiting weekends. The purpose was to maintain a healthy balance between collaboration and personal space, ensuring both productivity and well-being within the group. In the collaboration with Politecnico di Milano, the group experienced different working hours and workplaces where resulted in staggered working hours and often online meetings due to working from home. This asynchronous collaboration promoted flexibility and individual responsibility, while coordination required increased focus on planning and structural collaboration.

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